

Q.1)

Ans) a

Exp) Option a is the correct answer.

Option a is correct: The term **Agrahara** refers to a **revenue-free or tax-exempt village or land grant given to Brahmins by kings or local rulers**, especially during the ancient and early medieval period (c. 600–1200 CE). These grants were made as a mark of religious merit and patronage of learning.

Key features of an Agrahara:

- The land (often an entire village) was **exempted from taxes and state dues**.
- It was usually granted to learned Brahmins proficient in the Vedas and sacred texts.
- Such settlements often became **centres of Sanskrit learning, Vedic study, and education**.
- The Brahmin beneficiaries often **enjoyed the right to collect these dues from the local people**.
- Agraharas were especially common under South Indian dynasties such as the Pallavas, Chalukyas, and Cholas.

Option b is incorrect: This option describes **Devadana** (also called **Devadaya**).

In this type of grant, **land was donated in the name of a deity or temple**, and the revenue generated from that land was used for temple maintenance, performance of daily rituals, festivals, and payment to temple staff such as priests and attendants.

Option c is incorrect: This refers to **Pallichchhanda**.

Pallichchhanda denotes **land grants made to Jain and Buddhist religious establishments** such as monasteries (viharas or pallis). It represents the **non-Brahmanical counterpart of Devadana**.

Option d is incorrect: This corresponds to **Shalabhoga**.

Shalabhoga refers to **land grants earmarked for the maintenance of educational institutions**, such as schools or centres of learning. The revenue from such land was used to support teachers, students, and infrastructure.

Source:) <https://ncert.nic.in/textbook/pdf/lehs102.pdf> Page: 41.

<https://ncert.nic.in/textbook/pdf/gess102.pdf> Page: 26.

<https://egyankosh.ac.in/bitstream/123456789/61927/1/Unit-13.pdf> Page: 204.

Subject:) History

Topic:) Ancient History

Subtopic:)

Q.2)

Ans) c

Exp) Option c is the correct answer.

The **Indian Councils Act 1861** was a key constitutional measure enacted by the British Parliament in 1861 during the **tenure of Lord Canning**, the **first Viceroy of India** after the Crown took over administration from the East India Company under the **Government of India Act, 1858**.

The Act was a significant constitutional measure enacted by the British Parliament **to reorganise governance in India after the Revolt of 1857**. It marked a **shift from strict centralisation to a limited form of administrative decentralisation**. **Statement I is correct:** The **Indian Councils Act 1861 empowered the Viceroy (Governor-General) to promulgate ordinances during emergencies without the concurrence of the Legislative Council**. Such ordinances had the force of law for a limited period (6 months).

This power remains a feature of the Indian Constitution today under Articles 123 (President) and 213 (Governor).

The **Act restored the legislative powers of the Presidencies of Bombay and Madras**. The Act initiated the **process of decentralization**. It **reversed the centralizing tendency** that had started with the **Regulating Act of 1773** and reached its peak with the **Charter Act of 1833** (which had **stripped Bombay and Madras of their legislative powers**).

Statement II is correct: This Act marked the **beginning of representative institutions** by associating Indians with the law-making process. It provided that the **Viceroy should nominate some Indians as non-official members** of his expanded council.

In 1862, Lord Canning nominated three Indians to his legislative council: the Raja of Benares, the Maharaja of Patiala, and Sir Dinkar Rao.

Source:) <https://egyankosh.ac.in/bitstream/123456789/19084/1/Unit-3.pdf> Page: 37.

A Brief History of Modern India (Spectrum) by Rajiv Ahir, Chapter: 26 - Constitutional, Administrative, and Judicial Developments, Page: 526.

Subject:) History

Topic:) Modern History

Subtopic:)

Q.3)

Ans) b

Exp) Option b is the correct answer.

Option a is incorrect: Formation of the Naujawan Bharat Sabha (1926)

The Naujawan Bharat Sabha was established in 1926 at Lahore by Bhagat Singh along with other revolutionary nationalists.

- It functioned as a **youth organization** aimed at **mobilizing young people, peasants and workers against colonial rule**, promoting patriotism, secularism, and revolutionary ideals, and creating a mass base for revolutionary activities, **particularly in Punjab**.
- **Option b is correct: Formation of the Swaraj Party (1923)**

The Swaraj Party was established in January 1923 by Chittaranjan Das and Motilal Nehru following the **suspension of the Non-Cooperation Movement by Mahatma Gandhi in 1922** after the **Chauri Chaura incident**.

- A section of Congress leaders believed that the nationalist movement should **enter the legislative councils** created under the reforms of the Government of India Act 1919 and obstruct colonial governance from within.
- This approach came to be known as the **“Council-Entry Programme”** and those advocating entry into legislative councils came to be known as the **‘Swarajists’**,
- The **Swarajists contested elections** to the central and provincial legislatures in 1923 and attempted to expose the limitations of constitutional reforms by blocking government measures, opposing budgets, and demanding greater self-government.

Option c is incorrect: Formation of the Hindustan Republican Association (1924)

The Hindustan Republican Association (HRA) was **founded in 1924** by revolutionaries such as **Ram Prasad Bismil, Sachindra Nath Sanyal and Jogesh Chandra Chatterjee**.

- Its objective was to **organize an armed revolutionary struggle against British rule** and establish a federal republic in India.
- The organization later gained prominence through revolutionary actions such as the **Kakori Train Robbery (1925)**.
- In 1928, it was reorganized into the **Hindustan Socialist Republican Association (HSRA)**.

Option d is incorrect: Establishment of the Simon Commission (1927)

The **Simon Commission** was **appointed in 1927** by the British Government under the **chairmanship of John Simon** to **review the working of the constitutional reforms** introduced by the **Government of India Act 1919**.

- Since **no Indian member was included**, it triggered widespread protests across the country with the slogan **“Simon Go Back.”**
- During protests against the commission in Lahore in 1928, **Lala Lajpat Rai** was seriously injured in a police lathi charge and later died.

Chronologically, the **sequence is:**

Swaraj Party (1923) → HRA (1924) → Bharat Naujawan Sabha (1926) → Simon Commission (1927)

Therefore, the **earliest event is the formation of the Swaraj Party (1923)**.

Source:) A Brief History of Modern India (Spectrum) by Rajiv Ahir, Chapter: 17 - Emergence of Swarajists, Socialist Ideas, Revolutionary Activities and Other New Forces & Chapter: 18 - Simon Commission and the Nehru Report.

<https://egyankosh.ac.in/handle/123456789/44285>

<https://digital.nios.ac.in/content/213en/Lesson-08.pdf>

<https://www.thehindu.com/opinion/op-ed/revolutionary-ideas-that-live-on/article26613250.ece>

Subject:) History

Topic:) Modern History

Subtopic:)

Q.4)

Ans) a

Exp) Option a is the correct answer.

The famous revolutionary pamphlet “The Philosophy of the Bomb” (1930) was written by Bhagwati Charan Vohra, a leading member of the Hindustan Socialist Republican Association (HSRA).

- The pamphlet defended the revolutionary use of violence against British colonial rule and articulated the ideological position of the HSRA.
- It was written as a direct response to the article “The Cult of the Bomb,” authored by Mahatma Gandhi in his journal Young India in 1930.
- In December 1929, revolutionaries from the HSRA attempted to bomb the train carrying Viceroy Lord Irwin (near Delhi) as a symbolic protest against British rule, though the Viceroy escaped unharmed.
- In the article “The Cult of the Bomb,” Gandhi strongly criticized revolutionary violence and argued that the use of bombs and assassinations was morally wrong and politically harmful to the Indian freedom struggle, advocating instead the path of non-violence (Ahimsa) and Satyagraha.
- In response, Bhagwati Charan Vohra’s pamphlet justified revolutionary action and criticized the exclusive reliance on non-violence. He defended their methods by saying that these violent methods were as important as Gandhiji’s non-violent methods.

Thus, “The Philosophy of the Bomb” was written in reply to Gandhi’s article “The Cult of the Bomb.”

Source:) <https://cmsadmin.amritkaal.nic.in/district-reopsitory-detail?8785>

<https://indianexpress.com/article/explained/bhagat-singh-revolutionary-terrorist-delhi-university-2779711/>

Subject:) History

Topic:) Modern History

Subtopic:)

Q.5)

Ans) c

Exp) Option c is the correct answer.

The Vijayanagara Empire (1336–1646) maintained a highly organized administrative hierarchy to manage its vast territories in South India. To ensure efficient governance and revenue collection, the empire was divided into several tiers, ranging from the central capital to the smallest village unit.

In ascending order (from smallest to largest), the hierarchy is as follows:

1. **Grama (Village):** This was the basic, smallest unit of administration. Each village was managed by a body of local representatives.
2. **Sthala (Sub-district):** A group of villages formed a Sthala. It is roughly equivalent to a modern-day sub-district or the Mughal Pargana.
3. **Nadu (District):** Multiple Sthalas were grouped together to form a Nadu. This unit is comparable to a modern district or the Mughal Sarkar.
4. **Mandalam (Rajya):** This was the largest administrative division, equivalent to a province, modern day state or the Mughal Suba. The empire was divided into several Mandalams (like Cholanmandalam or Jayamkonda Cholanmandalam), each usually governed by a member of the royal family or a high-ranking noble known as a Mandaleshvara.

Therefore, the correct ascending order (smallest → largest) is:

Sthala → Nadu → Mandalam

Hence, **option c) is the correct answer.**

Source:) <https://egyankosh.ac.in/bitstream/123456789/22074/1/Unit-20.pdf> Page: 21.

History of Medieval India by Satish Chandra, Chapter: 9 - The Age of Vijayanagara and the Bahmanids, and the Coming of the Portuguese (Circa 1350–1565), Page: 168.

Subject:) History

Topic:) Medieval History

Subtopic:)

Q.6)

Ans) c

Exp) Option c is the correct answer.

The **All-India Kisan Sabha (AIKS)** was **established in 1936** at the **Lucknow Session of the Indian National Congress, 1936** as an organised platform to mobilise peasants and articulate their demands against oppressive agrarian practices such as high rents, illegal cesses, forced labour, and zamindari exploitation.

I. Swami Sahajanand Saraswati became the **first President** of the AIKS. He was a prominent peasant leader from Bihar and had earlier founded the Bihar Provincial Kisan Sabha (1929), which played a major role in mobilising peasants.

II. N. G. Ranga, an important **peasant leader from Andhra**, was **appointed the first General Secretary** of the AIKS. He later emerged as one of the foremost advocates of farmers' rights in India.

III. Also, a kisan manifesto was issued and a **periodical under Indulal Yagnik started**.

Source:) <https://thewire.in/history/at-home-in-the-august-house-n-g-rangas-soul-was-with-the-peasant>

<https://kisansabha.org/wp-content/uploads/2019/07/hannanbook.pdf> Page: 14.

<https://egyankosh.ac.in/bitstream/123456789/21059/1/Unit-28.pdf> Page: 8.

Subject:) History

Topic:) Modern History

Subtopic:)

Q.7)

Ans) d

Exp) Option d is the correct answer.

Option a is correct: The **Ilbert Bill (1833)** was introduced to **remove racial discrimination in the colonial judicial system**. Under the existing legal arrangement, **Indian district magistrates and sessions judges were not permitted to try European British subjects in criminal cases**, even if they were of equivalent rank to European judges.

The **bill proposed to empower Indian judges to exercise the same criminal jurisdiction over Europeans**.

Option b is correct: The **bill was introduced in 1883 during the viceroyalty of Lord Ripon**, who is often regarded as one of the most liberal British administrators in India.

The measure was **drafted and placed before the Imperial Legislative Council by Sir Courtenay Ilbert**, the Law Member of the Council.

Lord Ripon supported the proposal as part of his broader policy of administrative reform and greater equality in the colonial system.

Option c is correct: The **proposal provoked intense agitation among Europeans living in India**, particularly British planters, merchants, and officials.

Many Europeans argued that Indian judges were not competent to try European defendants, revealing the deep racial prejudices prevalent in colonial society.

Public meetings, petitions, and campaigns were organized against the bill, and this agitation came to be described by some historians as a **"White Mutiny."**

Because of this pressure, the **government eventually modified the bill before its enactment in 1884**.

Option d is incorrect: The Indian National Congress was established in 1885, whereas the Ilbert Bill controversy took place in 1883–84.

Therefore, the Congress did not exist at the time of the controversy and **could not have passed any resolution supporting the bill.**

Although the episode did help expose racial discrimination in colonial administration and influenced the growth of political consciousness among educated Indians, it preceded the formation of the Congress by about two years.

Source:) <https://ncert.nic.in/textbook/pdf/hess208.pdf> Page: 94.

<https://egyankosh.ac.in/bitstream/123456789/22018/1/Unit-14.pdf>

<https://egyankosh.ac.in/bitstream/123456789/19084/1/Unit-3.pdf>

<https://egyankosh.ac.in/bitstream/123456789/19933/3/Unit-9.pdf>

Subject:) History

Topic:) Modern History

Subtopic:)

Q.8)

Ans) a

Exp) Option a is the correct answer.

Statement I is correct: The Communist Party of India (CPI) was indeed **formed in 1920, in Tashkent** (now in Uzbekistan) by **M.N. Roy, Abani Mukherji** and others. This was done under the aegis of the Comintern (Communist International) to organize a revolutionary struggle against British imperialism. In 1925, the **Indian Communist Conference at Kanpur** formalised the foundation of the CPI.

Statement II is correct: Leaders such as **M. N. Roy, S. A., Muzaffar Ahmad**, and **P. C. Joshi** were central to the growth and consolidation of the communist movement in India.

Statement III is incorrect: The Communist Party of India **did not support the Quit India Movement (1942).**

Initially, the CPI had opposed the Second World War, considering it an imperialist conflict. However, after Nazi Germany invaded the Soviet Union in 1941, the party changed its stance and began supporting the Allied war effort, describing it as a **“People’s War” against fascism**. Consequently, the CPI opposed the Quit India Movement and instead urged cooperation with the British war effort.

Knowledge base:

1. The Meerut Conspiracy Case (1929) was initiated by the British colonial government to curb the spread of communist ideology and labour activism in India.

A number of trade unionists and communist leaders, such as **S. A. Dange, Muzaffar Ahmed, and Shaukat Usmani** were **arrested and tried** on charges of conspiring to overthrow British rule through revolutionary means. The case drew international attention and highlighted the growing influence of communist and trade union movements in India.

2. Communists played an important role in mobilizing workers and peasants through mass organizations such as:

- **All India Trade Union Congress** – founded in 1920, which became a major platform for labour organization and industrial strikes.
- **All India Kisan Sabha** – established in 1936 to organize peasants and address agrarian issues such as high rents, indebtedness, and exploitation by landlords.

Through these organizations, the communist movement significantly influenced labour and peasant politics in the late colonial period.

Source:)

A Brief History of Modern India (Spectrum) by Rajiv Ahir, Chapter: 17 – Emergence of Swarajists, Socialist Ideas, Revolutionary Activities, and Other New Forces, Page: 363.

<https://indianexpress.com/article/explained/explained-history/100-years-of-cpi-how-indias-communist-movement-came-to-be-10439181/>

<https://www.thehindu.com/news/national/carrying-the-red-flag-into-the-next-century/article70437256.ece>

<https://indianexpress.com/article/opinion/columns/100-years-of-the-communist-party-of-india-its-role-in-freedom-struggle-and-independent-india-9744395/>

<https://frontline.thehindu.com/the-nation/origins-of-communist-party-of-india-in-tashkent/article32952125.ece>

Subject:) History

Topic:) Modern History

Subtopic:)

Q.9)

Ans) d

Exp) Option d is the correct answer.

Option d is correct: Karle Caves (Maharashtra)

The **Karle (Karla) Caves** were **constructed around the 2nd century BCE during the Satavahana period**.

They represent one of the earliest and largest surviving Buddhist rock-cut caves in India.

Karla cave complex is famous for its **"Grand Chaitya"**. The Grand Chaitya hall is of apsidal vault roof type with many pillars and a stupa at the back.

The monument is associated with **Hinayana Buddhism**, and inscriptions indicate donations from merchants and guilds connected with the western Deccan trade routes.

Option a is incorrect: Dashavatara Temple, Deogarh (Uttar Pradesh)

The **Dashavatara Temple at Deogarh** was **built during the Gupta period (6th century CE)**.

It is **one of the earliest structural Hindu temples** made of stone and is **dedicated to Vishnu**.

The temple is notable for its **Nagara-style architecture**, square sanctum, and sculptural panels depicting themes such as **Anantasayana Vishnu**.

Option b is incorrect: Chaukhandi Stupa (Uttar Pradesh)

The Chaukhandi Stupa **marks the spot where Gautama Buddha** is believed to have **met his first five disciples** before delivering his first sermon at Sarnath.

The original stupa structure dates to the **Gupta period (around the 5th century CE)**.

The octagonal tower seen today was added in the **16th century** during the Mughal period by **Govardhan** to commemorate the visit of **Humayun**.

Option c is incorrect: Udayagiri Caves (Madhya Pradesh)

These are a group of **20 rock-cut caves** (primarily Hindu Vaishnava, with one Jain) **near Vidisha in Madhya Pradesh**.

These caves were **excavated in the early 5th century CE** during the **reign of Chandragupta II**.

They represent one of the earliest Hindu rock-cut cave complexes, with notable sculptures like the **Varaha (boar) incarnation of Vishnu relief**.

Hence, **among the given sites, Karle Caves (Maharashtra), constructed during the 2nd century BCE, is the earliest**.

Source:) <https://indianexpress.com/article/lifestyle/destination-of-the-week/5-historic-caves-in-madhya-pradesh-you-must-visit-to-step-into-the-past-10252254/>

<https://theprint.in/feature/around-town/karle-cave-complex-buddhist-donations/2756881/>

<https://www.telegraphindia.com/my-kolkata/places/sarnath-in-uttar-pradesh-a-sneak-peek-into-the-rich-cultural-heritage-of-buddhism-in-india/cid/1951082>

Forum IAS RedBook – Indian Art & Culture, Chapter: 3 - Indian Architecture and Sculpture

Subject:) Art and Culture

Topic:) Architecture

Subtopic:)

Q.10)

Ans) a

Exp) Option a is the correct answer.

Statement I is correct: The Constitution of India was formally adopted on 26 November 1949 by the Constituent Assembly of India after nearly three years of deliberations. However, instead of bringing it into force immediately, the Assembly decided that the Constitution would commence on 26 January 1950.

- During the interim period, some provisions such as those relating to citizenship, elections, and provisional parliament came into effect on 26 November 1949.
- The entire Constitution became operational on 26 January 1950, marking the transformation of India into a sovereign democratic republic and the end of the Government of India Act, 1935 as the governing framework.

Statement II is correct: At the Lahore Session of the Indian National Congress in 1929, presided over by Jawaharlal Nehru, the Indian National Congress adopted the historic Purna Swaraj Declaration, rejecting the earlier demand for Dominion Status and proclaiming complete independence (Purna Swaraj) as the ultimate goal of the national movement.

- Following this resolution, the Congress called upon Indians across the country to observe 26 January 1930 as "Independence Day", with public pledges to work for complete independence from British rule. Hence, on 26th January 1930, Indians across the country observed Purna Swaraj Day, committing themselves to the goal of complete self-rule and rejecting dominion status under British governance.
- For several years thereafter, 26 January continued to be commemorated by nationalists as the symbolic day of India's freedom struggle.

Why Statement II explains Statement I

When India finally became independent in 1947, the leaders of the Constituent Assembly of India wanted the commencement of the Constitution of India to retain a historical connection with the freedom movement.

Therefore, even though the Constitution had been adopted on 26 November 1949, they deliberately postponed its enforcement until 26 January 1950 to commemorate the historic observance of the Purna Swaraj Day on 26th January, 1930.

Thus, Statement II provides the historical reason behind the choice of 26 January for the commencement of the Constitution.

Therefore, Both Statement I and Statement II are correct and Statement II explains Statement I.

Source:) <https://www.pib.gov.in/FeaturesDeatils.aspx?NoteId=157078&ModuleId=2®=3&lang=1>

<https://indianexpress.com/article/explained/everyday-explainers/january-26-india-republic-day-history-reason-date-9128464/>

<https://knowindia.india.gov.in/my-india-my-pride/journey-of-indian-republic.php>

Subject:) History

Topic:) Modern History

Subtopic:)

Q.11)

Ans) b

Exp) Option b is the correct answer.

Ancient Indian philosophical thought developed through a rich and systematic tradition known as the Darshanas (schools of philosophy), which sought to understand the nature of reality, knowledge, and liberation (moksha). These schools broadly emerged during the later Vedic period and were refined over centuries through rigorous debate and textual traditions.

They are classified into two major categories:

- **Orthodox (Astika) schools** - These accept the authority of the Vedas. There are six classical systems: Nyaya, Vaisheshika, Samkhya, Yoga, Mimamsa, and Vedanta.
- **Heterodox (Nastika) schools** - These reject the authority of the Vedas. Prominent among them are Buddhism, Jainism, and Charvaka (Lokayata).

Information in row I is incorrect: Nyaya:

The Nyaya school of philosophy was **founded by Gautama, not Jaimini**. Jaimini is actually associated with the Mimamsa school.

- Nyaya primarily deals with logic (Anvikshiki) and epistemology, and it recognizes four valid means of knowledge (Pramanas): **Perception** (Pratyaksha), **Inference** (Anumana), **Comparison** (Upamana), and **Testimony** (Shabda).
- Nyaya holds that **correct knowledge obtained through valid pramanas leads to liberation (moksha)**.

Since the founder is incorrectly matched, this pair is incorrect.

Information in row II is correct: Vaisheshika

The Vaisheshika school was **founded by Kanada**. It is known for its **atomistic theory of the universe (Paramanuvada)**. It postulated that all objects in the physical universe are reducible to a finite number of atoms. It holds that the smallest, indivisible, indestructible part of the world is an atom (anu). It argued that all physical things are a combination of the atoms of earth, water, fire, and air.

Information in row III is incorrect: Samkhya

The Samkhya school was indeed **founded by Kapila**, but the **core idea given is wrong**.

It **does not deny Purusha**; rather, the entire philosophy is built on the **interaction between Purusha and Prakriti**. It is **dualistic philosophy** that views **reality to be composed of purusa** (consciousness or spirit) and **prakriti** (nature), both of which are independent of the other.

Liberation occurs when **Purusha realizes its distinction from Prakriti**.

Information in row IV is correct: Yoga

The **Yoga school was systematized by Patanjali** in the **Yoga Sutras**.

Its central teaching is the **control of mind and body to attain liberation through Ashtanga Yoga (Eightfold Path)**:

1. Yama (ethical restraints)
2. Niyama (observances)
3. Asana (posture)
4. Pranayama (breath control)
5. Pratyahara (withdrawal of senses)
6. Dharana (concentration)
7. Dhyana (meditation)
8. Samadhi (complete absorption)

The aim is **chitta-vritti-nirodha** (cessation of mental fluctuations), **leading to liberation**.

Knowledge Base:

Philosophical Systems	Seers	Works
<i>Sāṃkhya</i>	Kapila	<i>Sāṃkhyasūtra</i>
<i>Yoga</i>	Patañjali	<i>Yogasūtra</i>
<i>Pūrva Mīmāṃsā</i>	Jaimini	<i>Mīmāṃsāsūtra</i>
<i>Uttara Mīmāṃsā</i> (<i>Vedānta</i>)	Bādarāyaṇa (<i>Vedavyāsa</i>)	<i>Vedāntasūtra</i> (<i>Brahmasūtra</i>)
<i>Nyāya</i>	Gautama	<i>Nyāyasūtra</i>
<i>Vaiśeṣika</i>	Kaṇāda	<i>Vaiśeṣikasūtra</i>

Source: <https://ncert.nic.in/textbook/pdf/keks102.pdf>

Forum IAS Red Book (Art & Culture), Chapter: 6 – Philosophy.

<https://egyankosh.ac.in/bitstream/123456789/8271/1/Unit-6.pdf>

Subject: Art and Culture

Topic:) Schools of Philosophy

Subtopic:)

Q.12)

Ans) c

Exp) Option c is the correct answer.

Option c is correct: Major Rock Edict XIII of Ashoka provides the most **explicit reference to his conquest of Kalinga during the Kalinga War** (c. 261 BCE).

- In this edict, Ashoka describes the **devastating consequences of the war**, noting that about 100,000 people were killed, many more died later, and around 150,000 were deported.
- The **inscription reveals Ashoka's profound remorse for the suffering** inflicted on the people of Kalinga. He particularly laments the distress caused to Brahmanas, Shramanas, householders, and other innocent people who were affected by the war.
- This moral reflection led Ashoka to reconsider the value of military conquest. Consequently, **Ashoka declared his commitment to the policy of Dhamma**, emphasizing ethical conduct, compassion, tolerance, and welfare of all beings.
- He renounced the policy of **Bherighosha** (conquest by war/physical force) and replaced it with **Dhammaghosha** (conquest by Dhamma).
- Thus, **Major Rock Edict XIII** is historically significant because it **records both the Kalinga War and Ashoka's transformation into a ruler guided by moral and humanitarian principles**.

Option a is incorrect: Major Rock Edict I:

This edict primarily deals with the prohibition and regulation of animal slaughter in the Mauryan Empire. Ashoka emphasizes reducing the killing of animals for royal feasts and encourages greater compassion toward living beings. It does not mention the Kalinga War.

Option b is incorrect: Major Rock Edict V:

This edict discusses the appointment of Dhamma Mahamatras, special officials responsible for promoting Dhamma, ensuring social welfare, and looking after various groups such as women, the elderly, and prisoners. It contains no reference to the Kalinga conquest.

Option d is incorrect: Major Rock Edict VIII:

This edict describes Ashoka's practice of Dhamma-yatras (moral tours), where the emperor traveled to meet people, interact with ascetics and Brahmanas, and promote ethical values. It contrasts these tours with earlier royal pleasure tours but does not refer to the Kalinga War or Ashoka's remorse.

Source:) <https://egyankosh.ac.in/bitstream/123456789/84170/3/Unit-5.pdf>

<https://www.nationalmuseumindia.gov.in/en/collections/index/21>

<https://egyankosh.ac.in/bitstream/123456789/64793/1/Unit15.pdf>

<https://timesofindia.indiatimes.com/etimes/trending/what-do-king-ashokas-18-rock-edicts-reveal-key-messages-and-insights/articleshow/114879974.cms>

Subject:) History

Topic:) Ancient History

Subtopic:)

Q.13)

Ans) b

Exp) Option b is the correct answer.

Lord Linlithgow served as the Viceroy of India from 1936 to 1943, holding the **longest tenure of any British Viceroy in India**. His viceroyalty coincided with a highly turbulent phase of the Indian national movement and the outbreak of World War II.

During his tenure:

- The **Government of India Act 1935** was implemented, leading to the establishment of provincial autonomy and the formation of Congress ministries in several provinces in 1937.
- In 1939, he **unilaterally declared India a participant in World War II** without consulting Indian leaders, which led to the **resignation of the Indian National Congress ministries**.
- His period also witnessed major political developments such as the **Lahore Resolution** (1940), the **Individual Satyagraha** (1940–41), the **Cripps Mission** (1942), and the **launch of the Quit India Movement** (1942).

Option I is correct: Adoption of the Lahore Resolution by the Muslim League (1940):

The **Lahore Resolution** was adopted in **March 1940** at the **Lahore session of the All-India Muslim League** under the leadership of **Muhammad Ali Jinnah**. It demanded the creation of independent states for Muslims in the north-western and eastern regions of India, which later evolved into the demand for Pakistan.

This event took place during Lord Linlithgow's tenure.

Option II is correct: Launch of Individual Satyagraha (1940):

After the **failure of the August Offer** (1940), **Mahatma Gandhi** launched the **Individual Satyagraha** in **October 1940** as a limited protest against India's forced participation in World War II without consultation with Indian leaders and also to assert the right to free speech against British war efforts during World War II. The first satyagrahi was Vinoba Bhave, followed by Jawaharlal Nehru.

This event also took place during Linlithgow's tenure.

Option III is incorrect: Arrival of the Cabinet Mission to India (1946):

The **Cabinet Mission** was sent by the **British Government** in **March 1946**, comprising **Lord Pethick-Lawrence**, **Sir Stafford Cripps**, and **A.V. Alexander**.

Its objective was to negotiate the transfer of power and devise a constitutional framework for India, including the formation of a Constituent Assembly and a federal structure with groupings of provinces.

However, by this time Lord Linlithgow had already demitted office (1943), and Lord Wavell was the Viceroy. Hence, **this event did not occur during Linlithgow's tenure.**

Option IV is incorrect: Proposal of the C. Rajagopalachari Formula (1944)

The **C. Rajagopalachari Formula** was proposed in 1944 by **C. Rajagopalachari**.

Purpose of the formula:

- To resolve the political deadlock between the Congress and the Muslim League.
- It suggested that after independence, a plebiscite should be held in Muslim-majority areas to decide whether they wished to form a separate state.

The proposal formed the basis for the **Gandhi-Jinnah Talks** between Mahatma Gandhi and Muhammad Ali Jinnah.

However, **in 1944 the Viceroy was Lord Wavell, because Lord Linlithgow's tenure ended in 1943.**

Hence, **this event falls outside Linlithgow's period.**

Source:) A Brief History of Modern India (Spectrum) by Rajiv Ahir, Chapter: Appendices - Governors-General and Viceroy of India: Significant Events in their Rule.

A New Look at Modern Indian History by B. L. Grover, Appendix 5 – List of Governors, Governor General and Presidents, Page: 508.

Subject:) History

Topic:) Modern History

Subtopic:)

Q.14)

Ans) a

Exp) Option a is the correct answer.

Buddhism, founded by Gautama Buddha around the **6th century BCE**, later underwent a major doctrinal division into **Hinayana (Theravada)** and **Mahayana traditions**.

- **Hinayana**, which spread mainly in South and Southeast Asia, gave rise to philosophical schools such as **Vaibhasika** and **Sautrantika**.
- In contrast, **Mahayana**, prevalent in North and Northeast Asia, developed schools like **Yogacara (Vijnanavada)** and **Madhyamika (Shunyavada)**.

Option I is correct: Sautrantika

The Sautrantika school is a **Buddhist school associated with the Hinayana tradition**. It emphasized the **authority of the Sutras (Suttas) rather than the Abhidharma texts**, hence the name “Sautrantika” (followers of the sutras). It developed important philosophical ideas such as **momentariness (kshanika-vada)** and influenced later Buddhist epistemology.

Option II is incorrect: Sthanakvasi

This is not a Buddhist sect. It is a sub-sect of Svetambara Jainism (a major branch of Jainism). It emerged as a reformist group that **rejected idol worship and temple rituals** and focused on internal spiritual practices like meditation, ethical conduct, and ascetic discipline.

Option III is correct: Vaibhasika

A prominent Buddhist school **belonging to the Hinayana tradition**. The Vaibhasikas believed in the doctrine that “**all dharmas exist**” (**sarvam asti**) across past, present, and future. They gave great importance to the Abhidharma literature, especially the Mahavibhasha.

Together, Sautrantika and Vaibhasika represent the major philosophical traditions of Hinayana Buddhism.

Option IV is incorrect: Terapanthi

The Terapanthi sect is another **reformist school within Jainism**, emerging from the **Svetambara tradition** in the 18th century under **Acharya Bhikshu**. It emphasizes strict discipline, a single religious authority (Acharya), and moral purification.

Option V is incorrect: Kapalika

The Kapalikas were an extreme **Shaiva sect devoted to the worship of Shiva** in his fierce forms and associated with **Tantric practices**. Kapalikas were known for their extreme asceticism, **use of skulls (kapala)**, **cremation-ground rituals**, and transgressive practices aimed at attaining spiritual power.

Option VI is correct: Yogacara

Yogacara, also known as **Vijnanavada** (the doctrine of consciousness) is an **important Mahayana Buddhist philosophical school**. It holds that reality is fundamentally mental and that external objects have no independent existence apart from consciousness.

Yogacara and Madhyamika are among the major philosophical sects of Mahayana Buddhism.

Source:) <https://ncert.nic.in/textbook/pdf/keks102.pdf> Page: 26.

<https://egyankosh.ac.in/bitstream/123456789/115658/3/Unit-6.pdf>

<https://indianexpress.com/article/upsc-current-affairs/upsc-essentials/knowledge-nugget-buddhism-teachings-paths-sects-upsc-exam-9956135/>

<https://egyankosh.ac.in/bitstream/123456789/38138/1/Unit-3.pdf>

<https://www.newindianexpress.com/cities/hyderabad/2019/Sep/03/digambar-jain-sanstha-celebrating-maha-paryushan-parv-this-week-2028574.html>

<https://egyankosh.ac.in/bitstream/123456789/20151/1/Unit-25.pdf>

Subject:) History

Topic:) Ancient History

Subtopic:)

Q.15)

Ans) c

Exp) Option c is the correct answer.

The **Vijayanagara Empire** (14th–17th century), **founded by Harihara and Bukka in 1336**, emerged as a major political and cultural power in South India.

- Its capital, Vijayanagara (modern Hampi), became one of the largest and most prosperous cities in the medieval world, known for its well-planned urban layout, thriving markets, strong military, and temple architecture.
- Much of our knowledge about the empire comes from the accounts of foreign travellers: **Niccolo de Conti, Abdur Razzaq, Domingo Paes, Fernao Nuniz, Duarte Barbosa.**

Option I is correct: Niccol de Conti

He was an **Italian traveller** who **visited Vijayanagara** in the **early 15th century** (c. 1420s) during the **reign of Devaraya I (140-1422 CE)**. His account highlights the immense size of the capital city, dense population, flourishing trade, and strong military power, indicating the empire's early consolidation under the Sangama rulers.

Option II is incorrect: Francois Bernier

He was a **17th-century French physician and traveller** who **stayed in India primarily during the reign of Aurangzeb**. His writings mainly deal with the Mughal Empire; its administration, land revenue system, and social conditions. By this time, **Vijayanagara had already declined after the Battle of Talikota (1565)**; hence he had no association with it.

Option III is correct: Domingo Paes

A **Portuguese traveller** who **visited the Vijayanagara empire** in the **early 16th century during the reign of Krishnadeva Raya (1509-1529)**.

His account is one of the most detailed and reliable descriptions of the empire. He vividly described the royal court, military organization, festivals like Mahanavami, bustling markets, and the immense wealth of the kingdom, highlighting its peak prosperity.

Option IV is correct: Fernao Nuniz

A **Portuguese chronicler** who **visited Vijayanagara in the 16th century**. Unlike Paes, he focused more on the historical narrative of the empire, including:

- Origin and rise of Vijayanagara
- Succession of rulers
- Internal conflicts

His writings complement Paes by providing a historical perspective rather than just a contemporary description.

Option V is correct: Abdur Razzaq

He was a **Persian envoy/ambassador and traveller** sent by the **ruler of Herat (Timurid Empire) to the court of Deva Raya II of the Vijayanagara Empire in the 15th Century (1440s)**. His account is **one of the earliest detailed foreign descriptions of Vijayanagara**. He famously described Vijayanagara as one of the greatest cities in the world at that time.

Source: <https://ncert.nic.in/textbook/pdf/lehs203.pdf>

History of Medieval India by SATISH CHANDRA- Unit 9. The Age of Vijayanagara and the Bahmanids, and the Coming of the Portuguese (Circa 1350-1565).

Tamil Nadu SCERT class 11(new)- Unit 12- Bahmani and Vijayanagar Kingdoms, Page 175.

Class 12 NCERT History (Themes in Indian History – Part II), Chapter: “Through the Eyes of Travellers: Perceptions of Society (c. 10th-17th century), Page: 115.

Subject:) History

Topic:) Medieval History

Subtopic:)

Q.16)

Ans) b

Exp) Option b is the correct answer.

Sardar Vallabhbhai Patel (1875-1950) was a prominent leader of the Indian freedom struggle, a skilled lawyer, and a statesman known as the **“Iron Man of India.”**

- He played a key role in **organizing peasants during the Kheda (1918) and Bardoli (1928) Satyagrahas**, demonstrating exceptional leadership and organizational skill.
- Patel was instrumental in the **integration of over 500 princely states into the Indian Union** after independence, earning him the reputation of a national unifier.

- He served as **India's first Deputy Prime Minister** and **first Minister of Home Affairs**, and contributed significantly to the Constituent Assembly.
- He was **posthumously awarded the Bharat Ratna** (India's highest civilian honor) in 1991.
- His birth anniversary is celebrated as **National Unity Day** (Rashtriya Ekta Diwas) in India.

Option a is correct: **Sardar Vallabhbhai Patel** was **put in charge of the Flag Satyagraha** (also known as **Jhanda Satyagraha**) in **Nagpur (1923)** by the Congress leadership.

- **He led and organized the movement**, mobilized volunteers, addressed meetings, and negotiated with authorities.
- The **satyagraha involved mass civil disobedience against British restrictions on hoisting the Indian national flag**, with thousands participating, and it eventually led to an agreement allowing peaceful processions.

Option b is incorrect: While **Sardar Patel's political journey began with the Kheda Satyagraha (1918)**, where he gave up his thriving legal practice to organize peasants against tax collection during a famine. However, **he did not receive the title "Sardar" then. The title was conferred upon him during the Bardoli Satyagraha of 1928.**

- In Bardoli, he led an organized strike against a 22% increase in land revenue.
- His leadership was so impactful that the **women of Bardoli began addressing him as "Sardar" (meaning Chief or Leader)**, a title Mahatma Gandhi later popularized globally.

Option c is correct: **Sardar Patel presided over the Karachi Session of the Indian National Congress (1931).** This session was held after the Gandhi-Irwin Pact.

- **It adopted two landmark resolutions:**
 - **Fundamental Rights:** Included civil liberties such as freedom of speech, equality before law, and protection of minorities.
 - **National Economic Programme:** Focused on socio-economic justice, including labour rights, state control of key industries, and reduction of inequality.
- These resolutions later significantly influenced the Directive Principles and Fundamental Rights in the Constitution.

Option d is correct: In the Constituent Assembly, **Sardar Patel served as the Chairman of the Advisory Committee on Fundamental Rights, Minorities, and Tribal and Excluded Areas.** This committee was crucial in designing safeguards for civil liberties and minority rights.

Additionally, he **also served as the chairman of the Provincial Constitution Committee.**

Source:) <https://www.thehindu.com/news/cities/puducherry/sardar-patels-150th-birth-anniversary-celebrated/article70225948.ece>

<https://www.thehindu.com/news/national/sardar-patels-vision-and-the-meaning-of-national-unity-today/article70221709.ece>

<https://indianexpress.com/article/explained/explained-history/vallabhbhai-patel-who-called-him-sardar-vitthalbhai-10421383/>

<https://www.egyankosh.ac.in/bitstream/123456789/32341/1/Unit-4.pdf>

<https://indianexpress.com/article/upsc-current-affairs/upsc-essentials/sardar-vallabhbhai-patel-national-unity-day-the-great-unifier-of-india-upsc-9644334/>

Subject:) History

Topic:) Modern History

Subtopic:)

Q.17)

Ans) a

Exp) Option a is the correct answer.

The **Gupta Empire** (c. 4th–6th century CE) is often regarded as the **“Golden Age of Ancient India”** because of remarkable achievements in politics, literature, science, art, and economic prosperity. The empire reached its greatest expansion under **Samudragupta** and **Chandragupta II**.

The administration of the Guptas was monarchical and moderately decentralized. For administrative convenience, the **empire was divided into provinces (Bhuktis) governed by Uparikas**, which were further divided into **districts (Vishayas)** administered by **Vishayapatis**. At the local level, village assemblies and headmen played an important role in governance.

Pair I is incorrect: Dutakas

In the administration of the Gupta Empire, **the landgrant inscriptions often mention Dutakas who were associated with the task of implementing gifts when gifts of land were made to brahmanas and others.**

In the Gupta administration, **High-ranking officials** were called **Dandanayakas**, and **Mahadandanayakas** were **high-ranking judicial or military officers.**

Pair II is correct: Sandhivigrahika

The **Sandhivigrahika** (or **Mahasandhivigrahika**) was the **Minister for Peace and War**. He was a high-ranking official responsible for sandhi (peace/treaties) and vighraha (war). He managed diplomatic correspondence with other states, initiated wars, and concluded alliances and treaties.

Pair III is incorrect: Mahapratihara

The **Mahapratihara** was the **chief of palace guards** and an important official connected with the royal establishment, controlling access to the king. While, the commander of infantry and cavalry were designated as Baladhikrita and Mahabaladhikrita.

Source:) <https://egyankosh.ac.in/bitstream/123456789/20161/1/Unit-33.pdf>

<https://egyankosh.ac.in/bitstream/123456789/67712/1/Unit-6.pdf>

Subject:) History

Topic:) Ancient History

Subtopic:)

Q.18)

Ans) c

Exp) Option c is the correct answer.

The **Delhi Sultanate** was a major medieval Indian polity that **ruled large parts of the Indian subcontinent from 1206 to 1526 CE** with its **capital at Delhi.**

- It was established after the decline of Muhammad of Ghor's conquests by his general **Qutb-ud-din Aibak**, who founded the Mamluk (Slave) Dynasty.
- The Sultanate was ruled by **five successive dynasties: the Mamluk (Slave), Khalji, Tughluq, Sayyid, and Lodi dynasties.**
- It played a crucial role in shaping administration, military organization, architecture, and Indo-Islamic culture in medieval India.
- The Sultanate **came to an end in 1526** after the **defeat of Ibrahim Lodi by Babur in the First Battle of Panipat**, which marked the beginning of Mughal rule in India.

Statement I is incorrect: The **Khurasan expedition scheme** and the **Qarachil campaign** were actually **undertaken during the reign of Muhammad bin Tughluq (1325-1351), not Iltutmish.**

- Muhammad bin Tughluq planned an ambitious expedition to conquer Khurasan, for which a large army was recruited, but the plan was later abandoned.
- He also launched the Qarachil expedition towards the Himalayan region, which ended disastrously due to harsh terrain and climate.

Iltutmish (1211-1236) did not launch expeditions into Khurasan or Qarachil. In fact, when Chengiz Khan reached the Indus chasing the Khwarezm prince, Iltutmish wisely avoided confrontation by refusing asylum to the prince.

Statement II is incorrect: The **practices of branding horses (dagh)** and **maintaining descriptive rolls of soldiers (chehra)** were introduced by Alauddin Khalji as part of his military reforms. These measures aimed to prevent corruption, ensure that soldiers maintained quality horses, and stop false musters. It was **not introduced by Firoz Shah Tughluq (1351-1388).**

Statement III is correct: **Balban (1246-1287) implemented the policy of "Blood and Iron"** to restore the declining authority of the Sultanate. He adopted a highly centralized and authoritarian style of governance, suppressing rebellions

ruthlessly and curbing the power of the nobility (especially the **Chahalgani**). He enhanced the prestige of the monarchy through practices like **Sijda (prostration)** and **Paibos (kissing the feet)**, thereby reinforcing the absolute authority of the Sultan.

Source:) Tamil Nadu New SCERT History Class XI- Unit 10- Advent of Arabs and Turks, Page 93.

Tamil Nadu Old SCERT History Class XI- Unit 16- Delhi Sultanate, Page 172.

Subject:) History

Topic:) Medieval History

Subtopic:)

Q.19)

Ans) a

Exp) Option a is the correct answer.

The **Carnatic Wars** were a **series of three wars fought in the mid-18th century between the British East India Company and the French East India Company for political and commercial supremacy in South India.**

- These wars were **mainly fought in the Carnatic region**, which was nominally under the Mughal Empire but had become politically fragmented by that time.
- The conflicts were closely linked to contemporary European wars, such as the War of the Austrian Succession and the Seven Years' War.
- Through military alliances with rival Indian claimants and decisive battles such as the **Battle of Wandiwash (1760)**, the British gradually defeated the French.
- Ultimately, the **Carnatic Wars ended with the Treaty of Paris**, which effectively ended French political ambitions in India and marked the beginning of British political dominance in the subcontinent.

Statement I is correct: The **First Carnatic War (1740–1748)** was essentially an **extension of the European conflict between the British and the French**, which was **caused by the Austrian War of Succession**.

When Britain and France went to war in Europe, their respective trading companies in India (the British East India Company and the French East India Company) also entered into hostilities in South India.

Statement II is correct: The **Treaty of Aix-la-Chapelle (1748)** marked the **end of the First Carnatic War**, as well as the **conclusion of the War of Austrian Succession in Europe**. Under the terms of the treaty, both **Britain and France agreed to restore their captured possessions**. Accordingly, **Madras**, captured by the French in 1746, was **handed back to the British**, while the French received their territories in North America (notably Louisbourg).

Statement III is incorrect: The **Third Carnatic War (1758–1763)** **ended with the Treaty of Paris, not the Treaty of Seringapatam**. The Treaty of Paris ended the global conflict known as the Seven Years' War. It allowed the French to retain a few trading settlements like Pondicherry but without fortifications or large armies, effectively ending French political ambitions in India.

The **Treaty of Seringapatam (1792)** was **signed after the Third Anglo-Mysore War** between the British and Tipu Sultan. and had no connection with the Carnatic Wars.

Source:) A Brief History of Modern India (Spectrum) by Rajiv Ahir, Chapter: 3 -Advent of the Europeans in India, Page: 44.

From Plassey to Partition and After (A History of Modern India) by SEKHAR BANDYOPADHYAY Chapter: 1 - Transition of the Eighteenth Century, Page: 71.

<https://www.egyankosh.ac.in/bitstream/123456789/76478/1/Unit-3.pdf>

Subject:) History

Topic:) Modern History

Subtopic:)

Q.20)

Ans) d

Exp) Option d is the correct answer.

Statement I is incorrect: The **Dharma Shastras** consist of the Dharmasutras and the Smriti Literature mainly. The Dharmasutras were mainly compiled between 600 BCE and 400 BCE. The Smriti literature was **largely compiled during the post-Mauryan period**, roughly between the **2nd century BCE and 3rd century CE**. This period witnessed the consolidation of social and legal norms in response to changing political and social conditions after the decline of the Mauryan Empire. Important texts such as the **Manusmriti, Yajnavalkya Smriti, and Narada Smriti** belong broadly to this phase.

Hence the Dharm Shastras as a whole were compiled earlier than the 3rd Century CE and not between 6th century CE and 10th century CE.

Statement II is incorrect: Dharma Shastras are not part of Shruti literature. In the traditional classification of Hindu sacred texts:

1. Shruti ("that which is heard") refers to **texts believed to be of divine revelation**. These are considered **apaurusheya** (not of human origin) and eternal. It includes the **Vedas and Upanishads**.

2. Smriti ("that which is remembered") refers to **texts that are human authored** and composed by sages. It includes the **Dharma Shastras, Puranas, and Epics**.

Hence, **Dharma Shastras belong to the Smriti tradition, not Shruti**. They were written by scholars and lawgivers to interpret Vedic principles and apply them to practical aspects of society such as law, morality, and social duties

The **Manusmriti** (also known as **Manava Dharmashastra**) is **traditionally attributed to the ancient lawgiver Manu** and is regarded as **one of the earliest and most influential Dharma Shashtra texts**. Composed in metrical verse, **it systematically discusses:**

- the duties of the four varnas,
- the stages of life (ashrama system),
- rules of marriage and inheritance,
- duties of kings and administration of justice,
- punishments and judicial procedures.

Because of its comprehensive treatment of social and legal norms, the **Manusmriti greatly influenced later legal traditions and commentaries within the Dharma Shashtra tradition**. It was the **first translated into English by William Jones** in 1794.

Statement III is correct: Dharma Shastras deal with a wide spectrum of social, legal, and ethical aspects, making them early treatises on law and society. Their subject matter includes:

- **Acharya** (code of conduct): daily duties, rituals, customs
- **Vyavahara** (legal procedures): courts, evidence, contracts, inheritance, punishment
- **Prayaschitta** (penance): expiatory rites for sins and violations
- They also elaborate on:
 - o **Duties based on varna** (social class) and **ashrama** (life stages)
 - o Role and responsibilities of the **king (rajadharma)**
 - o Maintenance of social order and moral discipline

Thus, **they functioned as normative guides for both individuals and the state**, shaping ancient Indian legal and social frameworks.

Source:) A History of Ancient and Early Medieval India by Upinder Singh, Unit :1 - Understanding Literary and Archaeological Sources, Page: 74.

<https://ncert.nic.in/pdf/publication/otherpublications/tioi101.pdf> Page: 17

<https://egyankosh.ac.in/bitstream/123456789/118354/1/Unit-1.pdf>

Subject:) Art and Culture

Topic:) Literature

Subtopic:)

Q.21)

Ans) b

Exp) Option b is the correct answer.

Budget deficits are central to analysing the **fiscal health of the government**. Among them, the relationship between **Fiscal Deficit and Primary Deficit** is particularly important, as it directly helps us estimate the **interest burden of past borrowings**.

First, we will have to calculate Fiscal Deficit. Fiscal Deficit represents the **total borrowing requirement of the government**.

Fiscal Deficit = Total Expenditure - (Revenue Receipts + Non-debt Capital Receipts)

Total Expenditure = Revenue Expenditure + Capital Expenditure = 70,000 + 30,000 = ₹ 1,00,000 crores

Total Receipts (excluding borrowings) = Revenue Receipts + Non-debt Capital Receipts = 50,000 + 10,000 = ₹ 60,000 crores

Fiscal Deficit = 1,00,000 - 60,000 = ₹ 40,000 crores

Now, using the relationship between Fiscal Deficit and Primary Deficit we can arrive at the Interest Payment component.

Primary Deficit = Fiscal Deficit - Interest Payments

Rearranging it we get,

Interest Payments = Fiscal Deficit - Primary Deficit

Interest Payments = 40,000 - 28,000 = ₹ 12,000 crores

Knowledge Base:

Deficit Type	Formula	What it shows	Key Insight
Revenue Deficit	Revenue Expenditure - Revenue Receipts	Day-to-day income vs expenditure	Govt unable to meet routine expenses
Fiscal Deficit	Total Expenditure - (Revenue Receipts + Non-debt Capital Receipts)	Total borrowing requirement	Overall financial health
Primary Deficit	Fiscal Deficit - Interest Payments	Deficit excluding interest payments	Present fiscal discipline
Effective Revenue Deficit	Revenue Deficit - Grants for Capital Creation	True consumption deficit	Quality of expenditure

Source:) <https://ncert.nic.in/textbook/pdf/leec105.pdf> - NCERT Class 12 - Introductory Macro Economics -Chapter 5 - Government Budget and the Economy - Page 71

Subject:) Economy

Topic:) Budgetary Economics

Subtopic:)

Q.22)

Ans) c

Exp) Option c is the correct answer.

The Reserve Bank of India (RBI) controls money supply using various **quantitative monetary tools**. These tools influence the **availability of credit, cost of borrowing, and liquidity** in the system. There are 2 major types of tools:

I. Expansionary which **increases** money supply and

II. Contractionary which **reduces** money supply

Option I is correct. A **reduction in Cash Reserve Ratio (CRR)** means banks are required to keep **less money as reserves with RBI**. This frees up more funds for lending, thereby **increasing credit creation and money supply**. It is a **direct expansionary measure**.

Option II is correct. When RBI **purchases government securities in the open market**, it pays money to banks and financial institutions. This increases **reserves in the banking system**, leading to **higher lending capacity and increased money supply**. This is a classic **Open Market Operation (OMO)** for expansion.

Option III is correct. A **reduction in the repo rate** lowers the cost at which banks borrow from RBI. This encourages banks to borrow more and extend more loans, thereby **increasing liquidity and money supply** in the economy.

Option IV is incorrect. When RBI **sells Dollars**, it absorbs liquidity from the market.

Banks, institutions or individuals pay money to RBI to buy dollars, reducing available funds and thereby **decreasing money supply**.

Source: <https://ncert.nic.in/textbook/pdf/leec103.pdf> - NCERT 12 - Introductory Macro Economics - Chapter 3 - Money and Banking - Money Creation by Banking System - Page 40

Subject: Economy

Topic: Money and Banking

Subtopic:

Q.23)

Ans) c

Exp) Option c is the correct answer.

Infrastructure financing in India increasingly relies on **public-private partnership (PPP) models** to balance risk, efficiency, and investment needs. Each model such as **EPC, HAM, and BOT** among others, differs in terms of **who finances, who builds, who bears risk, and how returns are generated**.

Pair I is correct. The **Hybrid Annuity Model (HAM)** is a mix of **EPC and BOT models**. Under this, the government typically funds **around 40% of the project cost during the construction period in installments**, while the remaining is arranged by the private player (during the construction period).

Under HAM, the private developer does not depend on toll revenue; instead, it receives **fixed annuity payments from the government for the remaining 60% part of the project cost**, linked to **performance and maintenance standards**. This reduces **traffic risk for the private sector**, making it a balanced risk-sharing model.

Pair II is correct. The **Engineering, Procurement and Construction (EPC)** model is fundamentally a **government-funded model**. Here, the private contractor is responsible only for **designing, building, and delivering the project**, but **does not invest its own money or operate the asset**. The government bears the **entire financial burden and operational risk**.

Pair III is correct. Under the **Build-Own-Transfer (BOT)** model, a private entity **finances, constructs, and operates** the infrastructure project for a specified **concession period**. During this period, it recovers its investment through **user charges (like tolls)** or annuity payments. After the concession period ends, the project is **transferred back to the government**. This model places **significant risk on the private sector**, especially demand or traffic risk.

Source: <https://www.pppinindia.gov.in/glossary> ;

<https://indianexpress.com/article/india/govts-big-infra-push-nod-to-road-projects-worth-rs-50655-crore-9491780/>

Subject: Economy

Topic: Infrastructure

Subtopic:

Q.24)

Ans) d

Exp) Option d is the correct answer.

Statement I is incorrect. Outstanding Debt of the Union Government as a percentage of GDP has steadily decreased between 2020-21 and 2024-25.

Outstanding Debt of Union Government (% of GDP)

2020-21 61.4

2021-22 58.8

2022-23 58.1
2023-24 57
2024-25 56.2

Statement II is incorrect. Arunachal Pradesh (around 57 % of GSDP) and not Tamil Nadu (around 30.3 % of GSDP) has the highest Outstanding Debt among State Governments as percentage of State GDP (% of GSDP).

Source:)

[https://sansad.in/getFile/annex/268/AU2635_fQvzb2.pdf?source=pqars#:~:text=\(b\)%20the%20year%2Dwise,State%20Governments%20\(%E2%82%B9%20lakh%20crore\)](https://sansad.in/getFile/annex/268/AU2635_fQvzb2.pdf?source=pqars#:~:text=(b)%20the%20year%2Dwise,State%20Governments%20(%E2%82%B9%20lakh%20crore))

Subject:) Economy

Topic:) Budgetary Economics

Subtopic:)

Q.25)

Ans) b

Exp) Option b is the correct answer.

Non-Banking Financial Companies (NBFCs) constitute an important segment of India's financial system, particularly in **expanding credit access beyond the traditional banking network**. However, their regulatory design reflects a **careful balance between operational flexibility and systemic stability**.

Statement I is incorrect. NBFC in general including deposit-taking NBFCs (NBFC-Ds) are **not permitted to issue cheques drawn on themselves**, nor are they integrated into the **payment and settlement system**. This restriction arises because **payment systems are systemically sensitive**, requiring a high degree of regulatory control. Therefore, such functions are **exclusively reserved for banks**, ensuring financial stability.

Statement II is correct. NBFCs are **not subject to the Cash Reserve Ratio (CRR)** requirement applicable to banks. CRR is a monetary policy tool used to regulate **liquidity created through demand deposits**, which NBFCs are not authorised to accept. As NBFCs do not participate in **money creation in the same manner as banks**, imposing CRR on them is neither necessary nor appropriate.

Statement III is incorrect. In terms of Section 45-IA of the RBI Act, 1934, no company can commence or carry on business of a non-banking financial institution without obtaining a certificate of registration from the Reserve Bank and without having a Net Owned Funds (NOF) of ₹10 crore with effect from October 01, 2022 (NBFCs seeking registration shall have NOF of ₹10 crore *ab initio*, and existing NBFCs have timeline up to March 31, 2027 to attain NOF of ₹10 crore).

Source:) <https://www.thehindu.com/news/cities/Delhi/banking-nbfc-gas-supply-brought-under-lok-adalats/article69870905.ece> ; <https://www.rbi.org.in/commonman/english/scripts/FAQs.aspx?Id=1167> ; <https://www.livemint.com/industry/banking/investors-relieved-as-rbi-s-proposed-norms-for-nbfc-leaves-breathing-space-11611551707877.html>

<https://www.rbi.org.in/commonman/english/scripts/FAQs.aspx?Id=1167>

Subject:) Economy

Topic:) Money and Banking

Subtopic:)

Q.26)

Ans) d

Exp) Option d is the correct answer.

The Agreement on Climate Change, Trade and Sustainability (ACCTS) is a **first-of-its-kind initiative** that aims to use **trade rules to address climate change and environmental challenges**. ACCTS tackles fossil fuel subsidies, promotes trade in environmental goods and services, and establishes innovative eco-labelling standards.

1. Framework and Consistency: While the ACCTS was launched and negotiated by a small group of "like-minded" countries (New Zealand, Costa Rica, Iceland, and Switzerland), it is **explicitly designed to be consistent with the rules**

of the World Trade Organization (WTO). It is considered an "open plurilateral" agreement, meaning it is open for any WTO member to join if they meet the standards.

2. Key Objectives:

- **Eliminating Tariffs:** It removes tariffs on hundreds of environmental goods (e.g., solar panels, wind turbines, energy-efficient machinery).
- **Fossil Fuel Subsidies:** It is the **first legally binding international agreement to contain specific prohibitions on fossil fuel subsidies**, marking a crucial step toward sustainable development.
- **Eco-labelling:** It seeks to create **common guidelines for voluntary eco-labelling to prevent "greenwashing"** and facilitate trade in sustainable products.

The "**Pathfinder**" **Approach:** Because reaching a consensus among all WTO members can be slow, these four countries developed the ACCTS as a "pathfinder" or template. The goal is to demonstrate how trade and environmental policies can be integrated, eventually encouraging broader adoption within the multilateral WTO system.

Hence, **the ACCTS is associated with the WTO.**

Source:) <https://www.iisd.org/articles/deep-dive/agreement-climate-change-trade-sustainability-accts>
https://www.wto.org/library/events/event_resources/envir_30062025/834_2667.pdf

Subject:) Environment

Topic:) Climate Change

Subtopic:)

Q.27)

Ans) c

Exp) Option c is the correct answer.

Statement I is incorrect: Methane hydrates (also called **gas hydrates or clathrates**) are **ice-like crystalline solids** where **methane molecules are physically trapped (encaged) in a lattice of water molecules** via van der Waals forces. There is **no chemical bond** between the water and the methane; **it is a physical entrapment**. If the lattice breaks, the gas simply escapes.

Statement II is correct: They are stable only under **specific thermodynamic conditions of high pressure and low temperature**, which are typically found in:

- **Deep ocean sediments** (continental margins at depths generally >500 m), and
- **Permafrost regions** (Arctic areas with permanently frozen ground).

These conditions define the **Gas Hydrate Stability Zone (GHSZ)**.

In India, **exploration under the National Gas Hydrate Programme (NGHP) has confirmed significant methane hydrate deposits of biogenic origin**, especially in:

- **Krishna-Godavari Basin** (east coast) – largest and most promising reserves,
- **Mahanadi Basin**, and
- **Andaman offshore region**.

These findings indicate substantial potential for future energy extraction.

Statement III is correct: Methane hydrates are considered a **major unconventional energy resource** because:

- **Methane is a cleaner-burning fuel compared to coal** (lower CO₂ emissions per unit energy), and
- Global estimates suggest **hydrates may contain more carbon than all known reserves of coal, oil, and natural gas combined**.

However, **extraction is technologically challenging** and poses **risks such as methane leakage** (a potent greenhouse gas) and seafloor instability.

Source:) <https://www.thehindubusinessline.com/specials/clean-tech/a-major-energy-source-that-india-has-failed-to-tap/article69282389.ece>
<https://timesofindia.indiatimes.com/science/a-hidden-arctic-world-methane-mounds-and-life-found-3-6-km-below-the-greenland-sea/articleshow/126165916.cms>

<https://www.pib.gov.in/PressReleaseIframePage.aspx?PRID=1653545#:~:text=Methane%20hydrate%20is%20formed%20when, and%20low%20temperatures%20in%20oceans>

<https://education.nationalgeographic.org/resource/natural-gas/>

Subject:) Environment

Topic:) Energy

Subtopic:)

Q.28)

Ans) c

Exp) Option is the correct answer.

The **World Network of Biosphere Reserves (WNBR)** was established in 1971 as **part of UNESCO's Man and the Biosphere (MAB) Programme**. It is an intergovernmental scientific program that **aims to establish a scientific basis for enhancing the relationship between people and their environments**.

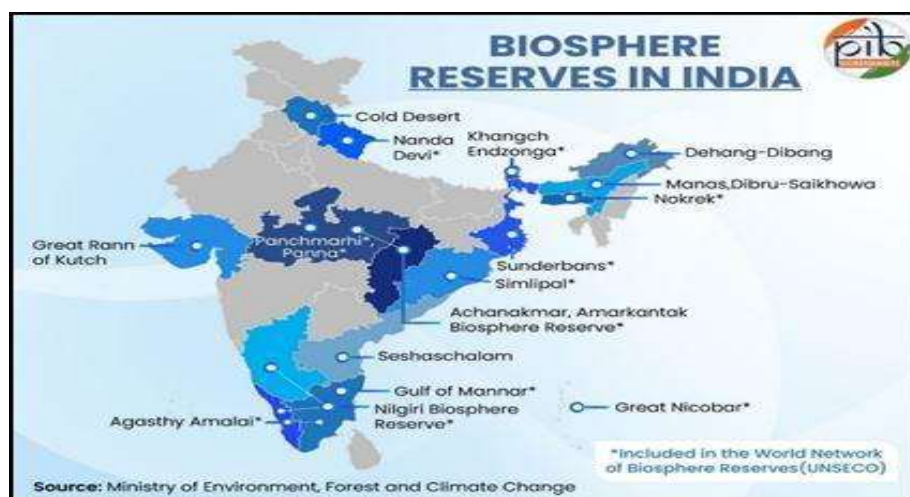
- **The Goal:** To balance the conservation of biological and cultural diversity with economic and social development.
- **The Concept:** Biosphere Reserves are "living laboratories" where scientists and local communities test sustainable development models.

Biosphere reserves in India:

India has 18 Biosphere Reserves covering 91,425 sq km, with **13 recognized by UNESCO under WNBR**.

The **first Indian Biosphere Reserve included in WNBR** was **Nilgiri (2000)**.

The **latest addition** is the **Cold Desert Biosphere Reserve (2025)**, which increased India's total to 13.



Option I is correct: Cold Desert Biosphere Reserve (Himachal Pradesh)

- It represents the **Trans-Himalayan cold desert region of Himachal Pradesh**, covering **Lahaul-Spiti** and adjoining high-altitude regions. It covers the **Pin Valley National Park** and its surroundings, **Chandratat and Sarchu & Kibber Wildlife Sanctuary**.
- It was **included in the UNESCO World Network of Biosphere Reserves (WNBR) in 2025** (latest Indian addition).
- This designation brings **India's total to 13 UNESCO Biosphere Reserves**.
- This is **India's first high-altitude cold desert biosphere reserve** and one of the coldest and driest ecosystems in UNESCO's WNBR.

Option II is correct: Panna Biosphere Reserve (Madhya Pradesh)

- Located in the **Bundelkhand region of Madhya Pradesh**, it encompasses **Panna National Park** and **Ken River basin**, which is central to the Ken-Betwa river-linking project.
- The **Panna Tiger Reserve** is one of the most important sites in the North Central Highlands for protecting the Bengal tiger.

- It was included in the UNESCO World Network of Biosphere Reserves in 2020.

Option III is incorrect: Manas Biosphere Reserve (Assam)

- It is located in the state of **Assam** and **includes Manas National Park**, which is a **UNESCO World Heritage Site** (1985), Project Tiger Reserve, and Elephant Reserve.
- It is famous for its role in **rhino reintroduction** and as a tiger reserve. It is home to the **Pygmy Hog** (world's smallest and rarest pig), Golden Langur, and Hispid Hare.
- Despite its high conservation value, the biosphere reserve remains a National Biosphere Reserve and a **UNESCO World Heritage Site**, but it has **not been inducted into the MAB (WNBR) network**.

Option IV is correct: Nokrek Biosphere Reserve (Meghalaya)

- It is located in the state of **Meghalaya** on the **Tura Range**, which forms part of the **Meghalaya Plateau**.
- Nokrek is the **highest peak of the Garo Hills** at 1,412 metres.
- Highly vulnerable and threatened faunal species in Nokrek include the **slow loris**, the **giant flying squirrel** and the **pig-tailed macaque**.
- It was **included in UNESCO WNBR in 2009**, being one of the earlier Indian biosphere reserves to receive this recognition.

Option V is correct: Agasthyamalai Biosphere Reserve (Kerala-Tamil Nadu)

- Situated in the **southern Western Ghats**, one of the world's major biodiversity hotspots.
- Noted for **high endemism**, especially in medicinal plants and flora, and cultural significance linked to Agastya hills.
- The reserve includes three wildlife sanctuaries, **Shendurney, Peppara and Neyyar**, as well as the **Kalakad Mundanthurai Tiger Reserve**.
- It was **included in WNBR in 2016** due to its ecological and conservation importance.

Knowledge Base: **Biosphere Reserves in India and their Status in WNBR**

Biosphere Reserve	Year of Designation	WNBR Inclusion Year	State(s)
Nilgiri	1986	2000	Tamil Nadu, Kerala, Karnataka
Nanda Devi	1988	2004	Uttarakhand
Nokrek	1988	2009	Meghalaya
Gulf of Mannar	1989	2001	Tamil Nadu
Sundarbans	1989	2001	West Bengal
Manas	1989	–	Assam
Great Nicobar	1989	2013	Andaman & Nicobar Islands
Simlipal	1994	2009	Odisha
Dibru-Saikhowa	1997	–	Assam
Dehang-Dibang	1998	–	Arunachal Pradesh
Pachmarhi	1999	2009	Madhya Pradesh
Khangchendzonga	2000	2018	Sikkim
Agasthyamalai	2001	2016	Kerala, Tamil Nadu
Achanakmar-Amarkantak	2005	2012	Madhya Pradesh, Chhattisgarh
Great Rann of Kutch	2008	–	Gujarat
Cold Desert	2009	2025	Himachal Pradesh
Seshachalam Hills	2010	–	Andhra Pradesh
Panna	2011	2020	Madhya Pradesh

Source: <https://www.pib.gov.in/PressReleasePage.aspx?PRID=2185715>

<https://static.pib.gov.in/WriteReadData/specificdocs/documents/2025/nov/doc2025113683301.pdf>

<https://www.unesco.org/en/articles/5th-wcbr-designates-cold-desert-indias-13th-unesco-biosphere-reserve>

<https://indianexpress.com/article/upsc-current-affairs/upsc-essentials/knowledge-nugget-upsc-unesco-biosphere-reserves-cold-desert-10285227/>

Subject:) Environment

Topic:) Protected Areas

Subtopic:)

Q.29)

Ans) b

Exp) Option b is the correct answer.

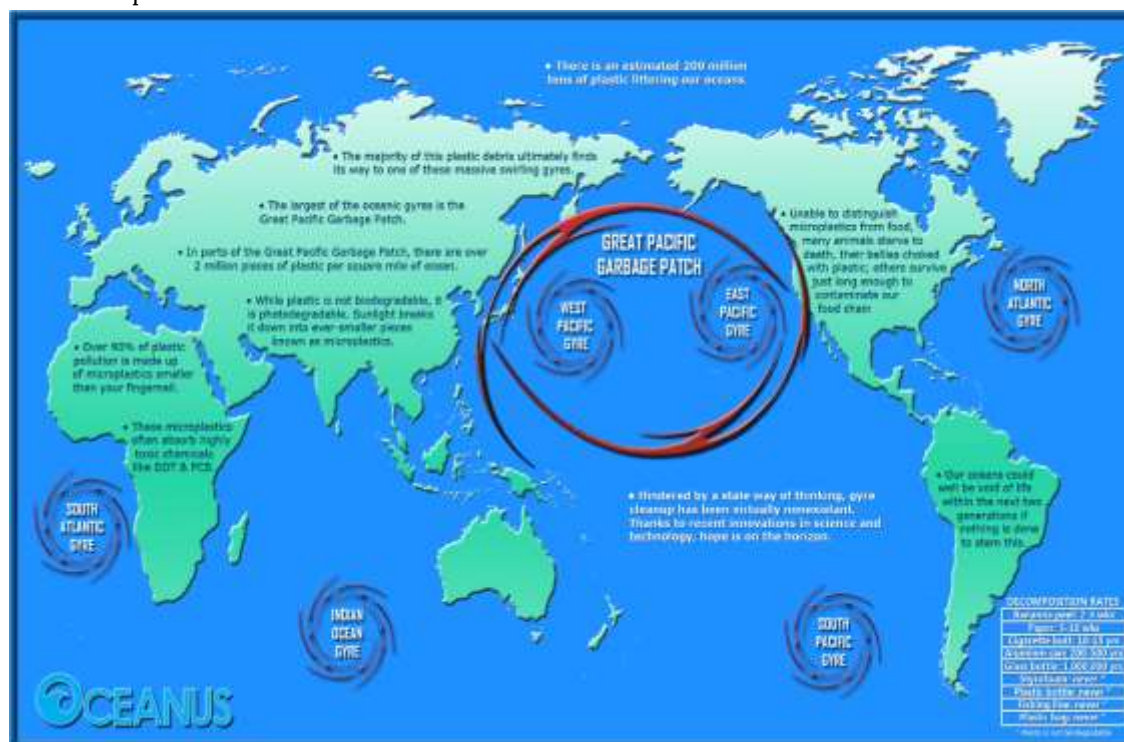
The **Great Pacific Garbage Patch (GPGP)** is a vast accumulation zone of marine debris located in the North Pacific Ocean, formed due to the North Pacific Subtropical Gyre, where circulating ocean currents concentrate floating waste. It is located halfway between Hawaii and California in the Pacific Ocean.

Core Composition:

- It is **primarily composed of plastics, especially:**
 - Microplastics** (tiny particles <5 mm formed due to breakdown of larger plastics)
 - Fragmented plastic debris** (bags, bottles, packaging material, etc.)
- Microplastics dominate in number, making the patch appear like a “**soup of plastic particles**” rather than a solid mass.
- Larger plastic items (e.g., fishing nets, ropes, crates) contribute significantly to the total mass, even though they are fewer in number.

Additional Key Facts:

- The **patch is not a visible island**; it is a **diffuse and dispersed zone of suspended debris**.
- It contains an estimated **~1.8 trillion pieces of plastic**.
- Major materials include polyethylene and polypropylene, which are lightweight and float easily.
- A significant portion of debris comes from abandoned, lost, or discarded fishing gear (ghost nets) along with land-based plastic waste.



Knowledge Base:



Source: <https://timesofindia.indiatimes.com/science/the-great-pacific-garbage-patch-is-becoming-home-to-some-aquatic-species/articleshow/126476618.cms>
<https://www.thehindu.com/sci-tech/science/the-science-quiz-features-of-the-pacific-ocean/article69316281.ece>
<https://indianexpress.com/article/cities/mumbai/bombay-hc-microplastics-marine-life-9603335/>
<https://reefci.com/2018/08/11/what-is-the-great-pacific-garbage-patch/>

Subject: Environment

Topic: Pollution

Subtopic:

Q.30)

Ans) a

Exp) Option a is the correct answer.

Decomposers are organisms that play a crucial role in ecosystem functioning by **breaking down dead plants, animals, and organic wastes into simpler inorganic substances** like carbon

dioxide, water and nutrients and the process is called **decomposition**. They **primarily include microorganisms** such as **bacteria and fungi**, which carry out decomposition through enzymatic (chemical) processes.

By converting complex organic matter into basic nutrients like carbon dioxide, water, nitrates, and phosphates, decomposers **ensure the recycling of nutrients back into the soil and environment**. This process, known as **mineralization**, maintains soil fertility and sustains the continuity of food chains and biogeochemical cycles.

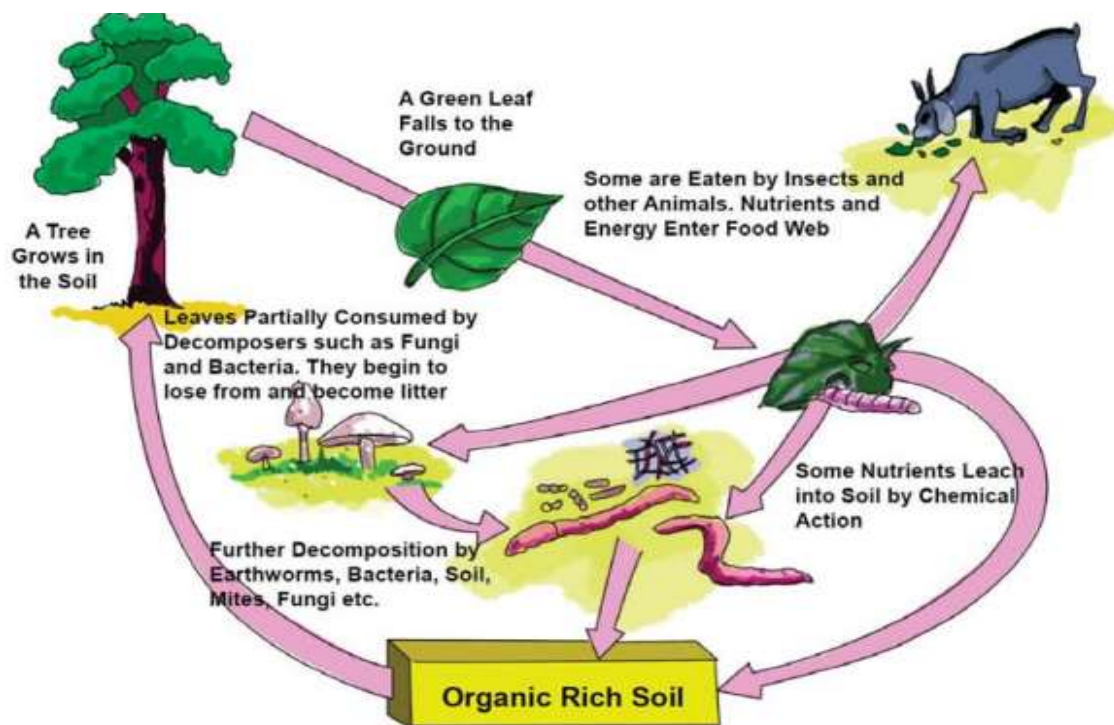


Fig. 2.9: Cycling of nutrients by the decomposers in the ecosystem

Option I is incorrect: Virus

Viruses are obligate intracellular parasites. They cannot grow or reproduce outside of a living host cell. In an ecosystem, they do not "eat" or break down dead organic matter to gain energy. Instead, they infect living organisms. While they can cause the death of an organism (which then becomes food for decomposers), the virus itself does not perform the chemical breakdown of tissues into nutrients. Therefore, they have no role as decomposers.

Option II is correct: Bacteria

Bacteria are the primary Decomposers (specifically, saprotrophs). They perform extracellular digestion, meaning they release powerful enzymes directly onto dead plant or animal matter. These enzymes break down complex organic compounds into simple inorganic nutrients (like nitrogen and phosphorus) which are then absorbed by the bacteria and also returned to the soil for plants to use. They are essential for nutrient cycling.

Option III is correct: Fungus

Fungi are also major Decomposers. Like bacteria, they are saprotrophic and use enzymes to break down tough organic materials that most other organisms cannot, such as cellulose and lignin found in wood. By turning dead wood and leaf litter into rich soil, they act as the "garbage disposals" of the forest. Because they break down matter at a molecular level, they are true decomposers.

Option IV is incorrect: Tapeworm

Tapeworms are parasitic flatworms (endoparasites) that live in the intestines of vertebrate hosts, including humans. They derive nutrition by absorbing pre-digested food directly through their body surface (tegument), rather than breaking down complex organic matter themselves.

Unlike decomposers, they do not secrete enzymes into dead organic material to convert it into simpler inorganic substances, nor do they contribute to nutrient recycling in ecosystems. Their role is purely parasitic (dependent on a living host for survival), hence they are not classified as decomposers in ecological food chains.

Option V is incorrect: Vultures

Vultures are Scavengers. They are the **first line of the "clean-up crew" in an ecosystem.** They consume large quantities of flesh from dead animals (carrion). While they help remove carcasses from the environment, they do not break down the organic matter into inorganic mineral nutrients that plants can absorb. They simply eat the dead meat, digest it as a consumer, and leave the remaining fragments for the actual decomposers to finish.

Therefore, **they are not typically classified as decomposers.**

Source:) <https://ncert.nic.in/textbook/pdf/jesc113.pdf> Page: 206.

<https://egyankosh.ac.in/bitstream/123456789/7901/1/Unit-13.pdf> Page: 55.

<https://egyankosh.ac.in/bitstream/123456789/58516/1/Unit2.pdf> Page: 27 and 33.

<https://education.nationalgeographic.org/resource/decomposers/>

Subject:) Environment

Topic:) Ecosystems

Subtopic:)

Q.31)

Ans) a

Exp) Option a is the correct answer.

The **GST Council** is the **apex federal decision-making body** responsible for **guiding and coordinating the implementation of the Goods and Services Tax (GST) regime in India.**

- The **primary role of the GST Council** is to **make recommendations on critical matters relating to GST**, such as the goods and services that may be subjected to or exempted from GST, model GST Laws, principles of levy, GST rates including the floor rates with bands, special rates for raising additional resources during natural calamities/disasters, special provisions for certain States, etc.

Statement I is incorrect: The **GST Council is a constitutional body.** It was **established under Article 279A** of the Indian Constitution, which was **inserted by the 101st Constitutional Amendment Act, 2016.**

- This amendment introduced the **Goods and Services Tax (GST) system in India** and provided for the establishment of the **GST Council as the apex institutional mechanism for cooperative fiscal federalism between the Union and the States.**
- A statutory body is created through an ordinary law passed by Parliament, whereas the GST Council derives its authority directly from the Constitution of India.

Statement II is incorrect: The **Union Minister of State in charge of Revenue or Finance** is a **member of the Council, but not the Vice-Chairperson.** According to Article 279A, the **Vice-Chairperson is elected by the members** of the GST Council from among themselves.

The constitutional composition of the Council is:

1. The **Union Finance Minister (Chairperson)**
2. The **Union Minister of State in charge of Revenue or Finance**
3. The **Minister in charge of Finance or Taxation** or any other Minister nominated by each State Government
4. **Any person nominated by the Governor of the State** where there is a proclamation of emergency under Article 356 of the Constitution of India.

A meeting of the Council is duly constituted when it is **presided over by the Chairperson** with **at least one-half of the Members of the Council being present.**

Statement III is correct: The Council does not work on a simple majority. **Every decision of the GST Council requires a three-fourths majority of the weighted votes of the members present and voting.**

The voting structure is:

- The vote of the **Central Government** has a weightage of **one-third (1/3) of the total votes cast.**
- The votes of all the **State Governments** combined have a weightage of **two-thirds (2/3) of the total votes cast.**

A **proposal is adopted** only when **at least three-fourths (75%) of the weighted votes support it.** This ensures cooperative federal decision-making between the Union and the States.

Knowledge base: Under Article 279A (5), five specific petroleum products: petroleum crude, high-speed diesel, motor spirit (petrol), natural gas, and aviation turbine fuel were temporarily kept out of the GST net.

The Constitution mandates that the **GST Council will recommend** the date from which GST shall be levied on these products. Until then, they continue to be subject to Central Excise and State VAT.

Source:) <https://www.thehindu.com/business/Economy/gst-council-members-voting-mechanism-share/article70006802.ece>

<https://gstcouncil.gov.in/>

<https://www.pib.gov.in/Pressreleaseshare.aspx?PRID=1575763®=3&lang=2>

<https://gstcouncil.gov.in/about-us-archive>

Subject:) Economy

Topic:) Taxation

Subtopic:)

Q.32)

Ans) b

Exp) Option b is the correct answer.

The **classification of MSMEs in India** is governed under the framework of the **Micro, Small and Medium Enterprises Development Act, 2006**, and is periodically revised by the Government to reflect changing economic conditions.

As per the **latest revised criteria (effective from 1 April 2025)**, **MSMEs are classified on the basis of both investment in plant & machinery/equipment and annual turnover**. Importantly, both conditions must be satisfied simultaneously for an enterprise to fall within a particular category.

Information in row I is correct: A **Micro Enterprise** is defined under the revised MSME classification (effective April 2025) as:

- Investment in plant and machinery/equipment $\leq$ ₹2.5 crore
- Annual turnover $\leq$ ₹10 crore

Information in row II is correct: A **Small Enterprise** is defined as:

- Investment in plant and machinery/equipment $\leq$ ₹25 crore
- Annual turnover $\leq$ ₹100 crore

Information in row III is incorrect: A **Medium Enterprise** is defined as:

- Investment in plant and machinery/equipment $\leq$ ₹125 crore
- Annual turnover $\leq$ ₹500 crore

The values given in the table (₹50 crore and ₹250 crore) correspond to the older classification (2020 criteria) and not the revised limits.

Source:) <https://msme.gov.in/know-about-msme>

<https://www.rbi.org.in/commonman/english/scripts/FAQs.aspx?Id=966>

<https://www.pib.gov.in/PressReleasePage.aspx?PRID=2098389®=3&lang=2>

Subject:) Economy

Topic:) Industrial Sector

Subtopic:)

Q.33)

Ans) c

Exp) Option c is the correct answer.

Option c is correct: **Capital goods** refer to those **durable man-made assets** that are **used as fixed inputs** in the **process of production of other goods and services**.

- These include **machine tools, industrial machinery**, process plant equipment, **construction & mining equipment**, electrical equipment, textile machinery, printing & packaging machinery etc.

- Unlike consumer goods, they are **not meant for direct consumption**, but rather play a crucial role in enhancing the productive capacity of an economy.
- A key feature of capital goods is that they are used repeatedly over a period of time and contribute to the creation of other goods and services, thereby generating income and economic output.
- In macroeconomic terms, they **form an essential component of capital formation** and are central to long-term economic growth.

Option a is incorrect: Goods that are directly consumed by individuals to satisfy wants and needs.

These are consumer goods, which are used for immediate satisfaction of human wants (e.g., food, clothing). They do not contribute to further production and hence are not capital goods.

Option b is incorrect: Goods that are intended for final consumption and not for further production processes.

These are final goods, which are purchased by end users for consumption or investment. Since they are not used as inputs in producing other goods, they do not qualify as capital goods.

Option d is incorrect: Goods that are used repeatedly over time to provide services without being transformed.

This refers to durable consumer goods, such as personal vehicles or household appliances. Although they are **long-lasting**, they are **used for personal utility rather than for producing other goods or services**, and hence are not capital goods.

Source:) <https://egyankosh.ac.in/bitstream/123456789/59891/3/Unit-4.pdf> page: 58.

<https://cii.in/PolicyAdvocacyDetails.aspx?enc=W8iaMhGfthAAbM31PZL4v+ikYwxrEyj6Ypan6pMAPECY5ye4pKrFBX+ffIO NCIEsbTVYMBLjqNdF8yld176tug==>

Subject:) Economy

Topic:) Microeconomics

Subtopic:)

Q.34)

Ans) c

Exp) Option c is the correct answer.

Bond yield refers to the **rate of return earned by an investor on a bond**, expressed as a percentage of its current market price. It indicates how much income a bond generates relative to its price.

- In simple terms, **bond yield = return an investor gets from holding the bond**.
- The most commonly referred yield is the **yield to maturity (YTM)**, which reflects the total return if the bond is held till maturity, including coupon payments and capital gains/losses.

Key Feature:

Bond yield and bond price have an inverse relationship:

- When bond prices fall → yields rise
- When bond prices rise → yields fall

Bond yields are influenced by:

1. **Monetary policy actions** (e.g., RBI repo rate changes, open market operations)
2. **Government borrowing** (higher supply of bonds → higher yields)
3. **Global factors** (e.g., US interest rates, capital flows, oil prices)
4. **Inflation expectations** (higher expected inflation → higher yields to compensate investors)

Importance:

Bond yields serve as a benchmark for **loan interest rates**, **investment returns**, and **macroeconomic signals**. Rising yields may indicate tightening liquidity or higher inflation, while falling yields signal easier monetary conditions or higher demand for safe assets.

Option I is correct: RBI increases the Policy Repo Rate

When the RBI hikes the repo rate, it **raises the cost of short-term borrowing for banks**. This signals **tighter monetary policy**, pushing up **interest rates** across the economy.

New bonds issued in the market will now offer higher interest rates (coupons) to match this new environment. Existing bonds with lower coupons become less attractive, leading to a fall in their market price. Since bond price and yield move inversely, this results in an increase in bond yields.

Option II is correct: Government increases Market Borrowing

If the Government of India decides to borrow significantly more to fund a higher fiscal deficit, it must issue a large volume of new Government Securities (G-Secs). This creates an excess supply of bonds in the market.

According to the basic laws of demand and supply, when supply exceeds demand, the price of the asset (the bond) falls.

As bond prices drop due to this oversupply, the bond yield increases.

Option III is incorrect: RBI conducts Open Market Purchases

This is the only scenario in the list that leads to a decrease in yields. When the RBI conducts large-scale open market purchases (OMO), it buys government securities from the market.

This increases liquidity and boosts demand for bonds. Higher demand pushes up bond prices. Since price and yield move in opposite directions, this action causes the bond yield to fall.

Thus, such actions are typically aimed at softening yields, not increasing them.

Option IV is correct: US Federal Reserve raises Interest Rates

When the US Fed raises rates, US Treasury bonds (considered the safest in the world) become more attractive to global investors. Foreign Portfolio Investors (FPIs) often pull their "hot money" out of emerging markets like India to reinvest in the US for better risk-adjusted returns. This mass selling of Indian bonds by foreign investors leads to a sharp drop in bond prices in India, which results in an increase in bond yields.

Option V is correct: Sharp increase in Global Crude Oil Prices

India is a major importer of crude oil. A spike in oil prices leads to "Imported Inflation," as the cost of transportation and production rises across the board. High inflation erodes the "real" return an investor gets from a fixed-income bond. To compensate for this loss of purchasing power and the risk of the RBI raising rates to fight that inflation, investors demand a higher return. This sentiment drives bond prices down and causes bond yields to increase.

Source:) : <https://www.thehindubusinessline.com/markets/india-bonds-seen-lower-as-elevated-oil-higher-us-yields-pinch-debt-sale-in-focus/article70778205.ece>

<https://www.thehindu.com/children/government-bonds-in-india-understanding-the-financial-market-component/article69400067.ece>

<https://www.thehindu.com/business/Economy/the-trump-turmoil-in-bond-markets/article69498711.ece>

Subject:) Economy

Topic:) Capital Market

Subtopic:)

Q.35)

Ans) a

Exp) Option a is the correct answer.

Statement I is correct: Digital Hub for Reference and Unique Virtual Address (DHRUVA) is designed as a Digital Public Infrastructure (DPI) aimed at creating a standardised, interoperable and digital addressing ecosystem across India.

- Its objective is to move beyond fragmented and descriptive addresses toward a uniform, geocoded, and machine-readable system, thereby improving service delivery, governance, logistics, and emergency response.
- Just as Aadhaar digitized identity and UPI digitized payments, DHRUVA aims to digitize and standardize "location" or "addressing" in India.

Statement II is correct: A key feature of DHRUVA is the concept of "Address-as-a-Service (AaaS)". Under this, users can generate and share tokenised virtual address identifiers (similar to UPI IDs, e.g., name@provider) instead of disclosing full physical addresses. These virtual addresses are mapped to actual geocoded locations, ensuring privacy, convenience, and ease of sharing, especially in e-commerce, delivery systems, and digital governance.

Statement III is incorrect: DHRUVA is not implemented by the Department of Telecommunications or under MeitY.

It is being **developed and anchored** by the **Department of Posts** under the **Ministry of Communications**, which is the nodal authority for addressing systems in India.

DHRUVA is built upon DIGIPIN (Digital Postal Index Number), which serves as its geospatial backbone and is the **National Addressing Grid**.

- DIGIPIN is a **nationwide geo-coded addressing system developed by the Department of Posts** in collaboration with IIT Hyderabad.
- It **divides India into approximately 4m x 4m grids** and **assigns each grid a unique 10-character alphanumeric code** based on latitude and longitude coordinates.
- This system simplifies addressing for public and private services, improving logistics efficiency and emergency response capabilities.

Source: <https://www.thehindu.com/news/national/what-is-the-india-posts-dhruva-framework-explained/article70373713.ece>

<https://www.pib.gov.in/PressReleaseIframePage.aspx?PRID=2132858>

<https://www.indiapost.gov.in/digipin>

Subject: Economy

Topic: Government Schemes

Subtopic:

Q.36)

Ans) b

Exp) Option b is the correct answer.

Inflation refers to a **sustained increase in the general price level of goods and services in an economy** over a period of time, which leads to a decline in the purchasing power of money.

Cost-Push Inflation:

Cost-push inflation refers to a situation where the **general price level in an economy rises due to an increase in the cost of production**. When the prices of key inputs, such as **raw materials, energy, wages, or taxes on production rise**, producers experience higher production costs. To maintain profitability, firms often **pass these higher costs on to consumers in the form of higher prices**, leading to inflation.

Cost-push inflation is therefore **supply-side inflation**, as it originates from factors that **reduce aggregate supply or increase production costs**, rather than from excessive demand in the economy.

In contrast, **demand-pull inflation** occurs when **aggregate demand in the economy exceeds the available supply of goods and services**. This usually happens due to factors such as **higher government expenditure, increased money supply, easy credit conditions, tax cuts, or higher household income**

Option I is correct: A sudden surge in global crude oil prices:

Crude oil is a critical input for transportation, manufacturing, fertilizers, plastics, and logistics. **When oil prices increase sharply** due to supply disruptions, production and transportation **costs across the economy rise**. Producers often pass these higher costs on to consumers, **leading to cost-push inflation**.

Option II is correct: An increase in the Minimum Support Price (MSP):

An **increase in MSP raises procurement prices of agricultural commodities** such as wheat and rice. Since food grains are key inputs for many food products, higher MSP can increase the cost of raw agricultural inputs, pushing up food prices and **contributing to cost-push inflation**.

Option III is incorrect: A substantial increase in Direct Benefit Transfers (DBT):

Direct Benefit Transfers increase the income and purchasing power of households. This **boosts aggregate demand** in the economy rather than increasing production costs.

Hence, **it may lead to demand-pull inflation, not cost-push inflation**.

Option IV is incorrect: A reduction in the repo rate by the RBI:

When the RBI reduces the repo rate, **borrowing becomes cheaper** for banks and businesses. This **increases liquidity and credit availability**, which tends to boost demand in the economy, **leading to demand-pull inflation, not cost-push inflation**.

Option V is correct: An increase in indirect tax rates (such as GST):

Higher indirect taxes on industrial inputs or raw materials **increase production costs for firms**. Businesses often pass this additional tax burden to consumers through higher prices, thereby **generating cost-push inflation**.

Source:) ForumIAS Economy Redbook (Second Edition), Chapter: 3 - Inflation and Business cycle, Page: 29-30.

<https://egyankosh.ac.in/bitstream/123456789/73894/1/Unit-8.pdf> Page: 83.

<https://frontline.thehindu.com/cover-story/understanding-inflation-the-real-causes/article65470160.ece>

<https://indianexpress.com/article/upsc-current-affairs/upsc-essentials/rising-inflation-a-growing-concern-for-policymakers-upsc-9706339/>

Subject:) Economy

Topic:) Inflation

Subtopic:)

Q.37)

Ans) b

Exp) Option b is the correct answer.

The **Unified Pension Scheme (UPS)** is a **pension framework introduced by the Government of India** as an **alternative to the National Pension System (NPS)** for Central Government employees, **with effect from 1 April 2025**.

It is designed as a **hybrid pension model, combining features of both:**

- the defined benefit nature of the **Old Pension Scheme (OPS)**, and
- the contributory structure of **NPS**.

The primary objective of UPS is to provide assured, predictable, and inflation-indexed pension benefits, addressing concerns related to market-linked uncertainty in NPS and ensuring long-term income security after retirement.

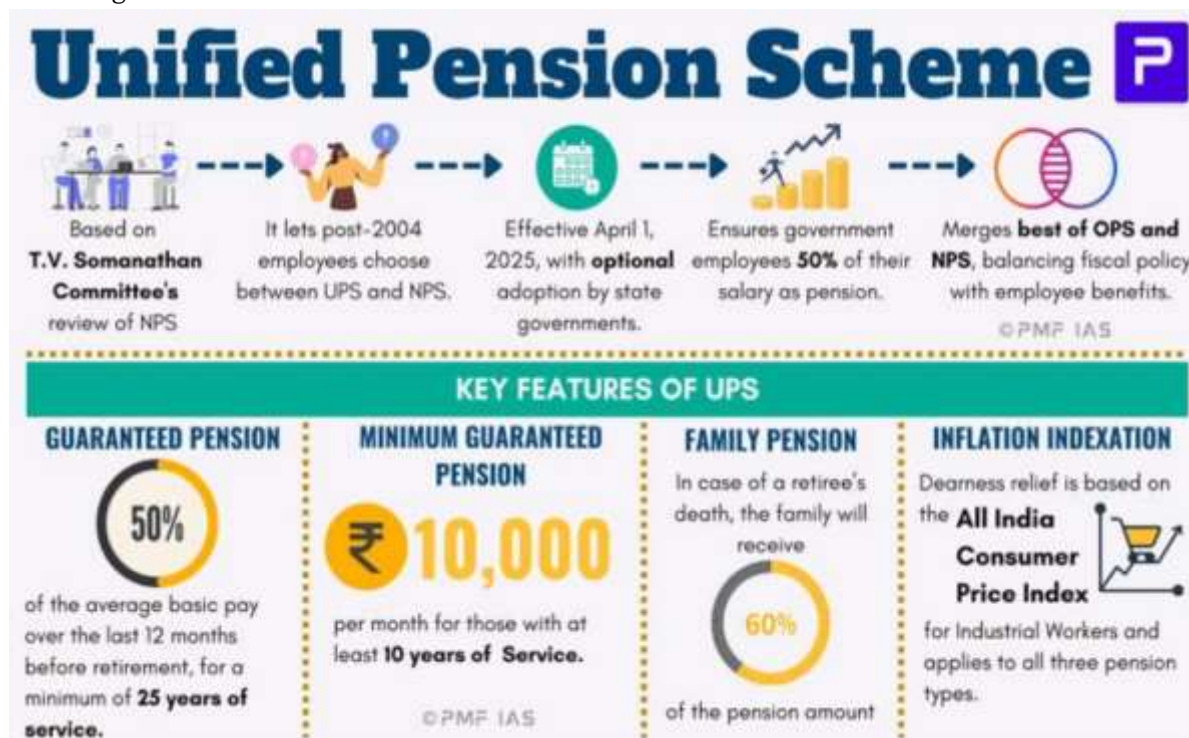
Statement I is incorrect: The **Unified Pension Scheme (UPS)** provides an **assured minimum pension of ₹10,000 per month**, but this benefit is **conditional upon completion of at least 10 years of qualifying service**. This provision ensures a basic income security even for employees with relatively shorter service duration, preventing very low pension outcomes.

Statement II is correct: The government has introduced a **one-time, one-way switch option** for employees. **UPS** provides flexibility by **allowing subscribers a one-time option to switch to the National Pension System (NPS)**.

However, **this choice is final and irreversible**, meaning once a subscriber opts out of UPS and moves to NPS, they cannot switch back to UPS again. This ensures administrative clarity and prevents repeated switching.

Statement III is correct: The scheme ensures social security for dependents by **providing a family pension equal to 60% of the pension** that was admissible to the deceased pensioner. The legally wedded spouse shall receive 60% of admissible payout drawn by the subscriber immediately prior to the demise.

Knowledge Base:



Source:) <https://pfrda.org.in/schemes/national-pension-system/unified-pension-scheme>

https://digitalscr.in/bzadiv/circulars/misc_circulars/uploads/UPS_010225.pdf

<https://www.thehindu.com/news/national/government-offers-one-time-switch-from-ups-to-nps/article69976785.ece>

<https://www.pib.gov.in/PressReleasePage.aspx?PRID=2168061®=3&lang=2>

<https://www.thehindu.com/news/national/ups-pfrda-notifies-assured-pension-scheme-for-government-employees/article69354699.ece>

<https://www.thehindu.com/news/national/unified-pension-scheme-how-does-it-differ-from-the-old-and-new-pension-scheme/article68571778.ece>

Subject:) Economy

Topic:) Government Schemes

Subtopic:)

Q.38)

Ans) d

Exp) Option d is the correct answer.

The **Wholesale Price Index (WPI)** is a key economic indicator in India that **measures the average change in prices of goods at the wholesale level** (i.e., before they reach the retail market). It **reflects price movements** from the **producer/supplier side of the economy**.

Statement I is incorrect: While the **Consumer Price Index (CPI)** base year was **recently revised to 2024 (from 2012)** in February 2026, the **Wholesale Price Index (WPI)** **continues to use the same base year**. The **base year** of the Wholesale Price Index (WPI) **in India is 2011-12**, which replaced the earlier base year of 2004-05.

Statement II is incorrect: WPI is **released by the Office of the Economic Adviser (OEA), Department for Promotion of Industry and Internal Trade (DPIIT), under the Ministry of Commerce and Industry**.

The **National Statistical Office (NSO)**, which falls under the Ministry of Statistics and Programme Implementation (MoSPI), is **responsible for releasing the CPI and the Index of Industrial Production (IIP)**.

Statement III is correct: The index is composed of three major groups: **Primary Articles, Fuel & Power, and Manufactured Products.**

In the WPI basket (2011-12 series), the **Manufactured Products group** is indeed the **largest component**, carrying a **weight of 64.23%** (approximately two-thirds). The other two major groups are **Primary Articles (22.62%)** and **Fuel & Power (13.15%)**.

Statement IV is correct: A fundamental difference between the two indices is that **WPI captures only goods** (traded in bulk at the wholesale level). **It does not include services.**

In contrast, the **CPI includes both goods and services** (such as education, healthcare, and transport) to reflect the actual cost of living for consumers.

Source:) <https://eaindustry.nic.in/>

<https://www.pib.gov.in/PressReleasePage.aspx?PRID=2214426®=3&lang=1>

<https://indianexpress.com/article/explained/everyday-explainers/what-are-wholesale-price-index-consumer-price-index-inflation-rates-7820831/>

<https://indianexpress.com/article/upsc-current-affairs/upsc-essentials/knowledge-nugget-consumer-price-index-cpi-base-year-2024-upsc-10530122/>

ForumIAS Economy Redbook (Second Edition), Chapter: 3 - Inflation and Business cycle, Page: 32.

Subject:) Economy

Topic:) Inflation

Subtopic:)

Q.39)

Ans) d

Exp) Option d is the correct answer.

Import Cover refers to the **number of months for which a country's existing foreign exchange (forex) reserves can pay for its imports** if no additional foreign currency inflows occur. It is an important indicator used to assess the **external sector stability and adequacy of foreign exchange reserves** of a country.

It is generally calculated as:

• **Import Cover = Total Foreign Exchange Reserves : Average Monthly Import Bill**

For example, if a country has **US\$600 billion in forex reserves** and its **average monthly imports are US\$60 billion**, the import cover would be **10 months**. This means the country can finance imports for **10 months even if no new foreign currency inflows occur**.

A **higher import cover** indicates a **stronger external position**, as the country has sufficient reserves to handle external shocks.

Significance:

- **Indicator of external stability:** Higher import cover implies a stronger ability to withstand external shocks (e.g., oil price rise, global crises).
- **Confidence for investors and rating agencies:** Adequate reserves signal lower risk of balance of payments (BoP) crisis.
- **Policy benchmark:** Traditionally, an import cover of around **8-10 months** is considered comfortable for developing economies like India.
- **Crisis buffer:** During sudden capital outflows or currency depreciation, countries with high import cover can continue financing essential imports

Source:) <https://timesofindia.indiatimes.com/business/india-business/rbi-report-shows-indias-forex-reserves-cover-stands-at-11-2-months-of-imports-with-a-slight-decline/articleshow/114763521.cms>

<https://www.indiabudget.gov.in/economicsurvey/doc/Infographics%20English.pdf> Page: 14.

<https://www.newindianexpress.com/opinion/2022/Nov/08/rupee-in-the-redcase-of-the-forex-blues-2516033.html>

<https://indiamacroindicators.co.in/economic-indicators/import-cover-in-months>

<https://www.thehindu.com/business/Economy/Import-cover-increases-to-10.9-months/article14499348.ece>

Subject:) Economy
Topic:) External Sector
Subtopic:)

Q.40)

Ans) c

Exp) Option c is the correct answer.

Option c is correct: Gross Domestic Product (GDP) measures the **total monetary value of all final goods and services produced within the geographical borders of a country** during a specific period (usually a year), **regardless of whether the producers are residents (citizens/nationals) or non-residents (foreigners).**

- GDP = Value of production inside the country

It includes income earned by foreign residents/companies from production within the domestic territory (option a), but excludes any income earned by the country's own residents from production abroad.

Gross National Income (GNI) measures the **total income earned by the residents** (nationals/citizens and resident entities) **of a country, regardless of where that income is generated** (inside or outside the country).

- The standard formula is: **GNI = GDP + Net Factor Income from Abroad (NFIA)**

Net Factor Income from Abroad (NFIA) = **Factor income earned by residents from abroad** (e.g., wages, profits, interest, dividends from foreign investments) **minus Factor income earned by non-residents from within the country** (what foreigners earn from production inside the country).

- **This net amount is added to GDP to get GNI.**
- Therefore, **net factor income earned from abroad by residents is included in GNI but excluded from GDP** (because GDP only counts domestic production, not foreign-earned income).

Option a is incorrect: Factor income earned by foreign residents within the domestic territory is **included in GDP** because GDP measures income generated **within the domestic boundaries**, irrespective of ownership. However, since this income accrues to non-residents, it is **deducted while calculating GNI**, which is based on residency.

Option b is incorrect: Net indirect taxes (indirect taxes minus subsidies) are part of the value of goods and services produced within the domestic economy, and hence are **included in both GDP and GNI**, as they do not involve any cross-border income flow.

Option d is incorrect: Profits earned by domestic firms from production within the domestic territory arise from domestic economic activity and are therefore **included in GDP**. Since these profits also accrue to residents, they are naturally included in GNI as well, making them common to both measures.

Source:) https://mospi.gov.in/sites/default/files/reports_and_publication/cso_national_accounts/Chapter%202.pdf
<https://www.cso.ie/en/interactivezone/statisticsexplained/nationalaccountsexplained/grossnationalproductgnpandgrossnationalincomegni/>

<https://sathee.iitk.ac.in/ncert-books/class-12/economics/introductory-to-macroeconomics/chapter-02-national-income-accounting/>

Subject:) Economy
Topic:) National Income Accounting
Subtopic:)

Q.41)

Ans) a

Exp) Option a is the correct answer.

The **Maldives** has **become the first country in the world to impose a generational ban on tobacco**, which came into effect on 1 November 2025 through an amendment to its Tobacco Control Act. Under this law, **any person born on or after 1 January 2007 is permanently prohibited from purchasing, selling, or using tobacco products** in the country.

It applies to **citizens as well as tourists** and covers **all forms of tobacco and vaping products**.

What is a Generational Ban on Tobacco?

A **generational ban** is a long-term public health policy in which a **particular generation is permanently prohibited from buying tobacco products**.

Unlike ordinary tobacco regulations, which only restrict the sale of tobacco to minors (for example, below 18 years of age), a generational ban **continues to apply even after those individuals become adults**.

In simple terms:

- People **born before the cut-off year** can legally buy tobacco.
- People **born on or after the cut-off year** will **never be allowed to purchase tobacco at any stage of their life**.

This means that **each new generation grows up without legal access to tobacco**, gradually eliminating tobacco consumption over time.

Why is it Important / Benefits

- **Prevents initiation:** Targets younger cohorts so that **future generations never start smoking**.
- **Gradual elimination:** As older users exit over time, **tobacco use declines naturally without abrupt disruption**.
- **Reduces disease burden:** Leads to a long-term decline in **tobacco-related illnesses** like cancer and cardiovascular diseases.
- **Breaks social transmission:** Stops the **intergenerational cycle of tobacco use**.
- **Lowers public health costs:** Reduces long-term **healthcare expenditure** on tobacco-related diseases.

Source:) <https://www.thehindu.com/news/international/maldives-to-phase-out-smoking-with-generational-ban/article69449527.ece>

<https://www.downtoearth.org.in/health/maldives-becomes-first-country-to-impose-generational-ban-on-tobacco>

Subject:) International Relations

Topic:) Countries in the News

Subtopic:)

Q.42)

Ans) a

Exp) Option a is the correct answer.

Eco-Sensitive Zones (ESZs) are **notified buffer areas around protected areas** (such as National Parks and Wildlife Sanctuaries) under the **Environment (Protection) Act, 1986**. Their primary objective is to minimise negative impacts of human activities on fragile ecosystems by acting as a “**shock absorber**” for protected areas.

Activities in ESZs are broadly **classified into three categories: prohibited, regulated, and permitted**, based on their environmental impact, with a focus on balancing conservation needs with sustainable livelihoods of local communities.

Option I is correct: Commercial mining → Prohibited

Mining, quarrying, and crushing units are strictly prohibited in ESZs due to their highly destructive impact on ecosystems, biodiversity, and groundwater systems.

Option II is incorrect: Felling of trees → Regulated

Tree cutting is not completely prohibited. It is allowed only with prior approval from competent authorities and in accordance with forest laws. Hence, **it falls under regulated activities**.

Option III is correct: Establishment of major hydroelectric projects → Prohibited

Large hydroelectric projects cause significant ecological disruption (deforestation, habitat fragmentation, alteration of river systems), and are therefore **not permitted in ESZs**.

Option IV is incorrect: Movement of vehicular traffic at night → Regulated

This is generally restricted to prevent disturbing wildlife and to reduce roadkill, but **it is not a total prohibition** (essential and local services are often exempt). Hence, **it falls under regulated activities**

Option V is incorrect: Establishment of hotels and resorts → Regulated

Tourism-related infrastructure is **allowed but strictly regulated**, with guidelines on construction, waste management, and environmental impact. Hence, **it is not prohibited, but falls under regulated activities**.

Source:) https://cpc.parivesh.nic.in/writereaddata/Guidelines_for_EcoSensitive_Zones_around_Protected_Areas.pdf
Page: 8.

<https://forest.rajasthan.gov.in/content/dam/raj/forest/ForestDepartment/PDFs/Public%20Information/Orders%20&%20Circulars/Guidelines%20for%20declaration%20of%20Eco-sensitive%20zones%20around%20National%20Parks%20and%20Wildlife%20Sanctuaries.pdf>

Subject:) Environment

Topic:) Conservations Efforts

Subtopic:)

Q.43)

Ans) b

Exp) Option b is the correct answer.

Monotremes are a primitive and **unique group of mammals** belonging to the **order Monotremata**. The term monotreme is derived from the Greek words “monos” (single) and “trema” (opening), referring to the presence of a single opening called the cloaca, which serves for excretion, reproduction, and egg-laying.

Monotremes are distinctive because **they are the only mammals which lay eggs** (oviparous) **instead of giving birth to live young**, a feature more typical of reptiles and birds. However, **like all mammals, they possess mammary glands and produce milk** to nourish their young, though they do not have nipples; the milk is secreted through pores in the skin and licked by the young.

They represent an ancient evolutionary lineage of mammals and display a mixture of reptilian and mammalian characteristics.

Today, **only five living species of monotremes remain on Earth, making them extremely rare. These include:**

- **One species of Platypus** (*Ornithorhynchus anatinus*).
- **Four species of Echidnas** (spiny anteaters): the short-beaked echidna and three species of long-beaked echidnas.

All of the **surviving members of the monotreme group** are **indigenous** to the island of **New Guinea and Australia**.

Option I is correct: Platypus

The **Platypus** (*Ornithorhynchus anatinus*) is a **classic example of a monotreme**. It lays eggs, has a duck-like bill, webbed feet, and produces milk to feed its young.

Option II is correct: Echidna

Echidnas (spiny anteaters) are **also monotremes**. They lay eggs and have spines covering their bodies. The female temporarily develops a pouch where the egg is incubated.

Option III is incorrect: Kangaroo

The **Kangaroo is not a monotreme. It belongs to the group marsupials**, which give birth to very underdeveloped young that continue to grow inside the mother's pouch.

Option IV is incorrect: Koala

The **Koala is also a marsupial, not a monotreme**. Like kangaroos, koalas give birth to tiny young that develop in a pouch.

Source:) <https://indianexpress.com/web-stories/miscellaneous/5-unique-egg-laying-mammals-youve-never-heard-of/9923053/>

<https://www.thehindu.com/sci-tech/science/the-animal-that-senses-electric-fields-tolerates-snow-and-has-mating-trains/article67244337.ece>

<https://timesofindia.indiatimes.com/etimes/trending/what-are-monotremes-the-rare-mammals-that-lay-eggs/articleshow/114930751.cms>

<https://www.worldatlas.com/articles/the-five-species-of-monotremes-living-today.html>

Subject:) Environment

Topic:) Biodiversity

Subtopic:)

Q.44)

Ans) b

Exp) Option b is the correct answer.

The **Wildlife (Protection) Act, 1972** is the primary legislation in India for the **protection of wild animals, birds, and plants**, and for ensuring ecological and environmental security. It was enacted to address the rapid decline of wildlife due to hunting, habitat destruction, and illegal trade. The '**Protected Areas**' covering the **Wildlife Sanctuaries, National Parks, Conservation/ Community Reserves and Tiger Reserves** are **notified under Chapter IV of the Wildlife (Protection) Act, 1972**.

Option I is incorrect: Biodiversity Heritage Sites (BHS)

- **Legal Origin: Biological Diversity Act, 2002.**
- **Definition:** These are well-defined areas that are unique, ecologically fragile ecosystems with a high diversity of wild as well as domesticated species, presence of rare and threatened species, or paleontological significance.
- **Key Distinction:** While they protect wildlife, **they are notified by State Governments in consultation with local bodies** (Panchayats/Municipalities) **under the Biodiversity Act, not the WPA.**
- **Example: Majuli in Assam and Nallur Tamarind Grove in Karnataka.**

Option II is correct: Wildlife Sanctuary (WLS)

- **Legal Origin: Wildlife (Protection) Act, 1972.** They are **declared by the State Government under the WPA, 1972.**
- **Definition:** An area of "adequate ecological, faunal, floral, geomorphological, natural or zoological significance."
- **Key Distinction:** In a Sanctuary, certain rights of people (like grazing or collecting minor forest produce) can be allowed by the Chief Wildlife Warden, provided they don't interfere with wildlife conservation. This is a primary statutory category of the WPA.

Option III is incorrect: Reserved Forest (RF)

- **Legal Origin: Indian Forest Act, 1927.** Reserved Forests are **defined and notified under the Indian Forest Act, 1927.**
- **Definition:** These are the **most restricted forests**. In a Reserved Forest, everything is prohibited unless specifically permitted.
- **Key Distinction:** The Indian Forest Act is a colonial-era law primarily focused on timber and forest produce management. While an RF provides a habitat for wildlife, it is not a "Protected Area" defined by the WPA 1972.

Option IV is correct: Tiger Reserve

- **Legal Origin: Wildlife (Protection) Act, 1972 (Inserted via the 2006 Amendment).**
- **Definition:** These areas are notified for the protection of tigers and their prey base. They consist of a **Core/Critical Tiger Habitat** and a Buffer/Peripheral area.
- **Key Distinction:** Before 2006, "Tiger Reserve" was an administrative designation under Project Tiger (1973). The 2006 amendment gave it statutory recognition and created the **National Tiger Conservation Authority (NTCA)**.

Option V is correct: Community Reserves

- **Legal Origin: Wildlife (Protection) Act, 1972 (Inserted via the 2002 Amendment).**
- **Definition:** These are **areas notified by the State Government on private or community land** where the community or an individual has volunteered to conserve wildlife and its habitat.
- **Key Distinction:** They act as buffer zones or connectors between established National Parks and Sanctuaries. They allow for conservation without displacing local communities.

Source:) <https://egyankosh.ac.in/bitstream/123456789/70834/1/Unit-3.pdf> Page: 50.

<https://www.pib.gov.in/PressReleaseIframePage.aspx?PRID=1982309>

<https://training.eforest.delhi.gov.in/ifa.aspx>

Subject:) Environment

Topic:) Conservations Efforts

Subtopic:)

Q.45)

Ans) c

Exp) Option c is the correct answer.

Electronic waste (e-waste) refers to **discarded electrical and electronic devices** such as mobile phones, computers, televisions, refrigerators, and other gadgets. It is one of the fastest-growing waste streams globally due to rapid technological advancement, short product lifespans, and increasing consumption.

When **e-waste is processed informally**, especially through improper dismantling, open burning, shredding, or acid-leaching, it releases a variety of hazardous substances into the environment (air, soil, water, and food chain). **These include:**

- **Heavy metals** (e.g., **lead, mercury, cadmium, chromium, antimony**)
- **Persistent organic pollutants and flame retardants** (e.g., **brominated flame retardants** like PBDEs and TBBPA)
- **Unintentional combustion by-products** (e.g., **dioxins and furans** formed during open burning of halogen-containing plastics)

These substances are **toxic, bio-accumulative**, and can cause **serious health effects** (neurological damage, cancer, reproductive disorders, etc.) in workers, nearby communities, and ecosystems.

Option I is correct: Chromium

Chromium is **commonly used in metal plating and protective coatings of electronic components**, connectors, and computer housings to prevent corrosion. Certain forms, particularly **hexavalent chromium (Cr⁶⁺)**, are **highly toxic and carcinogenic**. When electronic components are dismantled or dumped in landfills, **chromium can leach into the soil and groundwater**.

Option II is incorrect: Parathion

Parathion is a **highly toxic organophosphate pesticide used in agriculture** to control insects on crops. Although it is an environmental pollutant and hazardous chemical, it is not used in electronic components or materials associated with e-waste. Hence, **it is not released during e-waste recycling or disposal**.

Option III is correct: Brominated flame retardants

Brominated flame retardants (BFRs) are **widely added to plastic casings, printed circuit boards, cables, and electronic components** to reduce the risk of fire. These compounds are **persistent organic pollutants** that can **accumulate in living organisms** and disrupt endocrine systems. During burning or heating of e-waste plastics, **BFRs may be released or transformed into other toxic compounds**.

Option IV is correct: Dioxins

Dioxins are **extremely toxic persistent organic pollutants (POPs)**. They are **not intentionally used in electronic devices** but are often **formed as by-products** when plastics containing brominated flame retardants or chlorinated compounds are burned, a practice common in informal e-waste recycling sites. Dioxins can cause **serious health effects**, including **immune, reproductive, and developmental problems**.

Option V is correct: Antimony

Antimony compounds, particularly antimony trioxide, are used as synergists in flame-retardant systems in plastics and electronic components. They enhance the effectiveness of brominated flame retardants used in circuit boards and plastic housings. During improper recycling or incineration of e-waste, **antimony can be released into the environment and may pose toxic and carcinogenic risks**.

Source:) <https://www.thehindu.com/sci-tech/health/the-cost-of-convenience-health-hazards-as-a-side-effect-of-using-digital-tools/article70116984.ece>

<https://www.thehindu.com/opinion/op-ed/indias-rising-e-waste-the-need-to-recast-its-management/article69568262.ece>

<https://www.thehindu.com/incoming/e-waste-collection-faces-gaps-as-government-sets-sights-on-recycling-for-precious-metals/article70108477.ece>

<https://www.fao.org/4/w5715e/w5715e05.htm>

Subject:) Environment

Topic:) Pollution

Subtopic:)

Q.46)

Ans) d

Exp) Option d is the correct answer.

Option d is correct: The **London Convention** (formally known as the **Convention on the Prevention of Marine Pollution by Dumping of wastes and Other Matter**) was **adopted in 1972** and **entered into force in 1975**.

- Its **primary objective** is exactly as described: to **promote the effective control of all sources of marine pollution** and to take all practicable steps to **prevent pollution of the sea by the dumping of wastes and other matter**.
- It is one of the first global conventions to protect the marine environment from human activities.

London Protocol 1996:

In 1996, the "**London Protocol**" was agreed to further **modernize the Convention** and, **eventually, replace it**. Under the Protocol all dumping is prohibited, except for possibly acceptable wastes on the so-called "reverse list". The Protocol entered into force in 2006.

Option a is incorrect: Basel Convention

It was **adopted in 1989** under the **United Nations Environment Programme (UNEP)**. Its objective is to **control and reduce the transboundary movement of hazardous wastes**, especially from developed to developing countries.

It also promotes environmentally sound management and disposal of hazardous waste.

However, it **deals with international movement and disposal of hazardous waste on land, not specifically with dumping wastes into the sea**.

Option b is incorrect: Helsinki Convention

The **Helsinki Convention**, formally the **Convention on the Protection of the Marine Environment of the Baltic Sea Area** is a regional agreement applicable only to countries bordering the Baltic Sea.

- It seeks to address the increasing environmental challenges from industrialisation and other human activities that were having a severe impact on the marine environment.
- The Convention includes the **protection of the Baltic Sea from all sources of pollution from land, air and sea**.

However, its **scope is geographically limited** and not specifically centered on the global regulation of dumping of wastes at sea.

Option c is incorrect: MARPOL Convention

- The **International Convention for the Prevention of Pollution from Ships (MARPOL)** was **adopted under the International Maritime Organization**.
- It is the main international convention covering **prevention of pollution of the marine environment by ships** from operational or accidental causes.
- The convention contains **six technical annexes** covering different types of ship-generated pollution.

However, **MARPOL regulates operational and accidental pollution from ships, not intentional dumping of wastes into the sea**, which is **specifically addressed by the London Convention**.

Knowledge Base:



Source: <https://www.imo.org/en/ourwork/environment/pages/london-convention-protocol.aspx>
<https://indianexpress.com/article/upsc-current-affairs/upsc-essentials/how-india-seeks-to-deal-with-persisting-problem-of-marine-pollution-10497545/>
<https://www.thehindu.com/opinion/open-page/microplastics-menace/article69444191.ece>
<https://indianexpress.com/article/explained/antiquities-abroad-indian-international-laws-metropolitan-museum-of-art-subhash-kapoor-8495034/>
<https://www.unep.org/helsinki-convention>
[https://www.imo.org/en/about/conventions/pages/international-convention-for-the-prevention-of-pollution-from-ships-\(marpol\).aspx](https://www.imo.org/en/about/conventions/pages/international-convention-for-the-prevention-of-pollution-from-ships-(marpol).aspx)
 Subject: Environment
 Topic: Pollution
 Subtopic:

- Q.47)**
Ans) a
Exp) Option a is the correct answer.
- In ecology, **primary productivity** refers to the **rate at which producers** such as plants, algae and certain bacteria **convert solar energy into chemical energy through photosynthesis**. This process forms the foundation of all food chains, as it determines the amount of energy available to higher trophic levels.
- **Primary productivity** is commonly **measured in terms of biomass or organic matter produced per unit area per unit time** (e.g., grams of carbon per square metre per year). It is **broadly divided into Gross Primary Productivity (GPP) and Net Primary Productivity (NPP)**.

Net Primary Productivity (NPP):

Net Primary Productivity (NPP) refers to the **rate at which plants accumulate organic matter (biomass) after accounting for the energy they use in respiration**. It therefore represents the **actual amount of energy stored in plant tissues** that becomes available to herbivores and other consumers in an ecosystem.

$$\text{NPP} = \text{GPP} - \text{R}$$

Where:

- **GPP (Gross Primary Productivity)** = total energy captured through photosynthesis
- **R (Respiration)** = energy used by plants for metabolic activities

Thus, NPP is the portion of energy that remains stored as plant biomass and supports the rest of the food web.

Option a is correct: Tropical rainforest:

Tropical rainforests exhibit the highest net primary productivity per unit area among terrestrial ecosystems. This is due to several favourable environmental conditions:

- **High and stable temperatures throughout the year**, which enhance metabolic and photosynthetic processes.
- **Abundant rainfall**, usually **exceeding 2000 mm annually**, ensuring continuous water availability for plant growth.
- **Intense solar radiation** near the equator, providing ample energy for photosynthesis.
- **Year-round growing season**, with almost no dormant period.
- **Dense and multilayered vegetation** structure, including canopy, understory, and ground vegetation, which maximizes light capture.

Because of these conditions, **tropical rainforests** have **extremely rapid biomass production** and **nutrient cycling**, leading to the **highest productivity per unit area**.

Option b is incorrect: Open Ocean:

The **open ocean** covers nearly 70% of the Earth's surface, so its **total global productivity is very large**. However, **productivity per unit area is relatively low** because:

- Surface waters in the open ocean are often nutrient-poor (especially nitrogen and phosphorus).
- Phytoplankton growth is limited in many regions due to nutrient stratification and limited upwelling.

Thus, **despite contributing significantly to global primary production** due to its vast size, the **open ocean does not have high productivity per unit area**.

Option c is incorrect: Desert:

Deserts have very low net primary productivity due to severe environmental constraints:

- **Extremely low rainfall**, often less than 250 mm annually.
- **High evaporation rates**, leading to water scarcity.
- **Sparse vegetation cover**, limiting photosynthesis.
- **Poor soil nutrient availability** in many desert regions.

Plants in deserts are usually **xerophytic** (drought-adapted), such as cacti and shrubs, and grow slowly. Hence, **deserts show very low biomass production per unit area**.

Option d is incorrect: Tundra:

Tundra ecosystems have very low net primary productivity due to several harsh environmental constraints:

- Extremely low temperatures for most of the year.
- Very short growing season, often only a few weeks in summer.
- Presence of permafrost, which restricts root penetration and nutrient cycling.
- Low decomposition rates, leading to poor nutrient availability.

As a result, **tundra vegetation consists** mainly of **mosses, lichens, grasses**, and small shrubs, **producing very low biomass per unit area**.

Knowledge Base: Gross Primary Productivity (GPP):

Gross primary production refers to the total amount of solar energy fixed into organic matter by primary producers through photosynthesis. The **total energy accumulated in the plant during primary productivity is known as Gross Primary Productivity**.

Source: <https://egyankosh.ac.in/bitstream/123456789/16250/1/Unit-5.pdf> Page: 17.

https://egyankosh.ac.in/bitstream/123456789/59046/1/Unit8_Ecosystem%20Cunctioning.pdf Page: 73.

<https://ncert.nic.in/textbook/pdf/lebo112.pdf> Page: 207.

Subject: Environment

Topic: Ecology

Subtopic:)

Q.48)

Ans) c

Exp) Option c is the correct answer.

Nitrogen fixation is a crucial biological process in the nitrogen cycle through which **atmospheric nitrogen (N_2) is converted into ammonia (NH_3) or other biologically usable nitrogen compounds.**

- Although nitrogen gas constitutes about 78% of the Earth's atmosphere, it is chemically very stable and cannot be directly utilized by most plants and animals.
- To become available for biological use, atmospheric nitrogen must first be "fixed" into reactive forms such as ammonia, ammonium, or nitrates.
- This conversion is primarily carried out by certain **microorganisms** known as **diazotrophs**, including **bacteria and cyanobacteria**.

There are **three types of nitrogen-fixing bacteria**, besides a group of algae. They are:

- a) **Symbiotic nitrogen-fixing bacteria**, e.g., **Rhizobium**
- b) **Associative nitrogen-fixing bacteria**, e.g., **Azospirillum**
- c) **Free living nitrogen-fixing bacteria**, e.g., **Azotobacter**, **Bacillus**, etc.
- d) **Free-living blue green algae**, e.g., **Anabaena**, **nostoc**

Option I is correct: Rhizobium

A **symbiotic nitrogen-fixing bacterium** found in root nodules of **leguminous plants** (e.g., pea, gram). It **converts atmospheric nitrogen into ammonia** in association with the host plant, receiving nutrients in return.

Option II is correct: Azotobacter

A **free-living, aerobic soil bacterium capable of fixing atmospheric nitrogen independently**. It plays an important role in enhancing soil fertility in non-leguminous crops.

Option III is correct: Anabaena

A **filamentous cyanobacterium (blue-green algae)** that **fixes nitrogen** through specialized cells called heterocysts, which provide an anaerobic environment necessary for nitrogenase activity. It also forms **symbiotic associations** (e.g., with **Azolla in paddy fields**).

Option IV is incorrect: Nitrosomonas

Nitrosomonas, a **chemoautotrophic nitrifying bacterium**, is **not a nitrogen-fixing organism**, but a **nitrifying bacterium involved in the nitrogen cycle**.

- It **participates in the process of nitrification**, which is the **conversion of ammonia (NH_3) or ammonium ions (NH_4^+) into nitrite (NO_2^-)**.
- This is the first step of nitrification, followed by **conversion of nitrite into nitrate (NO_3^-)** by **Nitrobacter**.
- The entire process is carried out under **aerobic conditions** by **chemoautotrophic bacteria such as Nitrosomonas and Nitrobacter**

Option V is correct: Clostridium

A **free-living, anaerobic bacterium** that fixes nitrogen in oxygen-deficient environments such as waterlogged soils. It represents non-symbiotic nitrogen fixation under anaerobic conditions.

Option VI is correct: Nostoc

A **cyanobacterium (blue-green algae) capable of nitrogen fixation** via heterocysts. It can exist freely or in symbiotic associations (e.g., with lichens, bryophytes, and some higher plants), contributing to nitrogen enrichment.

Source:) <https://www.thehindu.com/sci-tech/science/is-nitrogen-the-building-block-of-life-a-latent-time-bomb/article69401531.ece>
<https://www.thehindu.com/children/the-green-peas-connection-to-science/article70641580.ece>
<https://www.thehindubusinessline.com/economy/agri-business/how-biofertilisers-are-changing-farmer-practices-in-india/article70739926.ece>
<https://ncert.nic.in/vocational/pdf/kefc105.pdf> Page: 86.
<https://egyankosh.ac.in/bitstream/123456789/94876/1/Unit-7.pdf>
<https://www.egyankosh.ac.in/bitstream/123456789/104824/3/Unit-5.pdf> Page: 92.
 Subject:) Environment
 Topic:) Ecology
 Subtopic:)

Q.49)

Ans) b

Exp) Option b is the correct answer.

The **Intergovernmental Negotiating Committee (INC)** was **established to negotiate a legally binding international instrument to end plastic pollution, including pollution in the marine environment.**

- The process was initiated after a **resolution titled “End Plastic Pollution: Towards an International Legally Binding Instrument”** was **adopted at the United Nations Environment Assembly (UNEA-5.2) of the United Nations Environment Programme (UNEP), in 2022.**
- The **5th session of the INC (INC-5)** was **held in Busan** from 25 November to 1 December 2024.
- The **objective of this session** was to **finalize the text of a global legally binding treaty to end plastic pollution.**
- The negotiations were particularly significant because the proposed agreement seeks to **tackle plastic pollution at a global scale**, including the growing problem of plastics accumulating in oceans and marine ecosystems.

However, **countries were unable to reach consensus during the Busan meeting.** As a result, the negotiations were not concluded at INC-5.

To continue the process, the **negotiations were resumed in 2025** through a follow-up session (INC-5.2) held in Geneva, where countries continued discussions to move toward finalizing the global plastics treaty. but it also **ended without agreement.**

Importantly, the **proposed treaty is designed to address the entire life-cycle of plastics, which includes:**

- Production of plastics and raw materials
- Product design and manufacturing
- Patterns of consumption and use
- Waste management and recycling systems
- Prevention of plastic leakage into terrestrial and marine environments

Thus, the **INC-5 negotiations specifically focused on creating a global legally binding agreement to end plastic pollution.**

Source:) <https://www.unep.org/inc-plastic-pollution>
<https://www.thehindu.com/news/international/un-sets-date-for-extra-session-to-finalize-plastics-treaty/article69289904.ece>
<https://www.thehindu.com/sci-tech/energy-and-environment/plastic-pollution-treaty-talks-end-without-deal/article69935907.ece>
 Subject:) Environment
 Topic:) Pollution
 Subtopic:)

Q.50)

Ans) b

Exp) Option b is the correct answer.

In an ecosystem, **organisms do not live in isolation** but **interact with one another in various ways** to obtain food, shelter, protection, and reproductive advantages.

Ecological interactions are broadly **classified based on whether the participating species benefit, are harmed, or remain unaffected. The major types include:**

- **Mutualism (+/+):** Both species benefit from the interaction.
- **Example:** fig tree and fig wasp, ants and Acacia plant.
- **Commensalism (+/0):** One species benefits while the other is neither harmed nor benefited.
- **Example:** barnacles growing on whales.
- **Parasitism (+/-):** One organism (parasite) benefits while the host organism is harmed.
- **Example:** tapeworm living in the human intestine.
- **Amensalism (-/0):** One species is inhibited or harmed, while the other species is **unaffected**.
- **Example:** The fungus **Penicillium** secretes antibiotics that inhibit bacteria.
- **Predation (+/-):** One organism kills and consumes another organism.
- **Competition (-/-):** Two species compete for the same limited resources, reducing the fitness of both.

Mutualism is a type of **ecological interaction in which both participating species benefit from the association**. It is denoted by the symbol (+/+), indicating a positive effect on both organisms involved.

In mutualistic relationships, each species provides **resources, protection, or services** that enhance the survival, growth, or reproduction of the other species.

Option I is correct: Fig tree and fig wasp - Mutualism

The **relationship between fig trees (genus *Ficus*) and fig wasps** is **one of the most classic examples of obligate mutualism in nature**.

- The female fig wasp enters the fig inflorescence (syconium) to lay its eggs.
- While doing so, it pollinates the fig flowers using pollen carried from another fig.
- The fig provides a protected site and food for the developing wasp larvae.

Thus, the **wasp gets a place to reproduce**, and the **fig gets pollinated, benefiting both organisms**.

Option II is incorrect: Tapeworm and human intestine - Parasitism

Tapeworms (such as *Taenia*) live inside the human intestine.

- The tapeworm obtains nutrients and shelter from the host.
- The human host is harmed, as the parasite absorbs nutrients and may cause disease.

Since **one organism benefits** while the **other is harmed**, **this interaction is parasitism, not mutualism**.

Option III is incorrect: Barnacles on the back of a whale - Commensalism

Barnacles often attach themselves to the skin of whales.

- The barnacles benefit by gaining transportation and access to food-rich waters.
- The whale is generally neither significantly harmed nor benefited.

This interaction is therefore **commensalism**, where **one species benefits** and the **other is unaffected**.

Option IV is correct: Stinging ants and Acacia plant - Mutualism

Certain Acacia plants have a well-known mutualistic relationship with stinging ants.

- The plant provides food (nectar and protein-rich Beltian bodies) and shelter (hollow thorns) for the ants.
- The ants protect the plant by attacking herbivores and removing competing vegetation nearby.

Thus, **both the plant and the ants benefit from the association**.

Knowledge Base:

Table 11.1 : Population Interactions

Species A	Species B	Name of Interaction
+	+	<i>Mutualism</i>
-	-	<i>Competition</i>
+	-	<i>Predation</i>
+	-	<i>Parasitism</i>
+	0	<i>Commensalism</i>
-	0	<i>Amensalism</i>

Source:) <https://www.egyankosh.ac.in/bitstream/123456789/12758/1/Unit-11.pdf> Page: 89.

<https://ncert.nic.in/textbook/pdf/lebo111.pdf> Page: 201.

Subject:) Environment

Topic:) Ecology

Subtopic:)

Q.51)

Ans) b

Exp) Option b is the correct answer.

Originally, the **Right to Property** was **one of the seven Fundamental Rights**. It was guaranteed under two specific provisions:

- Article 19(1)(f):** Guaranteed every citizen the right to acquire, hold, and dispose of property.
- Article 31:** Guaranteed that no person shall be deprived of his/her property except by authority of law.

The Shift in 1978:

The **44th Constitutional Amendment Act, 1978**, enacted by the Morarji Desai-led Janata Party government, **made a significant structural change to this right:**

- It **repealed Article 19(1)(f) and Article 31 from Part III** (Fundamental Rights).
- It **inserted a new Article 300A in Part XII** under the heading "**Right to Property**."

By moving the right to Article 300A, it ceased to be a Fundamental Right and became a constitutional right.

The primary implications are:

- No Direct Supreme Court Writ:** A citizen cannot move the Supreme Court directly under Article 32 for its violation (though they can still approach a High Court under Article 226).
- Protection against Executive Action:** It still protects a person against the arbitrary seizure of property by the executive, but the Parliament can regulate or curtail it by ordinary law rather than a constitutional amendment.

Source:) Mastering Indian Polity by Forum IAS, Chapter: 10 - Fundamental Rights, Page: 156.

<https://www.hindustantimes.com/india-news/right-to-property-is-a-human-and-constitutional-right-sc-101735844271693.html>

<https://legallaffairs.gov.in/sites/default/files/chapter%203.pdf>

Subject:) Polity

Topic:) Constitutional Amendments

Subtopic:)

Q.52)

Ans) c

Exp) Option c is the correct answer.

The **Governor of a State** is the **nominal executive head**, who ordinarily acts on the aid and advice of the Council of Ministers (Article 163). However, the Constitution also recognizes that in certain situations, the Governor may act in his/her discretion, i.e., independently of ministerial advice.

The **discretionary powers** of the governor **can be divided into constitutional discretion and situational discretion**.

- **Constitutional (explicit) discretionary powers** → Those expressly provided in the Constitution, such as reserving a Bill for the President (Articles 200–201).
- **Situational (implicit) discretionary powers** → Those that arise from constitutional practice and political circumstances (e.g., hung assembly), but are **not explicitly mentioned in the Constitution**.

Option a is incorrect: Appointment of the Chief Minister when no party has a clear majority in the State Legislative Assembly:

In a hung assembly, the Governor exercises judgment in inviting a leader to form the government. However, this discretion arises from conventions and practical necessity, not from any explicit constitutional provision declaring it as a discretionary power.

Hence, **it is not Constitutional discretion but a Situational Discretion**.

Option b is incorrect: Dismissal of the Council of Ministers when it loses the confidence of the Legislative Assembly.

The Council of Ministers is collectively responsible to the Legislative Assembly under **Article 164(2)**. If it loses the majority, it must resign. The Governor's role in dismissal flows from this principle, but the Constitution does not expressly vest this as a discretionary power.

Hence, **this too is a Situational Discretion of the Governor**.

Option c is correct: Reservation of certain bills passed by the State Legislature for the consideration of the President.

Under **Articles 200 and 201**, the Governor may reserve a Bill for the President's consideration. In certain cases (e.g., bills derogating from the powers of the High Court), reservation is mandatory, while in others it is left to the Governor's discretion.

This is a clear instance of discretionary power explicitly provided in the Constitution.

Option d is incorrect: Dissolution of the Legislative Assembly if the Council of Ministers has lost its majority.

Normally, dissolution is carried out on the aid and advice of the Council of Ministers (Article 174). However, the Governor can dissolve the legislative assembly and call for fresh elections under certain circumstances, such as when the government has lost its majority. But the Governor's role is shaped by conventions, but the Constitution does **not explicitly classify this as a discretionary power**.

Hence, **it is not Constitutional discretion but a Situational Discretion**.

Source:) Mastering Indian Polity by Forum IAS, Chapter: 15 – Executive Branch, Page: 308.

<https://indianexpress.com/article/explained/explained-law/how-supreme-court-has-defined-contours-of-gubernatorial-power-over-the-years-10378232/>

<https://www.thehindu.com/news/national/citing-article-356-supreme-court-asks-why-it-cant-review-governor-delaying-bills/article69980136.ece>

Subject:) Polity

Topic:) State Executive

Subtopic:)

Q.53)

Ans) a

Exp) Option a is the correct answer.

Option a is correct: The S.R. Bommai case is one of the most significant judgments in Indian Constitutional law regarding Center-State relations and the misuse of Article 356.

1. Judicial Review: The Supreme Court categorically held that **the President's proclamation imposing President's Rule (under Article 356) is not beyond judicial review**. While the "satisfaction" of the President is subjective, the material on which that satisfaction is based can be examined by the court to see if it is relevant or **mala fide** (in bad faith).

2. Floor Test: The court ruled that the majority of a government must be tested on the floor of the Assembly (Floor Test) and not in the Governor's chambers.

3. Power of the Court: If the court finds the proclamation unconstitutional, it has the power to restore the dismissed state government and revive the dissolved or suspended Legislative Assembly.

4. Secularism as Basic Structure: The court also held that **Secularism is a part of the "Basic Structure" of the Constitution**. Therefore, a state government acting against secularism can be a valid ground for action under Article 356. Thus, this case transformed Article 356 from a **political weapon into a constitutionally controlled power, ensuring accountability and reinforcing cooperative federalism**.

Option b is incorrect: Kesavananda Bharati v. State of Kerala (1973)

This case propounded the Basic Structure Doctrine, holding that Parliament cannot alter the basic structure of the Constitution; it did not deal with Article 356 or judicial review of the President's Rule.

Option c is incorrect: Minerva Mills v. Union of India (1980)

This case **reaffirmed the Basic Structure Doctrine and emphasized the harmony between Fundamental Rights and Directive Principles**; it has no direct connection with Article 356.

Option d is incorrect: The Maneka Gandhi v. Union of India (1978) case is a landmark judgment of the Supreme Court of India that fundamentally reshaped the country's constitutional law. It is best known for expanding the scope of Article 21 (Right to Life and Personal Liberty) and establishing the principle that any law depriving a person of their liberty must be "fair, just, and reasonable". It is not associated with originally establishing judicial review of the President's Rule.

Knowledge Base: **President's Rule (Article 356)**

President's Rule, also known as **State Emergency** or **Constitutional Emergency**, refers to the suspension of state government and the imposition of direct Central Government rule in a state.

Grounds for Imposition: According to the Constitution, President's Rule can be proclaimed under two circumstances:

- **Article 356:** If the President is satisfied (based on a report from the Governor or otherwise) that a situation has arisen in which the **government of a state cannot be carried on in accordance with the provisions of the Constitution**.
- **Article 365:** If a state **fails to comply with or give effect to any direction from the Centre**, it is lawful for the President to hold that a situation has arisen where the state government cannot be carried on constitutionally.

Parliamentary Approval and Duration

- **Approval:** A proclamation must be approved by **both Houses of Parliament within two months** from the date of its issue.
- **Majority:** It requires a **Simple Majority** (majority of members present and voting) in both houses.
- **Duration:** Once approved, it continues for **six months**. It can be extended for a maximum period of **three years**, with approval every six months.
- **The 44th Amendment Restriction:** Beyond one year, the rule can only be extended if:

1. A Proclamation of National Emergency is in operation.

2. The Election Commission certifies that elections to the Legislative Assembly cannot be held.

Source:) Mastering Indian Polity by Forum IAS, Chapter: 14 – Emergency, Page: 254.

<https://indiankanoon.org/doc/60799/>

<https://indianexpress.com/article/explained/what-was-the-bommai-judgment-which-the-sc-relied-on-in-its-article-370-ruling-9064090/>

<https://www.thehindu.com/news/national/how-is-presidents-rule-imposed-explained/article69650172.ece>

Subject:) Polity

Topic:) Judiciary

Subtopic:)

Q.54)

Ans) d

Exp) Option d is the correct answer.

Option d is correct: Article 42 of the Constitution is **part of the Directive Principles of State Policy (DPSPs)** under **Part IV (Articles 36–51)**. These principles are non-justiciable, meaning they are not enforceable by courts, but they guide the State in framing laws and policies to ensure social and economic justice.

Article 42 specifically directs the State to:

1. **Secure just and humane conditions of work** – ensuring fairness, safety, and dignity for workers.
 2. **Provide maternity relief** – protecting women during pregnancy and after childbirth, including leave and other benefits.
- [Article 42. **Provision for just and humane conditions of work and maternity relief.**–The State shall make provision for securing just and humane conditions of work and for maternity relief.]

Option a is incorrect: Fundamental Rights–Article 16

- Article 16 **guarantees equality of opportunity in matters of public employment**.
- It focuses on non-discrimination in government jobs on the basis of religion, race, caste, sex, descent, place of birth and residence.
- While it promotes fairness in employment, it does not provide any specific welfare benefits such as maternity relief or humane working conditions.

Option b is incorrect: Directive Principles of State Policy–Article 43

- Article 43 **directs the State to promote cottage industries in rural areas** and support local craftsmanship.
- Its focus is on rural economic development rather than labour welfare or gender-specific protections.
- Maternity relief or humane working conditions fall outside the scope of Article 43, making it incorrect.

Option c is incorrect: Fundamental Rights–Article 15

- Article 15 **prohibits discrimination on grounds of religion, race, caste, sex, or place of birth**.
- It indirectly supports gender equality by forbidding discrimination on the basis of sex, but it does not mandate any specific welfare measures like maternity relief or working conditions.
- Thus, while socially relevant, Article 15 cannot be the source of maternity benefits.

Source:) Mastering Indian Polity by Forum IAS, Chapter: 11 – Directive Principles of State Policy, Page: 173.

<https://indiankanoon.org/doc/111604/>

Constitution of India.

Subject:) Polity

Topic:) Directive Principles of State Policy

Subtopic:)

Q.55)

Ans) b

Exp) Option b is the correct answer.

Under the **Rules of Procedure of the Lok Sabha**, the **Speaker nominates a Panel of Chairpersons (not more than ten members)** from among the members of the House.

- In the **absence of both the Speaker and the Deputy Speaker**, any one member from this panel presides over the **sitting of the House**.
- A member of this panel, when presiding, exercises all the powers of the Speaker for the duration of that sitting.
- A Chairperson so nominated, holds office until a new Panel of Chairpersons is nominated, unless he/she resigns earlier from the Panel or is appointed a Minister or elected as Deputy Speaker.
- **If a member of the panel is also absent**, the **House can designate any other member to preside temporarily**.
- However, it is important to note that **members of the panel cannot preside when the office of the Speaker or Deputy Speaker is vacant**.
- In such a situation, the duties of the Office of the Speaker are **performed by such Members of the Lok Sabha as the President may appoint for the purpose**.

Source:) Mastering Indian Polity by ForumIAS, Chapter: 16 - LEGISLATIVE BRANCH, Page: 355.

<https://cdnbbsr.s3waas.gov.in/s380537a945c7aaa788ccfcd1b99b5d8f/uploads/2024/07/20240716890312078.pdf>

(Article 95 Page 42)

<https://sansad.in/ls/faq>

Subject:) Polity

Topic:) Parliament

Subtopic:)

Q.56)

Ans) c

Exp) Option c is the correct answer.

Article 19(1)a of the Constitution of India **guarantees to all citizens the right to freedom of speech and expression**. This Fundamental right guarantees all citizens the **right to express their views and opinions** about any kind of issue and can be done through any medium, such as by speaking, writing, printing, picturization, or through a movie.

However, **this right is not absolute**. **Article 19(2)** explicitly **allows the State to impose reasonable restrictions on this right** through law, but only **on the following specified grounds**:

- the **sovereignty and integrity of India**, (Option I is correct)
- the security of the State,
- **friendly relations with foreign States**, (Option III is correct)
- **public order**, (Option II is correct)
- **decency or morality**, (Option V is correct)
- in relation to contempt of court,
- defamation, or
- incitement to an offence.

Option IV is incorrect: In contrast, **health** is **not included among the grounds under Article 19(2)**. It is a ground specifically mentioned under **Article 25**, which deals with the **freedom of religion**, where the State can regulate religious practices in the interest of **public order, morality and health**.

Hence, **Options I (Sovereignty and integrity of India), II (Public order), III (Friendly relations with foreign States), and V (Decency or morality) are valid grounds** for reasonable restrictions on the freedom of speech and expression under Article 19(2), whereas **IV (Health) is not a ground under Article 19(2)**.

Therefore, the correct answer is option (c) only four.

Source:) Mastering Indian Polity by Forum IAS, Chapter: 10 - Fundamental Rights, Page: 123.

<https://indiankanoon.org/doc/1218090/>

<https://indianexpress.com/article/explained/first-amendment-to-constitution-challenged-supreme-court-8252509/>

Subject:) Polity

Topic:) Fundamental Rights

Subtopic:)

Q.57)

Ans) b

Exp) Option b is the correct answer.

Fundamental Rights are **enshrined in Part III** of the Constitution of India (**Articles 12–35**) and form the cornerstone of the Indian democratic framework. The framers of the Constitution drew inspiration from the **United States Constitution's Bill of Rights** to ensure that the basic freedoms and liberties of individuals are protected.

Part III is often described as the **"Magna Carta of India"** because it contains a comprehensive and extensive list of **justiciable rights**, which can be enforced in courts to **protect citizens against the arbitrary action of the State**. These rights aim to secure liberty, equality, and dignity, while also maintaining a balance with the sovereignty and integrity of the nation.

Statement I is incorrect: While the **primary purpose of FRs is to protect individuals from the arbitrary power of the State, several rights are also available against the actions of private individuals.**

- **Article 15(2):** Prohibits discrimination in access to shops, hotels, public places, etc., even by private individuals.
- **Article 17:** Abolition of Untouchability (enforceable against anyone).
- **Article 23:** Prohibition of traffic in human beings and forced labor.
- **Article 24:** Prohibition of employment of children in factories.

If these rights are violated by a private person, the aggrieved individual has a constitutional remedy.

Statement II is incorrect: All Fundamental Rights are not self-executory. While **many rights are directly enforceable** (e.g., Articles 14 and 21), **some require enabling legislation for their effective implementation** and prescribe punishments for their violation.

- **Article 35** specifically empowers Parliament (and not State Legislatures) to make laws to give effect to fundamental rights like Article 17 (Protection of Civil Rights Act) and Article 23.

Statement III is correct: Fundamental Rights are not sacrosanct or permanent. Parliament has the power to amend, curtail, or even repeal them under Article 368 through a Constitutional Amendment. However, **this power is not unlimited, any such amendment is subject to the "Basic Structure" doctrine.**

In the landmark **Kesavananda Bharati case (1973)**, the Supreme Court propounded the Basic Structure doctrine, which holds that:

- **Parliament cannot alter the essential features of the Constitution.**
 - Fundamental Rights, especially their core essence (like rule of law, equality), form part of this Basic Structure.
- Therefore, while FRs are amendable, their basic framework and spirit cannot be destroyed.

Knowledge base: Fundamental Rights possess both negative and positive character, reflecting a dual nature:

- **Negative Rights** – These **restrain the State from acting in a certain manner**, thereby protecting individual liberty (e.g., **Article 14** – prohibits arbitrary State action).
- **Positive Rights** – These **confer certain privileges or entitlements on the individuals**, requiring the State to take active steps for the benefit of individuals (e.g., **Article 21A** – obligates the State to provide free and compulsory education to children).

Source:) Mastering Indian Polity by Forum IAS, Chapter: 10 - Fundamental Rights, Page: 103.

<https://egyankosh.ac.in/bitstream/123456789/57872/1/Unit4.pdf>

https://digital.nios.ac.in/content/338hi/338_Introduction_To_Law_Eng_L19.pdf

Subject:) Polity

Topic:) Fundamental Rights

Subtopic:)

Q.58)

Ans) c

Exp) Option c is the correct answer.

The power to issue constitutional writs is an important mechanism for the protection of Fundamental Rights in India. The **Constitution empowers the Supreme Court under Article 32 and the High Courts under Article 226 to issue writs for the enforcement of Fundamental Rights and for other legal purposes.**

The Indian Constitution has adopted the concept of writs from the English common law system. There are **five types of writs** that can be issued by the courts: **Habeas Corpus, Mandamus, Prohibition, Certiorari, and Quo Warranto.**

Pair I is correct: Habeas Corpus

Literal Meaning: "To have the body."

- **Purpose:** This is the most fundamental writ for protecting individual liberty. It is a **court order directed to a person or authority who has detained another person**, commanding them to produce the detainee before the court.
- **Mechanism:** The court examines the **"cause and legality" of the detention**. If the detention is found to be illegal, the court orders the immediate release of the person.

- **Scope:** It can be issued against both public authorities and private individuals. However, it is not issued if the detention is; by a competent court, lawful, for contempt of court, or if the person is outside the court's jurisdiction.

Pair II is correct: Mandamus

Literal Meaning: "We command."

- **Purpose:** It is a command issued by a higher court to a public official, public body, corporation, inferior court, or government, asking them to perform a public duty that they have failed or refused to perform.
- **Mechanism:** It is used to ensure that a public duty is carried out by the person or body legally bound to do it.
- **Scope:** It is only issued for public duties. It cannot be issued against a private individual, the President of India, or State Governors in their official capacity.

Pair III is incorrect: Prohibition

Literal Meaning: "To forbid."

- **Purpose:** It is issued by a higher court (Supreme Court or High Court) to a lower court or quasi-judicial body to prevent it from exceeding its jurisdiction.
- **Why the pair was incorrect:** In the description, it was stated as "to quash an order already passed." This is factually incorrect for Prohibition. Prohibition is preventive, it is issued while the case is still pending to stop the lower court from proceeding. The act of "quashing" an order after it has been finalized is the function of the writ of Certiorari.
- **Scope:** It is available only against judicial and quasi-judicial authorities. It cannot be issued against administrative authorities, legislative bodies, or private individuals.

Pair IV is correct: Quo-Warranto

Literal Meaning: "By what authority (or warrant)."

- **Purpose:** This writ is issued to inquire into the legality of a person's claim to a public office. It is used to prevent the illegal "usurpation" (wrongful seizure) of a public office by an individual.
- **Mechanism:** The court asks the person by what authority they are holding the office. If the claim is not well-founded, the person is ousted from that office.
- **Scope:** It can be issued only in cases of a substantive public office of a permanent character created by a statute or by the Constitution. Unlike other writs, it can be sought by any interested person, even if they are not the directly aggrieved party.

Source:) Mastering Indian Polity by Forum IAS, Chapter: 10 - Fundamental Rights, Page: 154.

<https://www.thehindu.com/news/national/can-the-ed-file-writ-petitions-before-courts-explained/article70560835.ece>

<https://www.egyankosh.ac.in/bitstream/123456789/9943/1/Unit-14.pdf> Page: 14-19.

Subject:) Polity

Topic:) Constitution of India

Subtopic:)

Q.59)

Ans) a

Exp) Option a is the correct answer.

Statement I is correct: Personality rights (also known as "Right of Publicity") refer to the right of an individual to control and protect the commercial use of their identity or persona.

- These rights typically cover attributes such as name, image, likeness, voice, signature, or other distinctive personal characteristics.
- The objective is to prevent the unauthorized commercial exploitation of a person's identity, for example using a celebrity's photograph, voice imitation, or likeness in advertisements without consent.

Statement II is incorrect: The Digital Personal Data Protection Act, 2023 primarily regulates the collection, processing, and protection of digital personal data and provides rights to individuals (Data Principals) regarding their personal data. However, the Act does not exclusively define or create a dedicated statutory framework for personality rights.

- In India, **personality rights are not codified in a single statute** and have largely evolved through judicial interpretation, often drawing from principles of privacy, intellectual property, and unfair commercial use.
- They are protected through a combination of **Common Law** (the tort of 'Passing Off'), the **Trade Marks Act, 1999**, and the **Copyright Act, 1957**.

The **legal foundation of personality rights in India is rooted in the Right to Privacy**.

- In the landmark Justice **K.S. Puttaswamy v. Union of India** (2017) case, the Supreme Court held that the **Right to Privacy is a Fundamental Right under Article 21** (Right to Life and Personal Liberty).
- Since a person's identity, image, and personal attributes are closely linked with dignity and privacy, courts have treated the unauthorized commercial use of such attributes as a **violation of an individual's privacy and personal autonomy**.

Source:) <https://www.thehindu.com/news/national/what-are-personality-rights-and-how-are-courts-shielding-indian-celebrities-explained/article70083371.ece>

<https://www.thehindu.com/opinion/op-ed/decoding-personality-rights-in-the-age-of-ai/article70322498.ece>

<https://www.thehindu.com/news/national/what-is-the-salman-khan-personality-rights-case-explained/article70567338.ece>

Subject:) Polity

Topic:) Laws/Acts

Subtopic:)

Q.60)

Ans) c

Exp) Option c is the correct answer.

The **Right against Exploitation**, enshrined in **Articles 23 and 24 of Part III** (Fundamental Rights) of the Constitution, **aims to eliminate all forms of human exploitation and ensure dignity of labour**. It prohibits practices such as **human trafficking, forced labour (begar), bonded labour, and child labour in hazardous occupations**.

This right is **available to both citizens and non-citizens** and reflects the constitutional commitment to social justice and protection of vulnerable sections, particularly women, children, and economically weaker groups.

Option I is correct: Under **Article 23**, the **Constitution explicitly prohibits traffic in human beings, begar, and other similar forms of forced labour**.

- **Begar** refers to **forced labour without payment**, historically practiced in many parts of India where landlords or authorities compelled people to work without wages.
- The expression "**other similar forms of forced labour**" has been interpreted broadly by the Supreme Court to include situations where a person is forced to work under coercion, threat, or economic compulsion.
- Even if some remuneration is paid, the labour may still be considered forced if the worker has no real freedom to refuse the work.

Thus, **this protection clearly falls under the Right against Exploitation**.

Option II is correct: According to **Article 24**, the **Constitution prohibits the employment of children below the age of fourteen years in factories, mines, and other hazardous occupations**.

Key points:

- The **constitutional prohibition does not extend to all forms of employment**.
- **Children may still legally work in non-hazardous activities**, subject to regulatory laws.
- Later legislation such as the **Child Labour (Prohibition and Regulation) Act, 1986** (as amended) expanded restrictions, but the constitutional provision itself is limited to factories, mines, and other hazardous occupations.

Option III is incorrect: The **abolition of untouchability is guaranteed under Article 17**, which forms part of the **Right to Equality** (Articles 14–18).

Key features:

- Article 17 abolishes untouchability and forbids its practice in any form.

- Enforcement of any disability arising from untouchability is punishable by law.
- The Protection of **Civil Rights Act, 1955** was enacted to implement this provision.

Although this is an important Fundamental Right, **it does not fall under the Right against Exploitation (Articles 23–24).**

Option IV is correct: Article 23 prohibits “traffic in human beings”, which includes the buying, selling, or movement of persons for exploitative purposes.

This concept covers practices such as:

- **Human trafficking**
- Forced prostitution
- Commercial sexual exploitation
- Sale and purchase of persons for labour or other exploitation

In India, laws such as the **Immoral Traffic (Prevention) Act, 1956** were enacted to combat such activities. The **trafficking of women and children for immoral or commercial purposes** is therefore a **direct violation of Article 23** and **falls within the scope of the Right against Exploitation.**

Knowledge Base: **Right against Exploitation**

Article 23: Prohibition of traffic in human beings and forced labour

(1) Traffic in human beings and *begar* and other similar forms of forced labour are prohibited and any contravention of this provision shall be an offence punishable in accordance with law.

(2) Nothing in this article shall prevent the State from imposing compulsory service for public purposes, and in imposing such service the State shall not make any discrimination on grounds only of religion, race, caste or class or any of them.

Article 24: Prohibition of employment of children in factories, etc.

No child below the age of fourteen years shall be employed to work in any factory or mine or engaged in any other hazardous employment.

Source:) Mastering Indian Polity by Forum IAS, Chapter: 10 – Fundamental Rights, Page: 140.

Constitution of India.

Subject:) Polity

Topic:) Fundamental Rights

Subtopic:)

Q.61)

Ans) c

Exp) Option c is the correct answer.

The **Deputy Speaker of the Lok Sabha** is the **second-highest presiding officer of the Lower House of Parliament (Lok Sabha) after the Speaker.**

- The office is provided under **Article 93** of the Constitution of India, which requires the Lok Sabha to elect both a Speaker and a Deputy Speaker from among its members.
- The Deputy Speaker is **elected by the Lok Sabha from among its members** and performs the duties of the Speaker when the Speaker is absent or when the office of the Speaker is vacant.
- **While presiding** over the House, the **Deputy Speaker exercises the same powers and authority as the Speaker**, including maintaining order and conducting proceedings.
- The Deputy Speaker also continues in office until the dissolution of the Lok Sabha unless he or she resigns or is removed by a resolution of the House.

Statement I is incorrect: The **President of India fixes the date for the election of the Speaker of the Lok Sabha** when a new House is constituted.

However, the **date for the election of the Deputy Speaker is fixed by the Speaker** of the Lok Sabha under the **Rules of Procedure and Conduct of Business in Lok Sabha**. Therefore, the President has no role in fixing the date for the election of the Deputy Speaker.

Under **Article 93** of the Constitution, the **Lok Sabha shall choose two members** to be respectively the **Speaker and the Deputy Speaker “as soon as may be.”**

The Constitution deliberately uses this flexible phrase and **does not specify any fixed time frame** for electing the Deputy Speaker. This has allowed situations where the election of the Deputy Speaker has been delayed.

Statement II is correct: According to Article 94 of the Constitution, the **Deputy Speaker may be removed from office by a resolution passed by a majority of all the then members of the Lok Sabha**, which is known as an **effective majority**. Additionally, **at least 14 days' notice must be given** before such a resolution can be moved in the House.

Statement III is correct: Traditionally, the Lok Sabha elected a Deputy Speaker soon after the formation of the House. However, **during the 17th Lok Sabha (2019–2024)**, the **post remained vacant throughout the entire term**, making it the **first time since Independence** that a full Lok Sabha term passed without electing a Deputy Speaker, marking the **longest vacancy of the office**. This raised constitutional and procedural discussions regarding the interpretation of the phrase “as soon as may be” in Article 93.

Source:) Mastering Indian Polity by ForumIAS, Chapter: 16 - LEGISLATIVE BRANCH, Page: 355.

<https://www.thehindu.com/opinion/lead/the-post-of-deputy-speaker-is-not-symbolic-or-optional/article69502193.ece>

<https://www.thehindu.com/news/national/the-importance-of-the-deputy-speaker/article69630621.ece>

<https://frontline.thehindu.com/politics/lok-sabha-deputy-speaker-vacancy-parliament-constitution-violation/article69913075.ece>

Subject:) Polity

Topic:) Parliament

Subtopic:)

Q.62)

Ans) d

Exp) Option d is the correct answer.

Statement I is correct: According to **Article 88** of the Constitution of India, Every Minister **and the Attorney-General of India shall have the right to speak in, and otherwise to take part in the proceedings of, either House, any joint sitting of the Houses**, and any committee of Parliament of which he may be named a member, but shall not by virtue of this article be entitled to vote.

However, the Attorney General is not a Member of Parliament; therefore, he does not possess the right to vote in the House. This provision enables the Attorney General to assist the Parliament with legal clarifications and constitutional advice whenever necessary.

Statement II is correct: The **Union Executive** consists of the **President**, the **Vice-President**, the **Prime Minister**, the **Council of Ministers**, and the **Attorney General for India**.

- The office of the Attorney General is provided under **Article 76** of the Constitution of India.
- The Attorney General is **appointed by the President of India** and serves as the **chief legal adviser to the Government of India**.
- Since he/she performs functions on behalf of the Government and assists the executive in legal matters, **he/she is considered a part of the Union Executive**.

Statement III is correct: The **Attorney General** is recognized as the **highest law officer of the Government of India**. His **primary duties include:**

- Giving legal advice to the Government of India on matters referred by the President.
- Performing legal duties assigned by the President under Article 76.
- Discharge the functions conferred on him/her by the Constitution or other law.

Source:) Mastering Indian Polity by ForumIAS, Chapter: 15 - EXECUTIVE BRANCH, Page: 300.

<https://www.egyankosh.ac.in/bitstream/123456789/52033/1/Block-3.pdf>

<https://indianexpress.com/article/explained/everyday-explainers/what-functions-of-attorney-general-for-india-8173828/>

Subject:) Polity

Topic:) Union Executive

Subtopic:)

Q.63)

Ans) a

Exp) Option a is the correct answer.

Statement I is correct: The **Advisory Jurisdiction** of the **Supreme Court** of India **empowers** the **President of India** to **seek the opinion of the Supreme Court of India on specific matters**. Article 143 provides two categories of references:

1. **Article 143(1):** On any question of law or fact of public importance which has arisen or is likely to arise.
2. **Article 143(2):** On any dispute arising out of any pre-Constitution treaty, agreement, covenant, engagement, sanad, or other similar instrument, which is **excluded from the Court's original jurisdiction. (matters mentioned under the proviso of Article 131 of the Constitution)**

In both situations, the **opinion of the Court is advisory** in nature and not a judicial pronouncement, and therefore not binding on the President.

Statement II is incorrect: The **opinion given by the Supreme Court under Article 143** is **purely advisory and not binding** on the President.

The President may follow or may not follow the Court's opinion. The purpose of this provision is mainly to provide the government with an authoritative legal opinion on complex constitutional or legal questions.

Statement III is incorrect: The **Supreme Court is not bound to render an opinion in every case**. The duty depends on the category of the reference:

- **Category 1 (Public Importance):** The **Supreme Court "may" tender its opinion**. It can refuse to provide an advisory opinion if it considers the question inappropriate or political in nature (e.g., the Court declined to answer the reference in the Ram Janmabhoomi case, 1993).
- **Category 2 (Pre-Constitution disputes):** In this specific case, the **Supreme Court "shall" (is bound to) tender its opinion to the President**.

Source:) Mastering Indian Polity by ForumIAS, Chapter: 17 - JUDICIAL BRANCH, Page: 460.

<https://www.sconline.com/blog/post/2025/05/18/article-143-supreme-court-advisory-jurisdiction/>

<https://indianexpress.com/article/legal-news/10-reasons-why-supreme-court-decided-to-answer-presidential-reference-10376405/>

<https://indianexpress.com/article/upsc-current-affairs/upsc-essentials/knowledge-nugget-presidential-reference-article-143-polity-upsc-10204401/>

Subject:) Polity

Topic:) Judiciary

Subtopic:)

Q.64)

Ans) b

Exp) Option b is the correct answer.

Option b is correct: **Anticipatory bail** is a **pre-arrest legal remedy** that **enables a person to seek bail in anticipation of arrest** when they have a **reasonable apprehension** that they may be **arrested for a non-bailable offence**.

- Under the **Code of Criminal Procedure, 1973**, this provision is contained in **Section 438**. The corresponding provision under the **Bharatiya Nagarik Suraksha Sanhita, 2023** is **Section 482**.
- A person who fears arrest may apply to the High Court or the Court of Session seeking a direction that in the event of such arrest, they shall be released on bail.
- If the court grants anticipatory bail, the **person is released on bail immediately upon arrest without being subjected to prolonged custody**.
- While granting anticipatory bail, the **court may impose certain conditions**, such as requiring the accused to cooperate with the investigation, not tamper with evidence, or not leave the country without prior permission.

- Thus, anticipatory bail functions as an important **safeguard of personal liberty**, protecting individuals from arbitrary or motivated arrests, while still ensuring that they remain available for investigation and trial.

Option a is incorrect: Regular Bail

Regular bail is **sought after a person has already been arrested** and is in police or judicial custody. The accused applies before a competent court seeking release during the pendency of investigation or trial. Since regular bail is available **only after arrest**, it does not apply in this case

Option c is incorrect: Interim Bail

Interim bail is a **temporary or short-term bail granted for a limited period while the court is considering an application for regular bail or anticipatory bail**. It operates as a temporary protection against arrest or continued custody.

Unlike other forms of bail, interim bail is not expressly defined in statute; rather, it has evolved through judicial practice and the inherent powers of courts.

Option d is incorrect: Default Bail

Default bail (also known as **statutory bail**) arises when the **investigating agency fails to complete the investigation and file the charge sheet within the prescribed statutory time limit**. In such a situation, the accused becomes entitled to bail as a matter of right.

- Under the Code of Criminal Procedure, 1973, this right is provided under Section 167(2).
- Under the Bharatiya Nagarik Suraksha Sanhita, 2023, the corresponding provision is Section 187(3).

However, default bail is related to delay in investigation after arrest, not to apprehension of arrest.

Source: <https://cdnbbsr.s3waas.gov.in/s3ec030b6ace9e8971cf36f1782aa982a7/uploads/2025/09/2025090822.pdf>
<https://m.thewire.in/article/law/supreme-court-rules-anticipatory-bail-cant-be-time-bound-to-investigation-stage>
<https://www.thehindu.com/news/cities/Delhi/anticipatory-bail-should-be-granted-in-exceptional-cases-delhi-hc/article69779483.ece>

Subject: Polity

Topic: Judiciary

Subtopic:)

Q.65)

Ans) d

Exp) Option d is the correct answer.

Option d is correct: The given provision is enshrined in **Article 50** of the Constitution of India, which reads: “**The State shall take steps to separate the judiciary from the executive in the public services of the State.**”

Article 50 is placed under **Part IV** of the Constitution, which deals with the **Directive Principles of State Policy (Articles 36–51)**. These principles are **not enforceable by any court (non-justiciable)**, but they are fundamental in the governance of the country and it is the duty of the State to apply them while making laws (**Article 37**).

Substantive significance of Article 50:

- It aims to ensure the **independence of the judiciary**, which is a **basic feature of the Constitution**.
- It seeks to remove executive control over judicial functions, particularly at the **subordinate judiciary level (district and lower courts)**.
- This separation helps ensure **impartial administration of justice**, free from executive interference.

Option a is incorrect: Part V

Part V of the Constitution of India (**Articles 52–151**) deals with “**The Union**” and contains provisions relating to the structure and functioning of the Union Government.

- It includes provisions relating to the **President, Vice-President, Council of Ministers, Parliament**, and the **Supreme Court of India**.
- While it does contain provisions regarding the higher judiciary (Supreme Court), it does not deal with the principle of separation of judiciary from the executive in public services.

Option b is incorrect: Part IV-A

Part IV-A of the Constitution of India **contains the Fundamental Duties** of citizens, which are listed under **Article 51A** of the Constitution of India.

- These duties were added by the **42nd Constitutional Amendment Act, 1976** to emphasize the moral obligations of citizens towards the nation, such as respecting the Constitution, safeguarding public property, and promoting harmony.

Option c is incorrect: Part III

Part III of the Constitution of India **deals with the Fundamental Rights (Articles 12–35)**.

- These rights guarantee essential civil liberties and protections to individuals, including the Right to Equality, Right to Freedom, and Right to Constitutional Remedies.
- The provisions in this part are **justiciable**, meaning they can be **enforced by courts**.

Source:) Mastering Indian Polity by ForumIAS, Chapter: 11 - DIRECTIVE PRINCIPLES OF STATE POLICY, Page: 174.

<https://www.mea.gov.in/images/pdf1/part4.pdf> Page: 24.

Subject:) Polity

Topic:) Directive Principles of State Policy

Subtopic:)

Q.66)

Ans) c

Exp) Option c is the correct answer.

The concept of “**holding office during the pleasure of the President**” in India is **derived from the Doctrine of Pleasure**, which originates from British constitutional practice. In the Indian constitutional framework, this doctrine means that certain constitutional authorities can be removed by the President without a fixed removal procedure such as impeachment.

- Several constitutional functionaries in India that hold office during the pleasure of the President include the **Governor of a State, the Attorney General for India, and the members of the Council of Ministers (including the Prime Minister)**.

Option c is correct: Governor of a State

The Governor of a State holds office during the pleasure of the President of India.

- According to **Article 156(1)** of the Constitution of India, “**The Governor shall hold office during the pleasure of the President.**”
- Although **Article 156(3)** prescribes a **term of five years, this tenure is not absolute**. The Governor may be removed earlier by the President without any formal impeachment procedure.

Thus, the **Governor is a constitutional functionary who serves at the pleasure of the President**.

Option a is incorrect: Judge of the High Court

A Judge of a High Court does not hold office during the pleasure of the President.

- **As per Article 217** “**Every Judge of a High Court shall be appointed by the President by warrant under his hand and seal, and shall hold office, in the case of an additional or acting Judge, as provided in article 224, and in any other case, until he attains the age of sixty-two years**”
- Hence, a High Court judge **enjoys security of tenure and** cannot be removed at the discretion of the President.
- The **removal procedure is similar to that of judges of the Supreme Court**, ensuring the independence of the judiciary from executive interference.

Therefore, **High Court judges do not serve at the pleasure of the President**.

Option b is incorrect: Members of the Union Public Service Commission

Members of the Union Public Service Commission also do not hold office during the pleasure of the President.

- **Under Article 316, members of the UPSC are appointed by the President and hold office for six years or until the age of 65 years, whichever is earlier.**
- Their **removal is governed by Article 317**, which provides specific safeguards:

- In cases of misbehaviour, the President must refer the matter to the Supreme Court, which conducts an inquiry.
- The member can be removed only after the Supreme Court reports that the charges are proved.
- For certain other grounds (such as insolvency, engaging in paid employment outside duties, or infirmity of mind or body), the President may remove the member.

Therefore, **UPSC members do not serve at the pleasure of the President.**

Option d is incorrect: Comptroller and Auditor General of India (CAG)

The Comptroller and Auditor General of India **enjoys one of the strongest constitutional safeguards to ensure independence.**

- Under **Article 148**, the **CAG is appointed by the President by warrant under his hand and seal but cannot be removed at the President's pleasure.**
- The CAG is provided with the **security of tenure** and holds office for six years or until age 65 (whichever earlier).
- The Constitution provides that the CAG can be removed only in the same manner and on the same grounds as a judge of the Supreme Court.

Source:) Mastering Indian Polity by the ForumIAS, Chapter: 15 - EXECUTIVE BRANCH

Constitution of India

<https://www.rajbhavanodisha.gov.in/view/governorRole.php>

https://digital.nios.ac.in/content/338en/338_Introduction_To_Law_Eng_L21.pdf Page: 323.

Subject:) Polity

Topic:) President of India

Subtopic:)

Q.67)

Ans) d

Exp) Option d is the correct answer.

The **legislative procedure** in the Parliament of India is **governed primarily by Article 107** of the Constitution, which deals with the **introduction and passing of Bills** and also lays down the general rule regarding the lapse of Bills on the dissolution of the Lok Sabha.

- According to Article 107, a **Bill pending in Parliament lapses on the dissolution of the Lok Sabha, except a Bill which is pending in the Rajya Sabha** but has **not been passed by the Lok Sabha.**
- However, the Constitution also provides an important exception under **Article 108**, which deals with joint sittings of Parliament to resolve a deadlock between the two Houses.

Option d is correct: When a deadlock arises between the two Houses of Parliament, the **President may notify a joint sitting of the Lok Sabha and the Rajya Sabha under Article 108** to resolve the disagreement.

- According to **Article 108(5)**, once the President has notified the intention to summon a joint sitting, the **Bill does not lapse** even if the Lok Sabha is dissolved before the joint sitting actually takes place.
- The joint sitting can still be held after the reconstitution of the Lok Sabha.

Thus, a **Bill for which the President has already issued a notification for a joint sitting continues to remain alive despite the dissolution of the Lok Sabha.**

Option a is incorrect: A Bill originating in the Lok Sabha and pending in the Lok Sabha:

A **Bill pending in the Lok Sabha lapses on its dissolution, irrespective of whether it originated in the Lok Sabha or was transmitted from the Rajya Sabha.** This is because the Lok Sabha, being a temporary House, ceases to exist, and all pending legislative business before it comes to an end.

Option b is incorrect: A Bill transmitted from the Rajya Sabha and pending in the Lok Sabha:

Even if the Bill originated in the Rajya Sabha, once it is pending in the Lok Sabha at the time of dissolution, it lapses. The stage of consideration in the Lok Sabha is crucial; pending business in this House does not survive its dissolution.

Option c is incorrect: A Bill passed by the Lok Sabha and pending in the Rajya Sabha:

Such a Bill also lapses on dissolution of the Lok Sabha, as it has not completed the full legislative process. Since the House in which the bill originated i.e. the Lok Sabha is dissolved, the Bill cannot proceed further and must be reintroduced.

Source:) Mastering Indian Polity by the ForumIAS, Chapter: 16 - LEGISLATIVE BRANCH, Page” 385-386.

Constitution of India.

<https://prsindia.org/articles-by-prs-team/unfinished-business>

Subject:) Polity

Topic:) Parliament

Subtopic:)

Q.68)

Ans) a

Exp) Option a is the correct answer.

The **provisions relating to the District Planning Committee (DPC)** are contained in **Article 243ZD** of the Constitution of India, which was inserted by the **Seventy-Fourth Constitutional Amendment Act, 1992** to strengthen decentralized planning at the district level.

Statement I is correct: As per Article 243ZD of the Constitution of India, there shall be constituted in every State at the district level a District Planning Committee to consolidate the plans prepared by the Panchayats and the Municipalities in the district and to prepare a draft development plan for the district as a whole.

Statement II is incorrect: The **Constitution does not provide that all members of the Committee shall be elected by and from among the elected members of Panchayats and Municipalities.**

- According to **Article 243ZD** of the Constitution of India, **not less than four-fifths (4/5th) of the total number of members** of the District Planning Committee shall be elected by and from among the elected members of the Panchayats and Municipalities in the district, in proportion to the ratio of the rural and urban population.

Thus, the **remaining members may be nominated or included in another manner as determined by the State Legislature.**

Statement III is incorrect: The **Constitution does not mandate that the Chairperson of the District Planning Committee shall be nominated by the Governor.**

- Under **Article 243ZD** of the Constitution of India, the **State Legislature determines the manner** in which the **Chairperson of the District Planning Committee is to be chosen.**

Therefore, the mode of selection of the Chairperson varies from state to state, and the Constitution itself does not prescribe nomination by the Governor.

Source:) Mastering Indian Governance by ForumIAS, Chapter :6 - LOCAL GOVERNANCE, Page: 90.

Constitution of India.

http://constitutionofindia.etal.in/article_243zd/

Subject:) Polity

Topic:) Local Self Government

Subtopic:)

Q.69)

Ans) b

Exp) Option b is the correct answer.

Question Hour is generally the **first hour of a sitting of both Houses of Parliament**, during which Members of Parliament (MPs) ask questions to Ministers regarding the administration and functioning of the Government.

- It is a parliamentary convention governed by the Rules of Procedure of the Lok Sabha and Rajya Sabha.
- The **primary purpose of Question Hour** is to **hold the executive accountable** to the legislature and ensure transparency in governance.
- Questions can be classified into three types: Starred, Unstarred, and Short Notice Questions** (on urgent matters).

- The **Speaker or Chairperson** controls the conduct of **Question Hour**, including the admissibility and order of questions.

Statement I is correct: **Question Hour** is **not explicitly mentioned in the Constitution** of India. It is a parliamentary device evolved through practice and is **governed by the Rules of Procedure and Conduct of Business of Lok Sabha and Rajya Sabha**.

Statement II is incorrect: The statement **reverses the definitions** of **Starred and Unstarred questions**:

- **Starred Questions:** These are distinguished by an **asterisk**. They **require an oral answer** on the floor of the House, and because the answer is oral, **supplementary questions can be asked**.
- **Unstarred Questions:** These require a **written reply**, which is laid on the table of the House. Since the **answer is written, no supplementary questions can follow**.

Statement III is incorrect: A Question may also be addressed to a Private Member (Under Rule 40 of the Rules of Procedure and Conduct of Business in Lok Sabha), provided that the subject matter of the question relates to some Bill, Resolution or other matter connected with the business of the House for which that Member is responsible. The procedure in regard to such questions is same as that followed in the case of questions addressed to a Minister with such variations as the Speaker may consider necessary.

Source:) Mastering Indian Polity by the ForumIAS, Chapter: 16 - LEGISLATIVE BRANCH, Page: 386.

<https://sansad.in/rs/questions/general-information>

<https://timesofindia.indiatimes.com/blogs/reels-and-realms/democracy-in-action-unpacking-the-significance-of-parliaments-question-hour/>

<https://sansad.in/lr/questions/questions>

<https://www.thehindu.com/news/national/wheres-the-minister-asks-speaker-says-question-hour-more-important-than-meetings/article70758445.ece>

Subject:) Polity

Topic:) Parliament

Subtopic:)

Q.70)

Ans) c

Exp) Option c is the correct answer.

Option c is correct: **Constitutional Morality** is a **dynamic and transformative judicial doctrine in India** that **goes beyond the literal text of the Constitution** to embrace its **underlying spirit and foundational values**.

1. Meaning and Scope

- **Spirit over Letter:** It refers to adhering to the Constitution's norms and ethical spirit rather than just its literal words.
- **Core Values:** It is anchored in the core **principles of the Constitution: Justice, Liberty, Equality, Fraternity, Secularism** as well as the **Rule of Law**.
- **Counter to Majoritarianism:** It acts as a **safeguard for individual rights and minority interests** against the "fleeting fancies" or prejudices of the majority (Social Morality).

2. Origin and Evolution

- **Dr. B.R. Ambedkar:** During the Constituent Assembly debates, Ambedkar emphasized that **constitutional morality is not a "natural sentiment" in India** but a value that must be cultivated. He argued that a democratic constitution is **only a "top-dressing" on essentially undemocratic soil**, requiring deliberate respect for constitutional forms to prevent the perversion of administration.
- **George Grote:** Ambedkar borrowed the term from this British historian, who defined it as a **"paramount reverence" for constitutional forms**.

3. Landmark Judicial Applications

The Supreme Court has increasingly used this doctrine to strike down regressive laws or practices:

- **Navtej Singh Johar Case (2018):** **Decriminalized consensual same-sex acts**, ruling that Constitutional Morality must trump Social Morality.

- **Sabarimala Case (2018): Struck down the ban on women entering the temple**, holding that gender equality is a core constitutional value that overrides discriminatory religious customs.
- **Joseph Shine Case (2018): Decriminalized adultery**, viewing it as a move toward protecting individual autonomy and gender dignity.

Option a is incorrect: This reflects literalism or textualism, where emphasis is placed only on the written provisions of the Constitution. Constitutional morality, however, requires going beyond the text to uphold its underlying principles and transformative vision, especially in evolving social contexts.

Option b is incorrect: This describes constitutional supremacy, meaning that the Constitution is the highest law and all authorities derive their power from it. While related, it is a structural principle, whereas constitutional morality is a normative and value-based concept guiding how that power is exercised.

Option d is incorrect: This refers to a narrow or restrictive approach to judicial review, limiting courts to the explicit text. In contrast, constitutional morality supports a broader interpretative approach, enabling courts to uphold constitutional values like dignity, equality, and freedom.

Source: https://nja.gov.in/Concluded_Programmes/2023-24/P-1373_PPTs/Session%201-%20Constitutional%20Morality.pdf

<https://www.thehindu.com/specials/text-and-context/constitutional-morality-the-origins-and-nuances-of-the-concept/article69229696.ece>

<https://www.thehindu.com/opinion/lead/the-contours-of-constitutional-morality/article70205715.ece>

<https://www.thehindu.com/news/national/constitutional-morality-finally-comes-down-to-justice-without-fear-or-favour-kapil-sibal-at-justice-unplugged-2026/article70689096.ece>

<https://www.iilsindia.com/blogs/constitutional-morality-the-pillar-of-democracy-and-justice-in-india/>

Subject: Polity

Topic: Constitution of India

Subtopic:

Q.71)

Ans) c

Exp) Option c is the correct answer.

Tropical Rainforests are **dense, evergreen forests** that occur in regions with **high temperatures** and **heavy rainfall throughout the year**, mainly **near the equatorial belt** between about 10°–15° latitude **north and south of the Equator**. These forests develop in areas where the **annual rainfall generally exceeds 2000 mm** and the **average temperature remains around 25–28°C** with very little seasonal variation.

Due to the consistently warm and humid climate, tropical rainforests support **extremely high biological productivity and biodiversity**, making them the most species-rich terrestrial ecosystems on Earth. They are characterized by **multi-layered vegetation** (emergent, canopy, understory, and forest floor), **dense canopy cover**, **abundant epiphytes** and **lianas**, and **evergreen broad-leaved trees**.

Option I is correct: Tropical rainforests receive **very heavy** (often >2000 mm annually) and **continuous rainfall**. Continuous rainfall **causes intense leaching**, which **washes away soluble minerals** such as calcium, potassium, and magnesium **from the soil**.

As a result, **rainforest soils** (commonly **lateritic or oxisols**) are **acidic and nutrient-poor**, despite the presence of extremely dense vegetation. The apparent paradox is explained by rapid nutrient cycling rather than soil fertility.

Option II is correct: The warm, humid, and moist conditions prevailing throughout the year create an **ideal environment** for the **growth of epiphytes** (plants growing on other plants for support, e.g., orchids) and **bryophytes** (non-vascular plants like mosses). These plants thrive on tree trunks and branches due to constant moisture availability.

Option III is incorrect: The rate of decomposition in tropical rainforests is very rapid, not slow. High temperature and moisture promote intense microbial activity, which quickly breaks down organic matter. However, nutrients released are immediately absorbed by plant roots, preventing their accumulation in the soil.

Option IV is correct: Due to **rapid decomposition** and immediate nutrient uptake, **most nutrients are stored in the living biomass** (trees, plants, vegetation) **rather than in the soil**. This makes the ecosystem highly efficient but also fragile, as removal of vegetation quickly leads to soil degradation.

Source:) <https://ncert.nic.in/textbook/pdf/kegy105.pdf>

<https://egyankosh.ac.in/bitstream/123456789/85708/1/Unit-9.pdf> Page: 175.

Subject:) Geography

Topic:) Vegetation [World]

Subtopic:)

Q.72)

Ans) d

Exp) Option d is the correct answer.

Aerosols are small particles suspended in the atmosphere. They are often not or barely visible to the human eye, yet their impact on climate, weather, health, and ecology are significant.

Statement 1 is incorrect: Aerosols can be released in the atmosphere by both natural as well as the anthropogenic sources. Atmospheric aerosols consist of solid/aqueous particles suspended in the atmosphere and are typically of sizes in the range 0.001–10 μm . Aerosols are generated from a wide range of natural and anthropogenic sources. Natural aerosols include sulfate aerosols that are formed from emissions of sulfur dioxide from volcanic eruptions or from dimethyl sulfide emissions from phytoplankton in the ocean, sea salt particles emitted at the ocean surface, or mineral dust aerosol that is emitted by the effects of wind erosion on arid land. Anthropogenic aerosols, i.e., those that are generated from human activities, include sulfate, nitrate, black carbon, and organic carbon aerosols from fossil-fuel burning, deforestation fires, and burning of agricultural waste.

Statement II is correct: Aerosols also can act as sites for chemical reactions to take place. The most significant of these reactions are those that lead to the destruction of stratospheric ozone. During winter in the polar regions, aerosols grow to form polar stratospheric clouds. The large surface areas of these cloud particles provide sites for chemical reactions to take place. These reactions lead to the formation of large amounts of reactive chlorine and, ultimately, **to the destruction of ozone in the stratosphere**.

Statement III is correct: A **carbon dioxide** molecule has a **lifetime** of about **100 years** in the atmosphere, while an **aerosol particle** has an **average life expectancy** of only about **10 days**. Thus, this statement is **correct** as an **aerosol lasts a much shorter time** in the atmosphere than carbon dioxide. They are generally **classified as Short-Lived Climate Pollutants (SLCP)**.

Statement IV is incorrect: If aerosol particles are not present in the atmosphere of earth, the Earth would be even warmer than it already is. Aerosol air pollution has made the planet with about 0.7° F (0.4 °C) cooler than it otherwise would be, according to the 2021 report by the Intergovernmental Panel on Climate Change (IPCC). Hence aerosol particles have a net warming effect on earth's atmosphere.

Source: <https://www.ncert.nic.in/ncerts/l/kech207.pdf>

<https://earth.gsfc.nasa.gov/climate/data/deep-blue/aerosols>

[https://earth.gsfc.nasa.gov/climate/data/deep-](https://earth.gsfc.nasa.gov/climate/data/deep-blue/aerosols#:~:text=Sea%20spray%2C%20mineral%20dust%2C%20smoke,from%20volcanoes%20or%20industrial%20emissions.)

[blue/aerosols#:~:text=Sea%20spray%2C%20mineral%20dust%2C%20smoke,from%20volcanoes%20or%20industrial%20emissions.](https://earth.gsfc.nasa.gov/climate/data/deep-blue/aerosols#:~:text=Sea%20spray%2C%20mineral%20dust%2C%20smoke,from%20volcanoes%20or%20industrial%20emissions.)

<https://climate.nasa.gov/news/215/just-5-questions-aerosols/>

Subject:) Geography

Topic:) Atmosphere

Subtopic:)

Q.73)

Ans) c

Exp) Option c is the correct answer.

Statement I is correct: India holds the first position in the world in the production of several minor oilseeds, particularly **castor seed, safflower, sesame and niger seed**. These crops are well suited to India's semi-arid and tropical climatic conditions. Also, India ranks **second in groundnut, third in rapeseed-mustard, fourth in linseed, and fifth in soybean**.



INDIA'S GLOBAL RANKING
IN PRODUCTION OF EDIBLE OIL AND OIL SEEDS

India ranks **4th globally** in Overall Production of Edible Vegetable Oil

Global Rank	Edible Oil/Oil Seed
First	Rice Bran Oil, Castor Seed, Safflower, Sesame and Niger
Second	Cottonseed Oil, Groundnut Seeds and Oil
Third	Coconut (shells and oil) and Sesame Seed Oil
Fourth	Rapeseed / Canola Oil
Fifth	Soybean, Soyabean Oil and Linseeds
Sixth	Linseed Oil

Source: Niti Aayog (As of August 2024)

Statement II is incorrect: Oilseeds do not occupy the highest share in terms of acreage or production value among agricultural crops in India. The food grains group (rice, wheat, coarse cereals and pulses) occupies the largest share of cropped area and production value in Indian agriculture.

- Oilseeds hold the **second-highest acreage and production value in Indian agriculture after food grains**.
- Nine major oilseeds, groundnut, soybean, rapeseed-mustard, sunflower, sesame, safflower, niger, castor, and linseed, occupy 14.3% of the gross cropped area.

Statement III is incorrect: Madhya Pradesh is a major oilseed-producing state, mainly due to its large production of soybean. However, it is **not the largest producer of total oilseeds in the country**.

- **Rajasthan is the largest producer of oilseeds in India**, largely because of its dominant production of rapeseed-mustard, which is one of the most important rabi oilseed crops.
- **Rajasthan, Madhya Pradesh, Gujarat, and Maharashtra** together contribute over 77.68% of India's total oilseed output.

Among all oilseed crops cultivated in India, **soybean contributes the largest share to the total oilseed production**. Soybean accounts for roughly one-third to nearly forty percent of India's total oilseed output. Major soybean-producing states include Madhya Pradesh, Maharashtra and Rajasthan.

Statement IV is correct: Oilseed cultivation in India is **predominantly rainfed**. Nearly three-fourths (around 75%) of the **total area under oilseed cultivation depends on rainfall rather than irrigation**. This makes oilseed production highly vulnerable to monsoon variability, which is one of the major reasons behind fluctuations in domestic edible oil production.

Source: <https://www.pib.gov.in/PressReleasePage.aspx?PRID=2200287®=3&lang=1>

<https://indiantradeportal.in/vs.jsp?lang=0&id=0.31.24100.29403>

<https://www.indiabudget.gov.in/economicsurvey/doc/stat/tab1.18.pdf>

<https://www.thehindubusinessline.com/economy/agri-business/lower-maize-prices-may-see-indian-farmers-switch-back-to-oilseeds-pulses-this-kharif/article70766347.ece>

<https://www.thehindu.com/news/national/ethanol-blending-creates-tension-between-food-energy-security-demands-survey/article70566569.ece>

Subject:) Geography

Topic:) Crops [India]

Subtopic:)

Q.74)

Ans) a

Exp) Option a is the correct answer.

The **coastline of India** was **recently re-measured** and revised by the Government of India using **modern geospatial techniques** and **high-resolution mapping**. This exercise was carried out by the **National Hydrographic Office** in collaboration with the **Survey of India**.

- **Earlier officially accepted coastline length of India: 7,516.6 km** (based largely on older measurements from the 1970s).
- **Revised coastline length: 11,098.81 km.**

The increase in length occurred because advanced GIS-based mapping captured several small coastal features such as bays, creeks, inlets, estuaries, and irregular coastal indentations that were not measured earlier.

- While **Gujarat remains the state with the longest coastline overall**.
- In the revised list, **Tamil Nadu has overtaken Andhra Pradesh** to become the state with the second-longest coastline in India.
- **Andaman & Nicobar Islands** have the **longest coastline overall in India**

Revised Coastline Length of the Given States

- **Tamil Nadu: 1068.69 km**
- **Andhra Pradesh: 1053.07 km**
- **Maharashtra: 877.97 km**
- **West Bengal: 721.02 km**

Thus, **Tamil Nadu has the longest coastline among the options provided in the question.**

Knowledge Base: **Revised Coastline Length of Indian States and Union Territories (Survey of India, 2023–24):**

State / UT	Old Coastline (km)	Revised Coastline (km)
Gujarat	1,214.70	2,340.62
Tamil Nadu	906.90	1,068.69
Andhra Pradesh	973.70	1,053.07
Maharashtra	652.60	877.97
West Bengal	157.50	721.02
Kerala	569.70	600.15
Odisha	476.40	574.71
Karnataka	280.00	343.30
Goa	160.50	193.95
Daman & Diu	–	54.38
Andaman & Nicobar Islands	1,962.00	3,083.50
Lakshadweep	132.00	144.80
Puducherry	30.60	42.65
Total (India)	7,516.60	11,098.81

Source:)

https://shipmin.gov.in/sites/default/files/Length%20of%20Indias%20Coastline%20Circular_0.pdf

<https://www.pib.gov.in/PressReleasePage.aspx?PRID=2198800®=3&lang=1>

Subject:) Geography

Topic:) Physiography of India

Subtopic:)

Q.75)

Ans) c

Exp) Option c is the correct answer.

Gene drive technology is an advanced **genetic engineering technique** that **alters the normal rules of inheritance in sexually reproducing organisms**.

- Under **normal Mendelian inheritance**, a **gene** has about a **50% probability of being passed from a parent to its offspring**.
- However, **gene drives are designed to bias this inheritance pattern**, ensuring that a **particular gene is transmitted to a much larger proportion of offspring**, sometimes approaching 100% inheritance.
- This enables the engineered gene to **spread through an entire population over successive generations**, even if the gene does not provide a survival advantage to the organism.
- Modern gene drives are often developed using the **genome-editing system CRISPR-Cas9**, which allows scientists to precisely cut and copy genetic sequences.

Statement I is correct: **Gene drives** work by ensuring that a **modified gene copies itself onto the homologous chromosome in reproductive cells**. As a result, **instead of only one copy of the gene being inherited, both chromosomes carry the engineered gene**, making it highly likely that the gene will be transmitted to the next generation.

Because of this mechanism, the **engineered gene spreads rapidly through a population over time**. Unlike conventional genetic modifications that follow normal inheritance patterns, gene drives are specifically designed to increase the inheritance probability of a targeted gene.

Statement II is correct: One of the **most important proposed applications of gene drive technology** is the **control of disease-transmitting insects, particularly mosquitoes that spread diseases such as Malaria, dengue, and Zika**.

Scientists are exploring **two major strategies using gene drives**:

- 1. Population suppression:** Gene drives can be designed to reduce the reproductive capacity of mosquitoes or produce mostly sterile or male offspring, gradually reducing the population.
- 2. Population modification:** Gene drives can introduce genes that make mosquitoes incapable of transmitting disease-causing pathogens, thereby interrupting the transmission cycle.

If successfully implemented, such approaches could significantly reduce the incidence of malaria and other mosquito-borne diseases.

Gene drive technology is primarily designed for sexually reproducing organisms, where genes are inherited from two parents through reproduction. The technology works by manipulating the inheritance process that occurs during sexual reproduction so that the engineered gene is copied and transmitted to most offspring.

In contrast, **asexually reproducing organisms reproduce without mating**, and the offspring are genetically identical to the parent. Since there is no mixing of parental genes or competition between gene copies during inheritance, the **mechanism exploited by gene drives does not operate effectively in such organisms**.

Source:)

https://epaper.thehindu.com/ccidist-ws/th/th_delhi/issues/51357/OPS/GTOBO1DPC.1+G4RBO1JRK.1.html

<https://www.science.org/content/article/controversial-gene-drive-strategy-could-make-mosquitoes-hostile-malaria-parasites>

<https://www.thehindu.com/sci-tech/science/mosquito-control-gene-drive-technology-benefits-risks/article67287784.ece>

Subject:) Science and Technology

Topic:) Biotechnology

Subtopic:)

Q.76)

Ans) a

Exp) Option a is the correct answer.

RNA interference (RNAi) is a **natural biological process** in which **small RNA molecules regulate gene expression by silencing specific genes at the RNA level**.

RNAi serves as an important **gene-regulation system in eukaryotic cells** and also acts as a **defense mechanism against viruses and transposable genetic elements**.

RNAi primarily operates through small regulatory RNAs such as **Small interfering RNA (siRNA)** and **microRNA (miRNA)**, which guide the **RNA-induced silencing complex (RISC)** to target mRNA molecules for silencing.

Statement I is correct: RNA interference (RNAi) is a **natural cellular mechanism** used to **regulate gene expression**. It acts as a **"gene silencing" process** where **specific RNA molecules** (like siRNA or miRNA) **bind to a target messenger RNA (mRNA)** and signal for its destruction. Since the mRNA is degraded, it cannot be translated into a protein, effectively "silencing" that specific gene.

It operates at the **post-transcriptional level** and plays an important role in processes like **defense against viruses, control of transposable elements, and regulation of normal cellular functions**.

Small interfering RNA (siRNA) and microRNA (miRNA) are key players in RNAi and **they have different origins**:

- **siRNA** is typically derived from long **double-stranded RNA (dsRNA)** precursors. These can be exogenous (e.g., from viruses or experimentally introduced) or endogenous. These dsRNA molecules are processed by the enzyme **Dicer (enzyme)** into short siRNA fragments that guide the silencing complex to target mRNA.
- **miRNA** is derived from endogenous **single-stranded RNA** transcripts. It is transcribed as a **primary miRNA (pri-miRNA)** that folds into a **double-stranded hairpin structure (pre-miRNA)** before being processed into mature miRNA.

Statement II is incorrect: RNA-based editing or RNA-targeted gene regulation refers to modern biotechnological approaches that **modify or regulate RNA molecules instead of DNA**. The **fundamental difference between RNA and DNA editing** is:

- **DNA Editing** (e.g., CRISPR-Cas9): Physically cuts and changes the DNA sequence. **These changes are permanent** and can be passed down to offspring.
- **RNA-based Editing/Interference**: Targets the "messengers" (mRNA) rather than the "blueprints" (DNA). Because the original DNA remains untouched, the **effect is temporary (transient) and reversible**. Once the treatment stops or the RNA molecules degrade, the cell resumes normal protein production.

One of the major advantages of RNA-based approaches is that their effects are **reversible and transient, reducing the risk of permanent off-target mutations**.

Source:) <https://www.thehindu.com/opinion/open-page/the-therapy-for-the-future/article70423280.ece>

<https://www.thehindu.com/sci-tech/science/rna-editing-promises-to-go-where-dna-editing-cant/article68836223.ece>

<https://www.newindianexpress.com/xplore/2025/Jan/01/the-pendulum-of-scientific-innovation>

Subject:) Science and Technology

Topic:) Biotechnology

Subtopic:)

Q.77)

Ans) b

Exp) Option b is the correct answer.

White phosphorus is a **toxic, wax-like, highly reactive allotrope of phosphorus that ignites spontaneously on exposure to atmospheric oxygen** (around $\sim 30^{\circ}\text{C}$), **burning at very high temperatures and producing dense white smoke of phosphorus pentoxide**.

- It is extensively used in military munitions (artillery shells, bombs, rockets, grenades) as an incendiary agent, smoke-screen generator, and illumination device.
- Its incendiary nature allows it to **ignite buildings, vegetation, and enemy targets**, while the thick smoke is used to obscure visibility and conceal troop movements.
- White phosphorus **adheres to surfaces, including human skin**, and continues burning until oxygen is cut off, causing severe chemical and thermal burns (often to the bone) and can result in multi-organ failure due to systemic toxicity.
- **Though not completely banned**, its **use is regulated under international humanitarian law**, particularly restricting its use in civilian-populated areas due to its indiscriminate and severe humanitarian impact.

Recent context:

White phosphorus has been **frequently in news (2026)** due to reports of its use in the **Israel-Lebanon conflict**. The Human Rights Watch accused Israel of using white phosphorus in new Lebanon attacks.

Hence, **it is most commonly associated with incendiary munitions in warfare**.

Source: <https://www.thehindu.com/news/international/human-rights-watch-accuses-israel-of-using-white-phosphorus-in-new-lebanon-attacks/article70721156.ece>

<https://www.thehindu.com/opinion/op-ed/the-legality-of-using-white-phosphorus/article67458154.ece>

<https://www.thehindu.com/news/international/israel-iran-war-live-march-9-2026-mojtaba-khamenei-chosen-as-next-supreme-leader-missiles-fired-world-news-updates/article70720987.ece>

Subject: Science and Technology

Topic: Defence

Subtopic:

Q.78)

Ans) a

Exp) Option a is the correct answer.

A **desert air cooler operates** on the **principle of evaporative cooling**, a physical process in which liquid water absorbs heat from the surroundings and evaporates into water vapour.

During evaporation, water requires energy known as **latent heat of vaporization**, which it draws from the surrounding air.

As a result, the air loses heat and its temperature decreases, **producing a cooling effect**.

In a desert cooler, warm ambient air is drawn through water-soaked cooling pads by a fan.

When this air passes over the wet surface, a portion of the water evaporates, absorbing heat from the air and thereby **reducing the air temperature** before it is blown into the room.

Statement I is correct: The **cooling effect of a desert cooler is more efficient on a hot and dry day than on a hot and humid day**.

On a hot and dry day, the **surrounding air has low relative humidity**, meaning it contains relatively little water vapour. Because of this, **the air has a greater capacity to absorb additional moisture**, allowing water to evaporate more readily from the cooler pads. **Greater evaporation leads to greater heat absorption, resulting in more effective cooling.**

However, **on a humid day**, the **air already contains a large amount of water vapour** and is closer to saturation. Under such conditions, the air cannot easily absorb additional moisture. As a result, the rate of evaporation decreases, which **significantly reduces the cooling efficiency of the desert cooler**.

Statement II is correct: The **rate of evaporation depends mainly on temperature and humidity**:

- **Higher temperature** increases the kinetic energy of water molecules, making it easier for them to escape from the liquid surface into the vapour phase.
- **Lower humidity** means the surrounding air has more capacity to hold additional water vapour, thereby **facilitating faster evaporation**.

Thus, **high temperature combined with low humidity creates the most favourable conditions for evaporation**.

Link between the statements:

Since desert coolers rely entirely on the evaporation of water to produce cooling, their efficiency directly depends on how rapidly evaporation occurs. Dry air allows faster evaporation, which leads to greater absorption of heat and stronger cooling.

Therefore, the **physical principle stated in Statement II correctly explains why a desert cooler works more effectively on a hot and dry day, as stated in Statement I.**

Thus, **Both Statement I and Statement II are correct and Statement II explains Statement I.**

Source:) <https://www.egyankosh.ac.in/bitstream/123456789/110531/1/Unit-3.pdf> Page: 70.

<https://egyankosh.ac.in/bitstream/123456789/29868/1/Unit-7.pdf> Page: 219.

Subject:) Science and Technology

Topic:) Physics

Subtopic:)

Q.79)

Ans) d

Exp) Option d is the correct answer.

Zoonosis refers to **infectious diseases** that are **naturally transmitted between vertebrate animals and humans**. These diseases are caused by a wide range of pathogens including **viruses, bacteria, parasites, and fungi**.

Zoonotic diseases typically **originate in animal reservoirs** such as mammals, birds, or other wildlife, and may **reach humans through direct contact with infected animals, consumption of contaminated animal products, or through vectors** such as mosquitoes and ticks. In some cases, once the pathogen enters the human population, it may further spread through **human-to-human transmission**.

- Almost **60% of all human infections** and **75% of newly emerging infectious diseases are of zoonotic origin**. Examples include: **Rabies, Nipah Virus Disease, Crimean-Congo Haemorrhagic Fever, Marburg Virus Disease, Kyasanur Forest Disease, Ebola Virus Disease, Avian Influenza, Plague, Leptospirosis, and Brucellosis**.
- Zoonotic diseases are significant from a public health perspective, as they account for a large proportion of new and re-emerging infectious diseases worldwide.
- Their prevention often requires an integrated **“One Health” approach**, which recognizes the close interconnection between human health, animal health, and the environment

Option I is correct: Kyasanur Forest Disease (KFD)

Kyasanur Forest Disease is a **tick-borne viral haemorrhagic fever** endemic to **parts of India’s Western Ghats region**, particularly Karnataka.

- It is **caused by the Kyasanur Forest Disease virus (KFDV)**, a member of the **Flavivirus genus** (family Flaviviridae).
- The disease was **first identified in 1957** in the **Kyasanur forest area of Karnataka**, which gave the disease its name.
- The **virus circulates in nature among small mammals** (such as rodents and shrews), **monkeys, and ticks**.

Transmission cycle:

- The **primary vector** is the **hard tick Haemaphysalis spinigera**. **Humans usually become infected through the bite of an infected tick** while entering forested areas.
- **Monkeys** (especially langurs and macaques) **act as amplifying hosts**; sudden deaths of monkeys in forests often serve as an early warning of outbreaks.
- **Humans are considered accidental or dead-end hosts**, meaning they do not significantly contribute to further transmission.

Because the disease circulates naturally between wild animals and ticks and occasionally infects humans, **it is classified as a zoonotic disease**.

Option II is correct: Rabies

Rabies is one of the **oldest known zoonotic viral diseases, causing severe encephalitis** (inflammation of the brain) that is **almost always fatal once symptoms appear**.

- It is **caused by the rabies virus**, which belongs to the genus *Lyssavirus* in the family *Rhabdoviridae*.

- The **virus primarily infects mammals**, including dogs, bats, foxes, raccoons, wolves, and other carnivores.

Transmission:

- The virus is **transmitted mainly through the bite or scratch of an infected animal**, as the virus is present in the animal's saliva.
- In many developing countries, **domestic dogs are the main reservoir**, responsible for the vast majority of human cases.
- After entering the body, the virus travels along peripheral nerves to the central nervous system, eventually causing fatal neurological symptoms.

Because rabies is naturally maintained in animal populations and transmitted to humans through animal bites, it is **considered a classic example of a zoonotic disease**.

Option III is correct: Crimean-Congo Haemorrhagic Fever (CCHF)

Crimean-Congo Haemorrhagic Fever is a **severe viral haemorrhagic fever** with a wide geographic distribution across Africa, Asia, the Middle East, and parts of Europe.

- It is **caused by the Crimean-Congo haemorrhagic fever virus (CCHFV)**, belonging to the genus **Orthonairovirus** of the family **Nairoviridae**.
- The virus is maintained in nature in a **tick-animal-tick cycle**.

Transmission cycle:

- The **principal vectors are ticks** of the genus *Hyalomma*.
- **Domestic animals** such as cattle, sheep, goats, and camels **often become infected** but usually **do not show severe symptoms**.
- **Humans can acquire infection through:**
 - o Bites of infected ticks,
 - o Contact with blood or tissues of infected livestock, especially during slaughtering,
 - o Human-to-human transmission through contact with infected blood or bodily fluids in healthcare settings.

Because the virus circulates among ticks and livestock and occasionally spills over to humans, **CCHF is clearly classified as a zoonotic disease**.

Option IV is correct: Marburg Virus Disease

Marburg virus disease is a **severe viral haemorrhagic fever closely related to Ebola virus disease**.

- It is **caused by the Marburg virus**, belonging to the genus *Marburgvirus* in the family *Filoviridae*.
- The **natural reservoir host** of the virus is the **Egyptian fruit bat**.

Transmission cycle:

- Initial human infections typically occur after prolonged exposure to bat colonies, especially in caves or mines inhabited by infected bats.
- Once introduced into the human population, the **virus can spread through direct contact with blood, secretions, organs, or other bodily fluids of infected individuals**.

Therefore, **Marburg virus disease is also categorized as a zoonotic disease**.

Source:) <https://www.emro.who.int/about-who/rc61/zoonotic-diseases.html> <https://www.thehindu.com/sci-tech/health/icmr-initiates-human-clinical-trials-for-improved-kyasanur-forest-disease-vaccine/article70606917.ece>
<https://www.thehindu.com/sci-tech/health/rabies-the-cruel-and-expensive-disease-of-indias-most-impovert/article70411044.ece>
<https://indianexpress.com/article/cities/ahmedabad/26-year-old-in-gandhinagar-village-diagnosed-with-crimean-congo-haemorrhagic-fever-10463004/>
<https://www.thehindu.com/news/international/ethiopia-says-three-dead-in-marburg-virus-outbreak/article70290938.ece>
https://ncdc.mohfw.gov.in/wp-content/uploads/2025/09/ZDPHI_Manual_2025-1.pdf

Subject:) Science and Technology

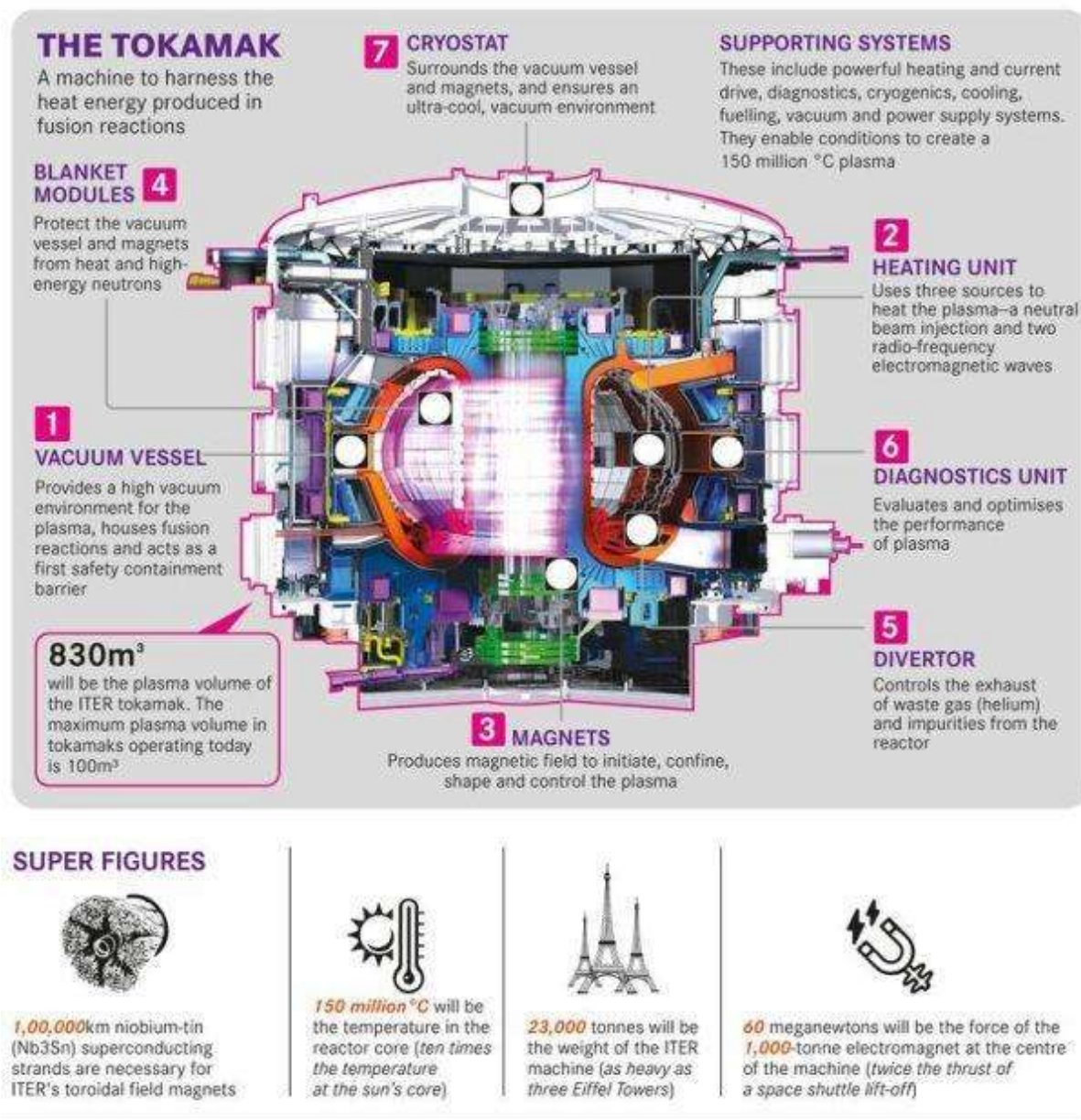
Topic:) Biology

Subtopic:)

Q.80)

Ans) c

Exp) Option c is the correct answer.



Statement I is correct: A Tokamak is a type of **experimental nuclear-fusion device** designed to **confine extremely hot plasma within a toroidal (doughnut-shaped) chamber**. Since the plasma reaches temperatures of tens of millions of degrees Celsius, **no solid material can contain it directly**. Therefore, **powerful magnetic fields** generated by external coils and plasma current are **used to confine and stabilize the plasma inside the chamber**. This method is known as **magnetic confinement**, which allows the plasma to remain suspended without touching the reactor walls.

Statement II is correct: Conventional nuclear power plants generate electricity through **nuclear fission**, in which **heavy nuclei** such as uranium or plutonium split into smaller nuclei, releasing energy. In contrast, **tokamak reactors** aim to **produce energy through nuclear fusion**, the **same process that powers stars like the Sun**.

- In fusion experiments, the most commonly used fuel consists of the **hydrogen isotopes deuterium and tritium**.

- When these nuclei fuse under extremely high temperature and pressure, **they form helium** and release a large amount of energy in the form of high-energy neutrons.
- **This process releases significantly higher energy than fission** per unit of mass and **produces significantly less long-lived radioactive waste than fission.**
- Tokamak reactors attempt to achieve and sustain these conditions for controlled fusion reactions.

Knowledge base: The **ITER** (International Thermonuclear Experimental Reactor) is one of the largest international scientific collaborations aimed at **demonstrating the feasibility of controlled nuclear fusion** as a future energy source. The project is being **constructed in France** and involves **seven major partners**, including the **European Union, India, China, Russia, the United States, Japan, and South Korea.**

ITER is designed to build the world's largest experimental tokamak reactor, intended to produce a large amount of fusion energy and demonstrate that fusion can generate significantly more power than the energy required to sustain the reaction. **India has played a key role in building some of its most critical infrastructure**

Source:) <https://www.thehindubusinessline.com/business-tech/the-pragya-tokamak/article70772940.ece>

<https://www.thehindu.com/sci-tech/science/china-east-fusion-reactor-beats-greenwald-plasma-limit-widens-path-to-power/article70500702.ece>

<https://www.energy.gov/science/doe-explainstokamaks>

<https://www.thehindu.com/sci-tech/science/worlds-largest-fusion-project-reaches-construction-milestone-with-indias-help/article69521130.ece>

<https://www.iter.org/machine/what-tokamak>

<https://forumias.com/blog/nuclear-fusion-technology-evolution-challenges-and-future-potential/>

Subject:) Science and Technology

Topic:) Nuclear Technology

Subtopic:)

Q.81)

Ans) d

Exp) Option d is the correct answer.

Climate-Resilient Agriculture (CRA) refers to a set of farming approaches designed to **adapt to climate variability, enhance productivity, and reduce environmental risks**

Statement I is correct. CRA incorporates advanced tools such as **soil microbiome analysis** to better understand soil health, nutrient cycling, and microbial activity. This helps farmers make informed decisions about crop management and improves long-term soil resilience.

Statement II is correct. CRA actively promotes the use of **biotechnology tools**. Techniques such as **genome editing, biofertilizers, and biopesticides** are important components of CRA, as they help develop climate-tolerant crops and reduce dependence on harmful chemical inputs.

Statement III is correct. CRA uses **precision nutrient management tools** like the **Leaf Colour Chart (LCC)** to optimize fertilizer application, especially nitrogen.

CRA integrates **climate information services** such as weather forecasts, early warning systems, and **risk management tools** like crop insurance. These help farmers make timely decisions and reduce vulnerability to extreme weather events.

Statement IV is incorrect. CRA does **not completely prohibit and eliminate chemical pesticides**. Instead, it focuses on **reducing and optimizing their use** through integrated pest management, biological alternatives, and sustainable practices.

Source:) <https://www.thehindu.com/sci-tech/science/why-does-india-need-climate-resilient-agriculture-explained/article70460248.ece>

<https://www.thehindubusinessline.com/economy/agri-business/why-climate-resilient-farming-must-become-a-priority-for-marginal-farmers/article70743925.ece>

https://www.icar-crida.res.in/assets_c/img/Books/statewise_prommising_climate_%20resilient_technologies_reports_13%20states&1UT/PCRT_Punjab_%20State.pdf

<https://epubs.icar.org.in/index.php/IndFarm/article/download/173369/63505/499855>

<https://www.ils.res.in/climate-resilient-agriculture-and-its-used/>

Subject:) Geography

Topic:) Agriculture

Subtopic:)

Q.82)

Ans) d

Exp) Option d is the correct answer.

Soybean is an important **oilseed and leguminous crop** in India, widely cultivated for its oil and protein content.

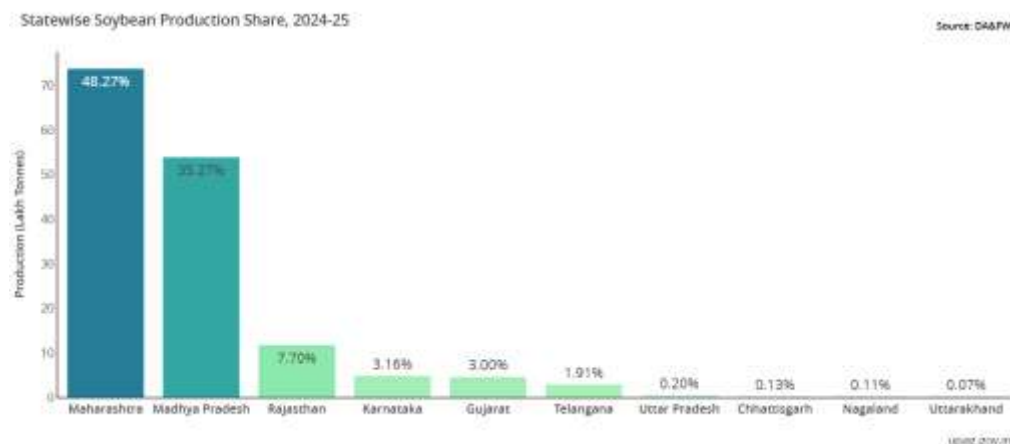
Option I is incorrect. Soybean is **not a rabi crop**, instead it is primarily a **kharif crop** grown during the monsoon season under warm and moist conditions. It requires temperatures around 25–30°C and adequate rainfall.

Soybean is a **leguminous crop**, which means it has the ability to fix atmospheric nitrogen through symbiotic bacteria (Rhizobium) present in its root nodules. This improves soil fertility and reduces the need for nitrogen fertilizers.

Option II is incorrect. In 2024-25, Maharashtra is the **largest producer of soybean in India**, contributing a significant share of over 48% of total production.

Maharashtra 73.7 Lakh Tonne (48.27 % of India's total soybean production)

Madhya Pradesh 53.8 Lakh Tonne (35.27 % of India's total soybean production)



Option III is incorrect. Soybean production in India has **not been consistently increasing after 2020**. It has shown fluctuations in the years 2023-24 and 2025-25, due to factors such as monsoon variability, changes in sowing area, and market conditions.

Source:) <https://www.thehindu.com/news/national/madhya-pradesh/trouble-in-soy-state/article70197763.ece>

<https://www.indiabudget.gov.in/economicsurvey/doc/stat/tab118.pdf>

<https://upag.gov.in/dash-reports/apymajorstates?rtab=Area%2C+Production+%26+Yield&rtype=dashboards>

https://agriwelfare.gov.in/Documents/TimeSeries_2nd_AE_2025_26_En.pdf

<https://ugcmoocs.inflibnet.ac.in/assets/uploads/1/51/1420/et/Lecture%2011%20-%20Text200320101003031717.pdf>

<https://egyankosh.ac.in/bitstream/123456789/45335/1/Block-2.pdf> under heading Processing of Soybean

Subject:) Geography

Topic:) Crops [India]

Subtopic:)

Q.83)

Ans) c

Exp) Option c is the correct answer.

The **Western Ghats** and **Eastern Ghats** are two major mountain systems of Peninsular India. The Western Ghats are majorly continuous and higher in elevation, while the Eastern Ghats are discontinuous and more eroded. These two ranges **converge at the Nilgiri Hills**, forming an important physiographic junction in southern India. Located at the tri-junction of Tamil Nadu, Karnataka, and Kerala, they are a key geographical and ecological region.

Source:) NCERT class 11-Indian Physical Environment-chapter-2-PHYSIOGRAPHY- page 13

Subject:) Geography

Topic:) Physiography of India

Subtopic:)

Q.84)

Ans) b

Exp) Option b is the correct answer.

The **Kaveri River** is one of the major peninsular rivers of India, originating in the Brahmagiri hills of the Western Ghats and flowing into the Bay of Bengal. Its basin spans Karnataka, Kerala, and Tamil Nadu. Important tributaries of the Kaveri include **Hemavati, Kabini, Bhavani, and Amravati**,

Musi, Tungabhadra and Ghataprabha are tributaries of the **Krishna River**.

Indravati is a tributary of the **Godavari River**

Source:) <https://ncert.nic.in/textbook/pdf/iess103.pdf> page 22

Subject:) Geography

Topic:) Drainage [India]

Subtopic:)

Q.85)

Ans) a

Exp) Option a is the correct answer.

In arid and semi-arid regions, **wind (aeolian processes)** plays a significant role in shaping landforms through **erosion** and deposition. Erosional landforms are created when wind removes or wears away materials, while depositional landforms form when transported sediments are deposited.

Statement I is incorrect. Esker is not a wind landform. Eskers are **glacial depositional features**, formed by sediments deposited by meltwater streams beneath glaciers.

Statement II is incorrect. Barchan is a depositional landform. Barchans are crescent-shaped sand dunes formed by wind deposition in desert areas

Statement III is correct. Inselbergs is an **erosional landform** formed by wind. **They** are **residual hills** rising abruptly from the level ground, formed due to long-term erosion of surrounding areas. They are relicts of an original plateau and are typical of desert landscapes

Statement IV is incorrect. Loess is a depositional landform, consisting of fine wind-blown silt deposited over large areas.

I

Source:) NCERT: Class 11; FUNDAMENTALS OF PHYSICAL GEOGRAPHY, chapter-LANDFORMS AND THEIR EVOLUTION, page 67-69

GC Leong: Chapter 7: Arid or Desert landforms: page 57-60

Subject:) Geography

Topic:) Landforms

Subtopic:)

Q.86)

Ans) a

Exp) Option a is the correct answer.

Statement I is correct. Lightning is caused by a **sudden electrostatic discharge** between oppositely charged regions within a cloud or between a cloud and the ground. This discharge releases a large amount of energy in the form of light and heat.

Statement II is incorrect. Ice in clouds is **not piezoelectric**, meaning it does not directly generate electric charge simply by being compressed. Although ice particles play a role in charge buildup through collisions, the mechanism is more complex (**involving processes like charge transfer and flexoelectric effects**), **not simple piezoelectricity**.

Source:) <https://www.thehindu.com/sci-tech/science/bending-ice-could-explain-how-lightning-is-born-in-thunderstorms/article70055649.ece>

<https://indianexpress.com/article/explained/explained-how-lightning-strikes-and-why-it-kills-7401198/>

<https://ncert.nic.in/textbook/pdf/hecu106.pdf> page 90

Subject:) Geography

Topic:) Climatology

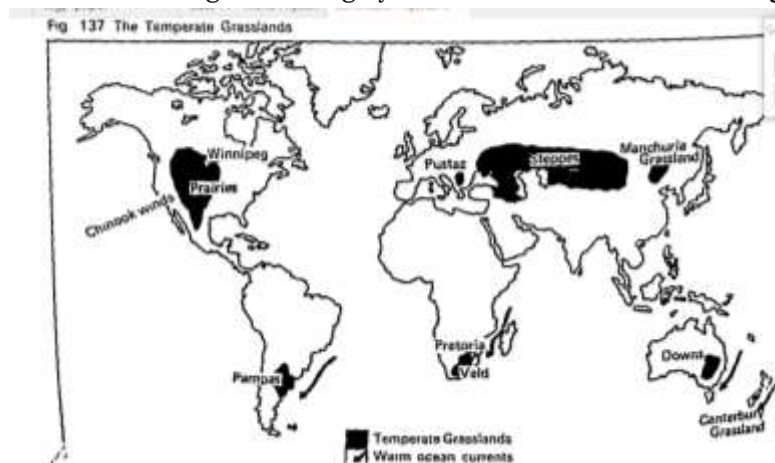
Subtopic:)

Q.87)

Ans) c

Exp) Option c is the correct answer.

Grasslands of the world are broadly classified into **tropical grasslands (savannas)** and **temperate grasslands** based on climate and location. **Temperate grasslands** are found in the **mid-latitude regions** of the world and are characterized by **moderate rainfall, hot summers, and cold winters**. They have vast stretches of grass with very few trees due to limited rainfall. These regions are highly suitable for **commercial farming and livestock rearing**.



Option a is incorrect. Llanos are tropical grasslands found in Venezuela and Colombia.

Option b is incorrect. Campos are tropical grasslands of Brazil.

Option c is correct. Temperate grasslands are found in mid-latitudes and include regions such as **Steppes (Eurasia)**, **Pampas (Argentina)**, **Prairie grassland (USA, Canada)**, **Veld (South Africa)**, **Downs (Australia)**, and **Pustaz (Hungary)**.

Option d is incorrect. Campos are tropical grasslands of Brazil.

Source:) Chapter 20: Temperate continental steppe climate, page-148

Subject:) Geography

Topic:) Vegetation [World]

Subtopic:)

Q.88)

Ans) d

Exp) Option d is the correct answer.

Waves represent the **transfer of energy, not the movement of water mass**. Water particles move in circular paths, while the wave energy travels across the surface. Waves play a crucial role in shaping coastlines through erosion, transportation, and deposition.

Statement I is correct. The **height of waves depends on wind strength, duration, and fetch (area over which wind blows)**. Stronger winds blowing for longer durations over larger areas transfer more energy to the water, resulting in higher waves.

Statement II is correct. As waves approach the shore, they **slow down due to friction with the sea floor**, and eventually break. At this stage, the **wave energy is released along coastlines**, contributing to coastal erosion and deposition.

Statement III is correct. Even after the wind stops, waves can continue to travel as **swell waves**. This happens because of **gravity**, which restores the displaced water and maintains the wave motion for long distances.

Source:) chapter 14-MOVEMENTS OF OCEAN WATER, page-116,117

<https://egyankosh.ac.in/bitstream/123456789/112060/3/Unit-11.pdf> 209-,211

Subject:) Geography

Topic:) Oceanography

Subtopic:)

Q.89)

Ans) a

Exp) Option a is the correct answer.

During the **hot weather season in India (March-May)**, several **local storms and winds** develop due to intense heating of the land.

Row I is incorrect. Towards the end of summer, there are pre-monsoon showers which are a common phenomena in Kerala and coastal areas of Karnataka (not UttarPradesh). Locally, they are known as mango showers since they help in the early ripening of mangoes.

Row II is incorrect. **Nor-westers** are dreaded evening thunderstorms in Bengal and Assam. Their notorious nature can be understood from the local nomenclature of 'Kalbaisakhi', a calamity of the month of Baisakh. These showers are useful for tea, jute and rice cultivation. In Assam, these storms are known as "Bardoisila". They are not associated with Punjab.

Row III is correct. Blossom Showers are **pre-monsoon local showers** of the hot weather season. They **occur mainly in Kerala** and nearby areas of the Western Ghats. These showers **help coffee flowers to blossom**, which is crucial for coffee cultivation.

Source:) <https://ncert.nic.in/textbook.php?kegy1=4-6> page 35

Subject:) Geography

Topic:) Monsoon

Subtopic:)

Q.90)

Ans) c

Exp) Option c is the correct answer.

The Moon shows the same face to the Earth at all times, a phenomenon explained by its rotational dynamics. This happens because the Moon's rotation period is equal to its orbital period, a condition achieved over time due to gravitational interactions with the Earth.

Option a is incorrect. **Orbital Precession** refers to the **slow change in the orientation of an orbit over time**, such as the gradual shift of the Moon's orbital path.

Option b is incorrect. **Axial Tilt** refers to the **tilt of a celestial body's axis relative to its orbital plane**, which is responsible for seasons on planets like Earth.

Option c is correct. Tidal Locking is the phenomenon where a body's rotation period becomes equal to its orbital period due to gravitational interactions. In the Moon's case, Earth's gravity slowed its rotation over time until it became synchronized, causing the **same side to always face Earth**.

Option d is incorrect. Lunar Libration refers to the **apparent oscillation or slight wobbling of the Moon**, which allows us to see slightly more than half (~59%) of its surface over time.

Source:) <https://science.nasa.gov/moon/tidal-locking/>

Subject:) Geography

Topic:) Earth and the Universe

Subtopic:)

Q.91)

Ans) a

Exp) Option a is the correct answer.

Butylated Hydroxyanisole (BHA) and **Butylated Hydroxytoluene (BHT)** are **synthetic compounds** widely used in the food industry for their **antioxidant properties**. Their primary function is to **prevent the oxidation of fats and oils**, which helps slow rancidity, maintain flavour, colour, and nutritional quality, and extend the shelf life of fat-containing processed foods such as snacks, baked goods, cereals, and animal fats.

Common Applications: They are widely used in processed foods that have high fat content, such as:

- Vegetable oils and butter.
- Potato chips and snack foods.
- Cereals and baked goods.
- Chewing gum.

Health Concerns: Both BHA and BHT are classified as possible human carcinogens and may **disrupt hormone function**. Long-term exposure has raised **concerns about liver and kidney toxicity**.

Regulatory Status: In India, the **FSSAI** (Food Safety and Standards Authority of India) permits their use but prescribes **strict maximum permissible limits** (e.g., up to **400 mg/kg** in specific edible oils) to ensure consumer safety.

Source:) <https://economictimes.indiatimes.com/news/new-updates/7-dangerous-ingredients-hidden-in-your-everyday-packaged-snacks/bha-and-bht-butylated-hydroxyanisole/toluene/slideshow/122922860.cms>

https://fssai.gov.in/upload/uploadfiles/files/Compendium_Nutra_29_09_2021.pdf Page: 114.

<https://theweek.com/health/food-additives-banned-united-states-european-union>

<https://www.newindianexpress.com/states/karnataka/2024/Mar/18/colours-of-danger>

Subject:) Science and Technology

Topic:) Biotechnology

Subtopic:)

Q.92)

Ans) c

Exp) Option c is the correct answer.

The **Doppler Effect** is a fundamental physical phenomenon that describes the **apparent change in the frequency or wavelength of a wave when there is relative motion between the source of the wave and the observer**.

- If the source and observer move closer, the **frequency appears higher (blueshift)**, while if they move apart, the **frequency appears lower (redshift)**.
- This principle is **extensively applied** across various fields such as **astronomy** (to study the motion of celestial bodies and expansion of the universe), **medical science** (in Doppler ultrasound for measuring blood flow), **meteorology** (in weather radars to track storms), and **navigation and defense systems** (in radar and sonar to determine the speed of moving objects).
- It is **named after Christian Doppler**, the physicist who **first proposed the concept of the Doppler effect**.

Option I is correct: Measurement of the velocity of an aircraft or a submarine

The **Doppler Effect** is applied in radar (for aircraft) and sonar (for submarines) to measure velocity. When waves are reflected from a moving object, the frequency of the reflected waves changes depending on whether the object is moving towards or away from the source; this frequency shift is used to calculate speed accurately.

Option II is correct: Determination of the expansion of the universe.

The **expansion of the universe** is studied through the **redshift of light from distant galaxies**. Due to the Doppler Effect, light from galaxies moving away from Earth shifts towards the red end of the spectrum, indicating their recessional velocity and thereby providing evidence for an expanding universe.

Option III is incorrect: Determination of the age of fossils

This is not an application of the Doppler Effect. Determining the age of fossils is done through **Radiometric Dating** (specifically **Carbon-14 dating** for organic remains).

This process involves measuring the ratio of stable isotopes to unstable, decaying radioactive isotopes within the sample. Because radioactive decay happens at a fixed, known rate (half-life), scientists can calculate how long ago the organism died.

This is a nuclear chemistry process and does not involve wave-frequency shifts or relative motion.

Option IV is correct: Measurement of blood flow in arteries

In medical science, this is called a **Doppler Ultrasound**. A handheld **transducer** sends **high-frequency sound waves into the body**. These waves **bounce off moving targets**, specifically **red blood cells**.

Because the blood cells are in motion, the **reflected sound wave returns to the transducer at a different frequency** than the one originally emitted. This **frequency shift** allows the machine to **create a visual representation of blood flow velocity and direction, helping doctors identify blood clots or narrowed heart valves**.

Source:) <https://indianexpress.com/article/explained/explained-sci-tech/doppler-weather-radars-india-10429115/>
<https://timesofindia.indiatimes.com/science/explained-what-is-a-doppler-radar-network-and-what-are-its-usages/articleshow/97035898.cms>

Subject:) Science and Technology

Topic:) Space

Subtopic:)

Q.93)

Ans) b

Exp) Option b is the correct answer.

Distributed Ledger Technology (DLT) is a **digital system of recording transactions** in which the **ledger is shared, synchronized, and maintained across multiple nodes** (computers) in a network, rather than being controlled by a single central authority.

Key Characteristics:

- **Decentralization:** It operates on the principle of **decentralization**, where each participant (node) holds a copy of the ledger, and **transactions are validated through consensus mechanisms** (such as **Proof of Work or Proof of Stake**).
- **Immutability:** Once a transaction is recorded and verified, it becomes tamper-resistant, enhancing trust and transparency.
- **No intermediary:** It **eliminates the need for intermediaries** like banks or clearing houses, reducing costs and increasing efficiency.
- **Transparency:** All participants in the network can see the history of transactions, which builds trust.
- **Blockchain:** It is the **most prominent and widely recognized form of DLT**, where data is stored in blocks linked in a chronological chain using cryptography.
- **Cryptocurrencies:** **Digital currencies** like Bitcoin and Ethereum **use blockchain** (a form of DLT) to securely record peer-to-peer transactions, ensure transparency, and prevent double-spending.

Hence, **DLT is most commonly associated with blockchain technology and cryptocurrencies, making option (b) correct.**

Source:) <https://www.pib.gov.in/PressReleasePage.aspx?PRID=2182023®=3&lang=2>

<https://indianexpress.com/article/upsc-current-affairs/upsc-essentials/knowledge-nugget-what-is-cryptocurrency-and-how-is-the-technology-associated-with-it-relevant-for-the-upsc-prelims-2025-exam-9798658/>

<https://ddnews.gov.in/en/from-certificates-to-property-records-blockchain-adoption-expands-across-india/>
Subject:) Science and Technology

Topic:) Information and Communication Technology

Subtopic:)

Q.94)

Ans) c

Exp) Option c is the correct answer.

Changes occurring in matter can broadly be **classified into physical changes and chemical changes**, based on whether the composition of the substance remains the same or a new substance is formed.

Physical Change:

A physical change is a **process in which only the physical properties of a substance**; such as **shape, size, colour or state are altered**, while its **chemical composition remains unchanged**.

Key features:

- **No new substance is formed**
- Generally **reversible**
- Only physical properties change

Examples:

- Melting of ice
- Sublimation of camphor
- Boiling of water

Chemical Change:

A chemical change is a process in which **one or more substances undergo a transformation to form new substances** with **different chemical compositions and properties**.

Key features:

- **New substance(s) are formed**
- Generally **irreversible**
- Involves **change in chemical composition**
- Often accompanied by heat, light, gas, or colour change

Examples:

- Rusting of iron
- Burning of magnesium ribbon
- Fermentation of sugar
- Digestion of food

Option a is incorrect: Rusting of iron:

This is a chemical change. When iron reacts with oxygen and moisture in the environment, it forms iron oxide (rust). The new substance has chemical properties completely different from metallic iron, and the process is irreversible under normal conditions.

Option b is incorrect: Burning of magnesium ribbon:

This is a chemical change. Magnesium reacts with oxygen on burning to form magnesium oxide, a new compound with properties distinct from magnesium metal. This reaction is highly exothermic and cannot be reversed to get back magnesium.

Option c is correct: Sublimation of camphor:

Sublimation is a physical change in which camphor directly changes from solid to vapour without passing through the liquid state. **No new substance is formed**, its **chemical composition remains the same**, and the **process is reversible** by

condensation. No chemical reaction occurs; only the state of matter changes., which is a **key characteristic of physical processes**.

Option d is incorrect: Fermentation of sugar:

This is a chemical change. Sugar is converted by yeast into ethanol and carbon dioxide. Both the products have chemical properties different from the original sugar, and the reaction is irreversible under normal conditions.

Source:) <https://ncert.nic.in/textbook/pdf/gesc105.pdf>

Subject:) Science and Technology

Topic:) Chemistry

Subtopic:)

Q.95)

Ans) d

Exp) Option d is the correct answer.

Glyphosate is a non-selective, broad-spectrum systemic herbicide widely used in agriculture and horticulture to control weeds and unwanted plants.

It works by **inhibiting an enzyme** in the **shikimic acid pathway**, which is essential for synthesizing certain amino acids in plants. Without these amino acids, plant growth stops, leading to the death of the weeds.

It is **not a fertilizer** (does not provide nutrients), **not an insecticide** (does not target insects), and **not a fungicide** (does not prevent fungal infections). Its effectiveness, low cost, and versatility have made it one of the **most widely used herbicides globally**.

Why is it in the News?

- **Health Concerns:** The International Agency for Research on Cancer (IARC), an arm of the WHO, **classified glyphosate as "probably carcinogenic to humans"** in 2015.
- **Agricultural Impact:** It is famously used in conjunction with **Genetically Modified (GM) crops**, specifically **"Herbicide-Tolerant" (Ht) varieties**. These crops are engineered to survive glyphosate, allowing farmers to spray entire fields to kill weeds without harming the crop itself.
- **Regulatory Status in India:** In India, the **use of glyphosate is officially restricted**. The **Ministry of Agriculture** issued a notification in 2022 stating **that it can only be applied by Pest Control Operators (PCOs) to prevent indiscriminate use**, particularly in non-designated areas like HtBT cotton fields.

Source:) <https://www.thehindu.com/sci-tech/energy-and-environment/survey-flags-hazardous-pesticideusein-south-asiamaharashtrasyavatmalcotton-belt-under-spotlight/article70388011.ece>

<https://www.thehindubusinessline.com/economy/agri-business/citing-health-risks-bks-asks-government-to-ban-glyphosate/article69367779.ece>

<https://npic.orst.edu/factsheets/glyphogen.html>

<https://www.iarc.who.int/featured-news/media-centre-iarc-news-glyphosate/>

Subject:) Environment

Topic:) Pollution

Subtopic:)

Q.96)

Ans) b

Exp) Option b is the correct answer.

The SRY (Sex-determining Region Y) gene is a key genetic factor in human sex determination. It is found on the Y chromosome. Its presence or absence decides whether the embryo develops male or female characteristics.

In simple terms, human embryos follow a default female pathway. The SRY gene acts like a switch. When it is present and active, it turns on male development.

The SRY gene, located on the Y chromosome initiates the formation of testes. These testes produce testosterone, which leads to male sexual development. This makes it the primary trigger for maleness.

Hence, the SRY gene, located on the Y chromosome, triggers development along the male pathway. In its absence, development proceeds along the default female pathway.

Knowledge base: SRY in an XX individual

Contrary to the general scenario, very rarely, a mutation can occur that transfers the SRY gene from the Y chromosome to the X. These mutations are called translocations. When a sperm bearing a translocation X chromosome fertilises an egg, an XX baby is born with the SRY gene.

In most cases, XX individuals with the SRY gene develop testes and become biological males. The testes produce the male hormone testosterone and initiate the development of other male sexual characteristics. However, they don't produce mature sperm because other Y chromosomal genes required for sperm formation are absent. Since these males are sterile, the translocation is not transmitted to future generations.

Source:) <https://www.thehindu.com/sci-tech/science/will-the-baby-be-a-boy-or-a-girl-sry-gene-claims-to-have-the-answer/article69196574.ece> ; <https://www.thehindu.com/sport/world-athletics-mandates-gene-test-for-female-category-eligibility/article69875401.ece> ; <https://www.thehindu.com/sport/athletics/athletics-afi-asks-females-to-do-sry-test-before-worlds/article69933697.ece>

Subject:) Science and Technology

Topic:) Biology

Subtopic:)

Q.97)

Ans) a

Exp) Option a is the correct answer.

Statement I is correct. Section 14 of the J&K Reorganisation Act, 2019 (as amended in 2023) provides for a total of 90 elected seats in the Legislative Assembly of J&K.

In addition, Sections 15, 15A and 15B provide that the LG may nominate a total of up to five members – two women, two Kashmiri migrants and one displaced person from Pakistan occupied Kashmir – to the J&K Assembly.

Statement II is correct and statement III is incorrect. The composition of the Legislative Assemblies of three Union Territories (UT) are governed by acts of Parliament. Section 3 of the Government of National Capital Territory of Delhi Act, 1991 provides for 70 elected members in the Delhi Assembly. There are no nominated MLAs in the Delhi Assembly. Section 3 of the Government of Union Territories Act, 1963 provides for 30 elected members in the Legislative Assembly of Puducherry. It also provides that the Union government may nominate up to three members to the Puducherry Assembly.

Source:) <https://www.thehindu.com/news/national/who-decides-nominations-to-ut-assemblies-explained/article69948834.ece>

<https://scroll.in/latest/1085452/j-k-lg-can-nominate-five-assembly-members-without-ut-governments-approval-centre-tells-high-court> ; <https://www.thehindu.com/news/national/jk-assemblies-nominated-seats-outside-the-realm-of-govt-l-g-has-statutory-powers-mha/article69917126.ece>

Subject:) Polity

Topic:) Legislature

Subtopic:)

Q.98)

Ans) b

Exp) Option b is the correct answer.

A Financial Emergency under **Article 360** is designed to protect the **financial stability and credit of India**. When such a situation arises, the Constitution strengthens the hands of the **Union executive**. The normal federal balance shifts, and the Centre gains the authority to **guide, control, and discipline financial management across States**.

Statement I is incorrect. The Constitution does not empower **Parliament** to take such action. Under **Article 360(4)a**, it is the power of the Union executive which can issue direction to State which may include a provision requiring all Money

Bills or other Bills to which the provisions of article 207 apply to be reserved for the consideration of the President after they are passed by the Legislature of the State. So this is the executive power of the Union and not the power of Parliament by law.

Statement II is correct. According to **Article 360(3) of the Constitution of India**, during the period Proclamation of financial emergency is in operation, the executive authority of the Union shall extend to the giving of directions to any State to observe such canons of financial propriety as may be specified in the directions, and to the giving of such other directions as the President may deem necessary and adequate for the purpose. Also, Notwithstanding anything in this Constitution—(a) any such direction may include—(i) any such direction may include—(i) **a provision requiring the reduction of salaries and allowances of all or any class of persons serving in connection with the affairs of a State.**

Source:) The Constitution of India – Article 360

Mastering Indian Polity by Forum IAS - 0th Edition – Chapter 14 – Emergency – Page 256

Subject:) Polity

Topic:) Emergency Provisions

Subtopic:)

Q.99)

Ans) c

Exp) Option c is the correct answer.

The **BHARATPOL portal** is a new initiative aimed at strengthening **international police cooperation**. It provides a **real-time digital platform** for Indian law enforcement agencies to connect with **INTERPOL's global network**. This helps in faster investigation, tracking of fugitives, and handling **cross-border crimes** such as drug trafficking, human trafficking, and arms smuggling.

The **Central Bureau of Investigation (CBI)** has developed the **BHARATPOL portal**. It functions as India's **National Central Bureau (NCB) for INTERPOL**.

Through this portal, the CBI enables **direct and seamless linkage of Indian agencies with INTERPOL across 195 countries**. It also allows access to **multiple INTERPOL databases** and supports features like **notices, alerts, and real-time information sharing**.

Knowledge Base:

- The **National Crime Records Bureau (NCRB)** works mainly on **crime data collection and analysis**. It manages databases like CCTNS.
- The **National Counter Terrorism Centre (NCTC)** is related to **counter-terrorism coordination**.
- The **Enforcement Directorate (ED)** investigates **financial crimes**, especially money laundering and foreign exchange violations.

Source:) <https://www.pib.gov.in/PressReleasePage.aspx?PRID=2090877®=3&lang=2>

<https://www.pib.gov.in/PressReleasePage.aspx?PRID=2099539®=3&lang=2>

Subject:) Polity

Topic:) Union Executive

Subtopic:)

Q.100)

Ans) a

Exp) Option a is the correct answer.

The constitutional scheme of **official language (Part XVII, Articles 343–351)** reflects a careful balance between **linguistic diversity and administrative uniformity**.

Statement I is correct. Under **Article 343(1)**, the Constitution provides **Hindi in Devanagari script as the official language of the Union**. However, recognising practical constraints, **Article 343(2) and (3)** allow the continued use of **English for official purposes**, even beyond the initial 15-year period, through parliamentary law, the **Official Languages Act, 1963**.

Statement II is correct. As per Article 345, the State Legislature is empowered to adopt any language(s) in use in the State or Hindi as its official language. This provision respects India's federal structure and linguistic diversity.

Statement III is incorrect. As per Article 348 (2) of the Constitution of India, the Governor of a State may, with the previous consent of the President, authorise the use of the Hindi language, or any other language used for any official purposes of the State, **in proceedings in the High Court having its principal seat in that State:**

Provided that nothing in this clause shall apply to any judgment, decree or order passed or made by such High Court.

Source:) The Constitution of India - Part XVII - Official Language

Subject:) Polity

Topic:) Official Language

Subtopic:)